

Materials in Radiative Cooling Technologies

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Radiative cooling (RC) is a carbon-neutral cooling technology that utilizes thermal radiation to dissipate heat from the Earth's surface to the cold outer space. Research in the field of RC has garnered increasing interest from both academia and industry due to its potential to drive sustainable economic and environmental benefits to human society by reducing energy consumption and greenhouse gas emissions from conventional cooling systems. Materials innovation is the key to fully exploit the potential of RC. This review aims to elucidate the materials development with a focus on the design strategy including their intrinsic properties, structural formations, and performance improvement. The main types of RC materials, i.e., static-homogeneous, static-composite, dynamic, and multifunctional materials, are systematically overviewed. Future trends, possible challenges, and potential solutions are presented with perspectives in the concluding part, aiming to provide a roadmap for the future development of advanced RC materials.

1. Introduction

Radiative cooling (RC) is a process that utilizes thermal radiation to transfer heat from an object to the external environment. It is well-established from thermodynamics that heat spontaneously flows from the hot to the cold body when hot and cold objects engage in radiation exchange. Thus, the enormous temperature difference between the Earth (about 300 K) and the cold universe (3 K) drives terrestrial objects to emit thermal energy into the expansive cosmic space, leading to RC.^[1] Additionally, infrared (2.5–25 μm), especially the infrared with the wavelength range of 8–13 μm , known as the “atmospheric transparency window,” plays a crucial role in RC. Within this specific spectral region, the atmosphere demonstrates an exceptionally high transmittance that coincides with the peak of blackbody radiation at ≈ 300 K. Heat escapes from the surfaces of matter in the form of thermal

radiation into the cold space, resulting in a temperature drop at night. On the other hand, since cooling demands typically peak during the day, overcoming solar heating under direct sunlight is crucial for achieving passive daytime RC (PDRC). Consequently, the PDRC technology needs precise spectral management, allowing the surface of objects to possess a low absorptivity within the solar irradiation wavelengths (0.3–2.5 μm) to minimize solar heat absorption, while having high emissivity to maximize heat dispersion into cold outer space.^[2] Although nighttime RC has a history spanning decades,^[3] the first PDRC was demonstrated in 2014 by Fan's group.^[2] Since then, the development of RC materials has emerged as a frontier in energy-efficient materials and has witnessed rapid

advancement. In recent years, a variety of materials have been proposed for efficient RC (Figure 1), including polymers^[4] and ceramics.^[5] Additionally, researchers have explored various structural designs, including multilayer structures,^[6] photonic crystals,^[7] hybrid metamaterials,^[8] and hierarchically porous arrays.^[9] Moreover, since RC needs to be suppressed in winter, progress has been made in developing dynamic electro-,^[10] thermo-,^[11] mechano-,^[12] and photochromic^[13] RC materials. Furthermore, various forms of RC materials have been developed, including film,^[8] coating,^[4c] paint,^[14] aerogel,^[15] glass,^[5b] and textile^[16] for vehicle and building space cooling,^[14,17] solar cell system cooling,^[18] personal thermal management,^[19] and even energy generation^[20] and water harvesting.^[21]

RC technology has been flourishing over the past decade, and many reviews have summarized the development of RC technology from different perspectives,^[1,22] such as principles, design, fabrication, applications, and commercialization. For example, some reviews have introduced the fundamental principles of RC,^[1,22a] which lay the foundation for the theoretical analysis and experimental design of radiative coolers. Others have examined the structure design and performance evaluation of RC devices,^[22b–g] which are crucial for establishing the structure–property relationship, designing effective RC devices, and comparing their cooling performance objectively. Meanwhile, several reviews have discussed the fabrication methods and processes of RC devices and structures,^[22h–j] such as micro-nano fabrication, coating, and electrospinning, and addressed the challenges and solutions in the manufacturing process. Additionally, some review papers have focused on the applications of RC technology,^[22k–q] especially in building^[22m–o] and human thermal

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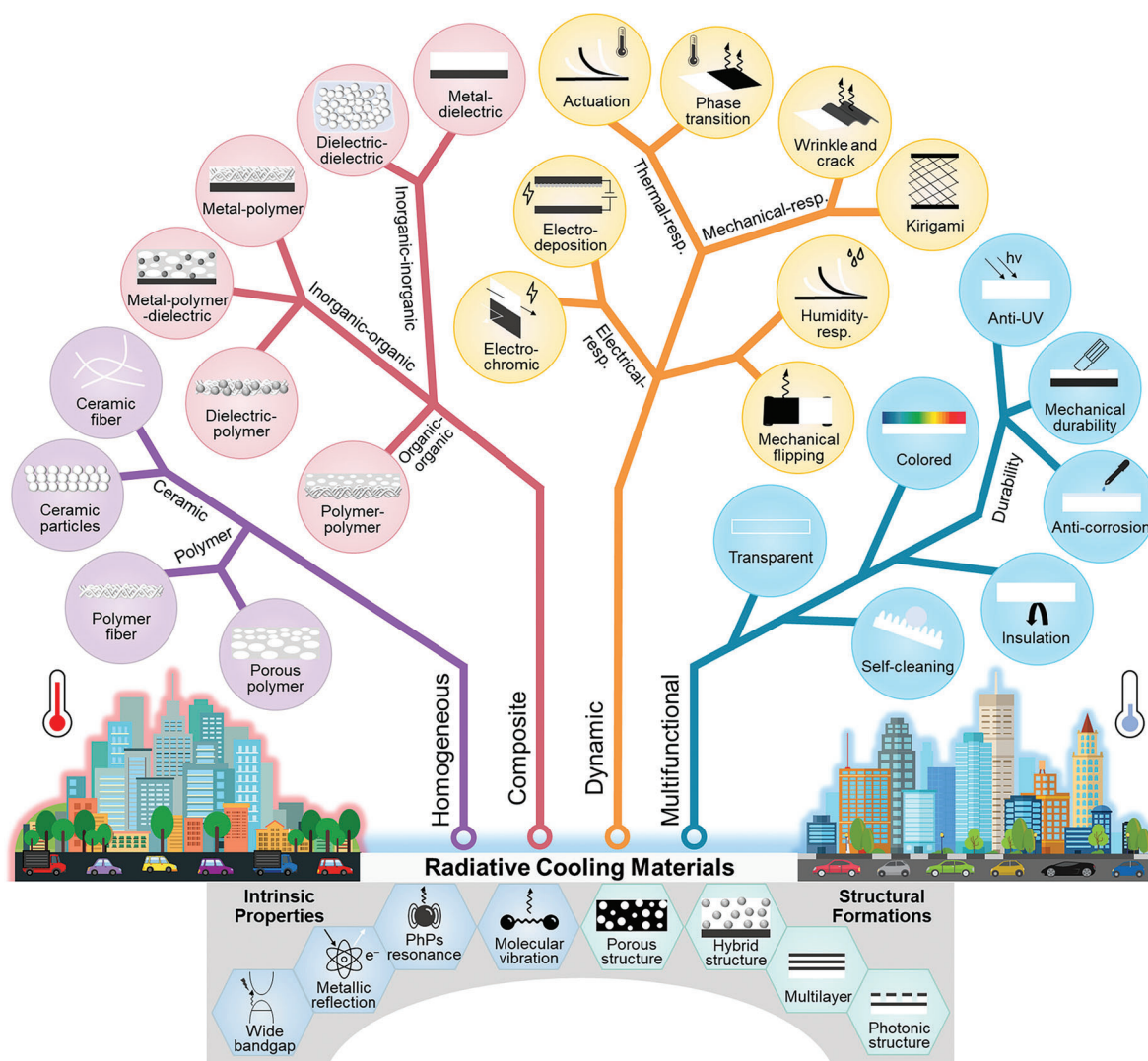


Figure 1. Schematic representation of the main categories of RC materials, including static-homogeneous, static-composite, dynamic, and multifunctional materials. RC materials serve as a crucial “bridge” for cooling terrestrial objects. The design strategy of these materials, including intrinsic properties and structural formations, serves as the pillar of the “bridge”.

management,^[22p,q] and investigated the effects of environmental factors on cooling performance in practical scenarios,^[1,22r] such as climate, atmospheric humidity, and wind speed. Lastly, a few reviews have summarized the commercialization progress and prospects of RC technology, highlighting the obstacles and challenges in commercialization.^[22s–u] However, to the best of our knowledge, most reviews have not systematically summarized the RC materials and their properties, which are key to the performance and efficiency of RC systems/devices. This leads to a lack of systematic and comprehensive understanding of different categories of RC materials. Therefore, in this review, we aim to fill this gap by providing a comprehensive overview of RC materials, including their design strategies, fabrication, properties, and challenges.

In this review, the recent advances in RC materials are overviewed from the perspective of material properties, design and utilization. First, we focus on the design strategy of RC ma-

terials, elucidating their intrinsic properties and structural manipulation. Following that, we comprehensively review the state-of-the-art RC materials research from the perspectives of compositions, mainly including static-homogeneous, static-composite, dynamic, and multifunctional coolers. The perspectives and recommendations of RC materials are concluded for future development. Through this review, we hope to offer a comprehensive and updated reference on RC materials and techniques, inspire more new ideas and innovations, and promote the application of RC technologies.

2. Design Strategy of Radiative Cooling Materials

Spectral control is exceptionally important to achieve high RC efficiency, which refers to the ability of materials to selectively emit thermal radiation without absorbing undesirable sunlight.^[2] Based on the wavelength band it emits, there are two designs

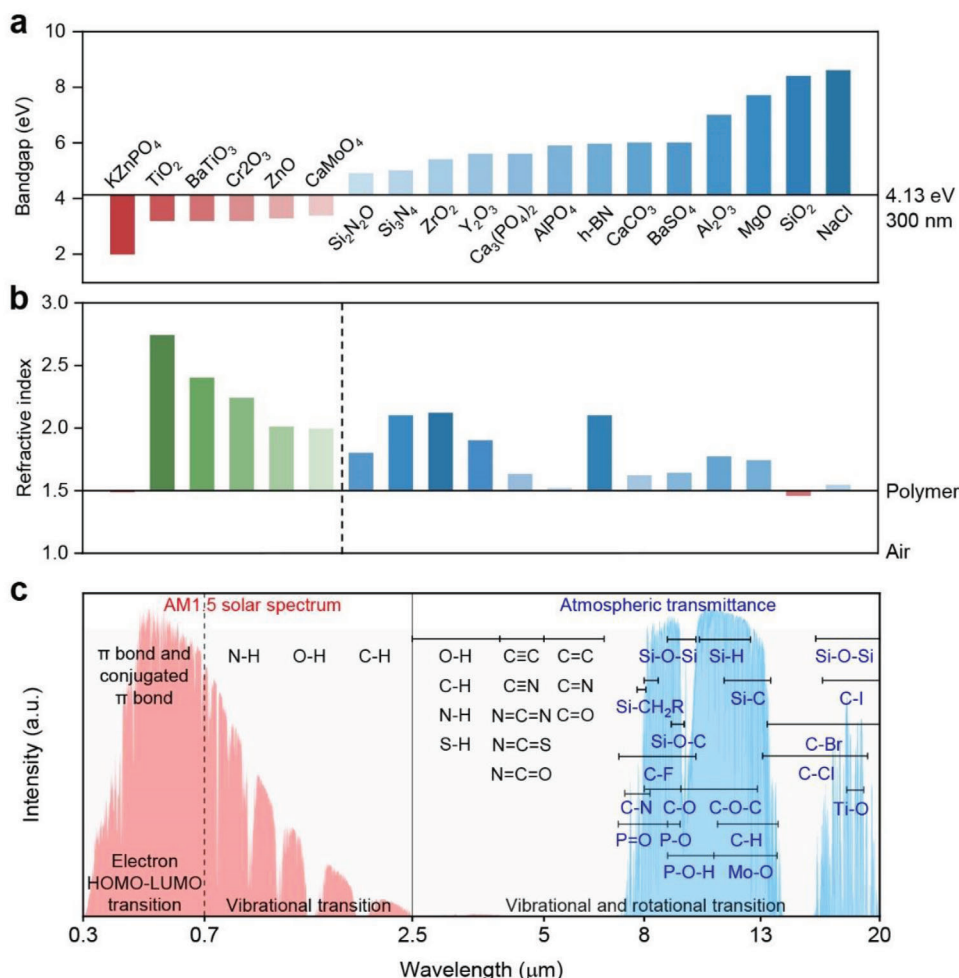


Figure 2. Fundamental intrinsic properties of materials. a) The band gap and b) the refractive index of typical inorganic materials. c) Spectra of chemical bonds or functional groups in the range of 0.3–20 μm. The normalized AM1.5 Global (AM1.5G) solar spectrum (red shaded area) and atmospheric transmission spectrum (blue shaded area) are indicated as references. Adapted with permission.^[24] Copyright 2020, Elsevier.

for RC materials: 1) A near blackbody design with a near-unity broadband emissivity ($\epsilon_{\text{Broadband}}$) spans from 2.5 to 25 μm. 2) A wavelength-selective design with a high long-wave infrared emissivity (ϵ_{LWIR}) within the atmospheric window of 8–13 μm. The broadband design will have a larger net cooling power when its surface temperature is above or close to the ambient temperature, while the wavelength-selective design allows for sub-ambient cooling.^[22u] The intrinsic properties and structural formations of RC materials are the critical factors to precisely control the optical properties (e.g., absorptivity and emissivity) over a broad range of wavelengths. These wavelengths span from ultraviolet (UV) to infrared, covering solar band (0.3–2.5 μm) and broadband infrared (broadband IR, 2.5–25 μm). Solar band consists of UV (0.3–0.36 μm), visible (vis, 0.36–0.78 μm), and near-infrared (NIR, 0.78–2.5 μm), while broadband IR includes mid-wave infrared (MWIR, 3–8 μm)^[23] and long-wave infrared (LWIR, 8–13 μm). Understanding the design strategies of RC materials, including intrinsic material properties and structural formations, is essential in guiding researchers to develop more

novel and effective materials to fully realize the potential of RC technologies.

2.1. Radiative Cooling Based on Intrinsic Material Properties

For optimal cooling performance, the intrinsic properties of RC materials are required to possess extremely low absorptivity in the solar spectrum (A_{sol}) to limit solar heating input, while having high emissivity for dissipating excess heat effectively to the sky and, eventually, to the cold outer space.^[22u] Here, we provide the summary of the key intrinsic properties of materials that are important for RC performance (Figure 2). In Figure 2a,b, we summarize the band gap and the refractive index of typical inorganic materials, respectively. Figure 2c illustrates the vibrational regions of chemical bonds or functional groups.^[24] Based on these intrinsic properties, we will elaborate on how they affect the spectral properties of RC materials across the whole spectral range (Figure 3a), focusing on the low A_{sol} of inorganic dielectrics, metals, and polymers, and the high emission of

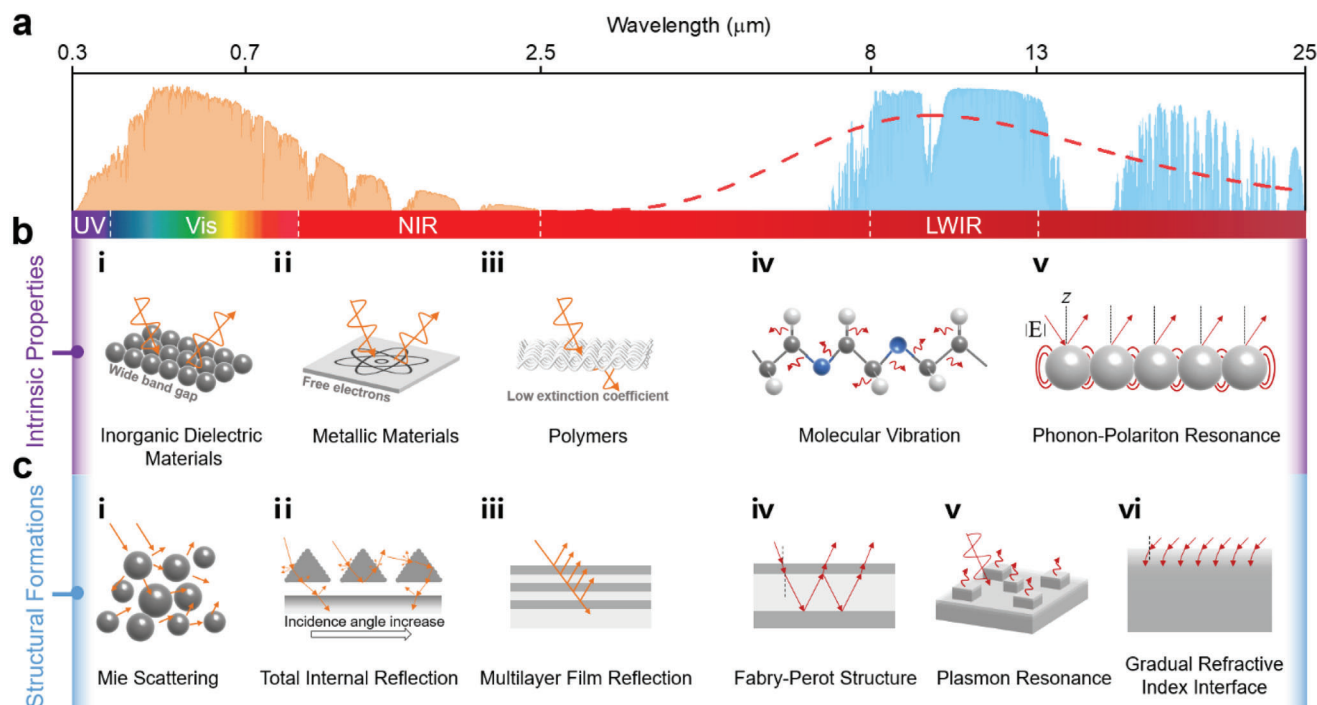


Figure 3. Summary of design strategies for representative RC materials. a) The normalized AM1.5G solar spectrum (orange shaded area) and the atmosphere transmission spectrum (blue shaded area), along with the blackbody radiation spectrum at ≈ 300 K (red dashed line). b) RC based on intrinsic material properties. Schematic of i) inorganic dielectric materials, ii) metallic materials, iii) polymers, iv) molecular vibration, and v) phonon-polariton resonance. c) RC through structural formations. Schematic of i) Mie scattering, ii) total internal reflection, iii) multilayer film reflection, iv) Fabry–Perot structure, v) plasmon resonance, and vi) gradual refractive index interface.

molecular vibrations and phonon-polariton resonances (Figure 3b–v).

2.1.1. Low Solar Absorption

Inorganic Dielectric Materials: Dielectric materials with wide band gaps exhibit minimal A_{sol} . Solar radiation includes UV, visible, and NIR light ($0.3\text{--}2.5\ \mu\text{m}$), with photon energies ranging from 0.49 to $4.13\ \text{eV}$.^[25] If the energy required to excite an electron from the valence band to the conduction band is larger than the photon energy of solar radiation, electrons will be unlikely excited by photons in the sunlight band, leading to low A_{sol} (Figure 3b–i). Figure 2a illustrates the band gaps of some commonly used inorganic dielectric materials. A candidate dielectric material, including but not limited to barium sulfate (BaSO_4 , $\approx 6\ \text{eV}$),^[26] aluminum oxide (Al_2O_3 , $\approx 8.7\ \text{eV}$),^[27] silicon oxide (SiO_2 , $\approx 8.4\ \text{eV}$),^[28] magnesium oxide (MgO , $7.6\ \text{eV}$),^[29] and silicon nitride (Si_3N_4 , $5\ \text{eV}$),^[30] possesses band gap exceeding the energy range of solar photons, which is advantageous for minimizing A_{sol} . Besides, in the visible range, most inorganic dielectric materials typically exhibit a higher n compared to polymers ($n \approx 1.5$) and air ($n \approx 1$); for example, materials like calcium carbonate (CaCO_3 , $n \approx 1.62$), BaSO_4 ($n \approx 1.64$), Al_2O_3 ($n \approx 1.77$), hexagonal boron nitride (h-BN, $n \approx 2.1$), titanium dioxide (TiO_2 , $n \approx 2.74$) (Figure 2b). This property enables effective sunlight scattering at interfaces constructed with air or polymers, resulting in a high solar reflectance (R_{sol}).

Metallic Materials: Metallic materials, such as smooth silver (Ag) and aluminum (Al) films, are regarded as metal mirrors for RC coolers due to their exceptional R_{sol} .^[31] The high reflectance of metals can be attributed to the abundance of free electrons. When sunlight illuminates the surface of a metal, the electromagnetic waves induce vibrations and acceleration in the free electrons (Figure 3b–ii). Consequently, the motion of these electrons generates a current, giving rise to an electromagnetic wave that possesses an identical frequency but an opposite direction to the incident electromagnetic wave, which serves as the reflected wave.^[32]

Polymers: Polymers exhibiting low extinction coefficients (k) in the solar spectrum are highly desirable in RC applications.^[4c] The k describes the degree of attenuation of electromagnetic waves in a medium. A material with a low k in the solar wavelengths will weakly attenuate the sunlight intensity when it transmits, leading to a low A_{sol} (Figure 3b–iii).^[33] From a micro perspective, the UV–vis of the material is intimately linked to the electronic within its molecules. Typically, the electrons of π (π) bonds are weakly bound and can be excited from the π to the π^* orbital by sunlight.^[34] When materials are illuminated with light energy that matches the electronic transitions in the molecules, the electrons absorb the light and then rise from the ground state (the highest occupied molecular orbital, HOMO) to a higher energy orbital (the lowest unoccupied molecular orbital, LUMO). Therefore, polymer molecules that contain π bonds or conjugated π bonds, such as CN, N=N, and C=S bonds, are not desired candidate for RC material due to their high A_{sol} .

Additionally, the NIR absorption of the material is closely related to the vibrational transitions of its chemical bonds. In the NIR region, certain overtones or combinations of fundamental stretching bonds, such as N—H, O—H, or C—H stretching vibrations (common in polymers), must be carefully considered to strike a balance to minimize A_{sol} . Polymers such as polydimethylsiloxane (PDMS), poly(vinylidene fluoride-co-hexafluoropropene) (P(VdF-HFP)), and polymethyl methacrylate (PMMA) are great candidates for RC technology where solar heat gain needs to be minimized.^[4c,9,31]

2.1.2. High Emission

Molecular Vibration: In general, broadband IR absorption is closely related to the vibrational and rotational transition of chemical bonds or functional groups of materials. Figure 2c illustrates the wavelength distribution of the absorption peaks corresponding to the vibrations of various chemical bonds.^[24] Organic polymers, with their diverse chemical bonds and functional groups, exhibit a wide range of vibrational behavior compared to inorganic materials. The bending and stretching vibrations of the chemical bonds of polymers typically occur within the spectral range of 400–4000 cm^{-1} (2.5–25 μm), such as C=O (1825–1725 cm^{-1} , 5.5–5.8 μm),^[16] C—O (1050–1310 cm^{-1} , 7.6–9.5 μm), and C—H (700–900 cm^{-1} , 11.1–14.3 μm).^[35] These vibrations resonate with the broadband IR, resulting in strong absorption and emission (Figure 3b-iv). Moreover, simultaneous resonance with two or more molecular vibrations enhances broadband IR absorption and emission.^[4d]

Phonon-Polariton Resonance: Phonon-polaritons (PhPs) are hybrid modes that result from the coupling of electromagnetic waves with lattice vibrations in polar dielectrics (Figure 3b-v), such as in silicon carbide (SiC), SiO_2 , and BaSO_4 . These materials exhibit a distinct plasmonic behavior in the broadband IR region within a material-specific spectral window known as the Reststrahlen band.^[36] Within the Reststrahlen band, the materials confine electromagnetic waves propagating along the material interface through surface PhPs, which can enhance the local density of states and the radiative decay rate of the materials, thus leading to robust emission capabilities.^[37] Furthermore, by manipulating the geometry and periodicity of material structures, the dispersion and polarization of PhPs can be precisely controlled to achieve selective and directional emission.^[8,37,38]

2.2. Radiative Cooling through Structural Formations

In addition to the aforementioned inherent optical properties of RC materials, the introduction of subwavelength structural design gives novel optical characteristics that precisely control the light across a wide spectral range. By precisely manipulating the geometric shape, size, and period, the reflection, transmission, and emission can be effectively regulated across different wavelength ranges. This enables the material to possess high R_{sol} while maintaining high emissivity. Mie scattering, total internal reflection, multilayered reflection, Fabry–Perot (F–P) structure, plasmon resonance, and gradual refractive index interference are generally adopted to enhance RC performance (Figure 3c-i–vi), with the details elaborated below.

2.2.1. Reducing Solar Absorption

Mie Scattering: Mie scattering arises when electromagnetic waves encounter scatterers of sizes comparable to or larger than the wavelength of the waves, leading to the elastic scattering of light on the surface or within the object.^[39] The properties of the scattered wave depend on the incident wave and the particle. Mie scattering generally impacts the spectral range from the near-UV to the NIR, and becomes more pronounced for longer wavelengths. In the presence of numerous scatterers, strong Mie scattering occurs, resulting in diffusing or backscattering that reflects a significant portion of the incident light, thereby achieving a high reflectivity (Figure 3c-i).^[38b]

Total Internal Reflection: Total internal reflection happens when light waves arrive at an interface from a medium with higher n (also referred to as a denser medium) to a medium with lower n (also known as a less dense medium). The light will be entirely reflected into the denser medium rather than being refracted into the less dense medium while the incident angle exceeds the critical angle (Figure 3c-ii). Leveraging total internal reflection, certain photonic structures (e.g., hollow cylindrical, pyramidal, and triangular structures) can achieve a high R_{sol} .^[40]

Multilayer Film Reflection: Multilayer film reflection exploits the optical interference phenomenon of multilayer dielectric films with disparate refractive indices to selectively reflect light (Figure 3c-iii). Various approaches are employed to design multilayer reflective film structures, including alternating high and low n , periodic film, and gradient film. The alternating high and low refractive index design yields a composite reflection coefficient that enables a broad reflection bandwidth and a high reflectivity.^[2] Periodic thin film design involves the repetition of a multilayer film structure with defined periodicity to form a Bragg reflector, thereby achieving high reflection within narrow or multiple bands.^[41] Gradient film design involves the gradual variation of n , resulting in a gradient multilayer reflective film structure.^[42] By optimizing the n and thickness of each film layer, the multilayer film structure can exhibit high reflectivity within specific wavelength ranges.

2.2.2. Enhancing Emission

Fabry–Perot Structure: The F–P resonance cavity (F–P resonator) is formed by three layers, comprising two mirrors separated by a lossless spacer (Figure 3c-iv), which can produce an absorption enhancement via the wave interference effect.^[43] When the incident light falls on the F–P structure, it undergoes multiple reflections and interferences between the two mirrors, leading to standing waves inside the cavity. Moreover, by adjusting the thickness or refractive index of the medium layer, the resonance wavelength can be tailored to facilitate spectral selective emission within a specific wavelength range. It is imperative to select suitable materials for the mirrors and medium. Typically, the mirror utilizes metals or semiconductors to ensure that light undergoes multiple reflections and interferences within the cavity. The lossless spacer is commonly based on semiconductors or dielectrics. The design of the F–P resonator can be categorized into planar and microstructure types. The planar structure is to stack two parallel mirrors directly together, which is easy to

fabricate and adjust, but its resonance peak is narrow and sensitive to incident angle variations.^[11a] The microstructure is to make the mirror or medium into small geometric shapes, such as spheres and cylinders. The microstructure F–P resonator is difficult to fabricate and adjust. However, its resonance peak is wide and insensitive to the incident angle variations.^[44]

Plasmon Resonance: Plasmon resonance refers to the phenomenon of light resonance coupling between the surface plasmon waves. Various approaches have been proposed to boost broadband IR radiation via plasmon resonances. For instance, bulk metals with micrometer-deep grooves were employed to enable spectral and angular control of thermal radiation by modulating the width, depth, and period of the grooves.^[45] Another approach involves a metal/dielectric/metal gap plasmonic cavity which capitalizes on the gap plasmon resonance (Figure 3c-v), to control the intensity and spectral characteristics of thermal radiation by manipulating the widths of the gap plasmon cavities.^[46] Therefore, by precisely designing and optimizing nano/microstructures, metamaterials/metasurfaces with plasmonic resonance effects can be obtained, which produce enhanced infrared absorption/emission.^[47]

Gradual Refractive Index Interface: The gradual refractive index (GRIN) interface represents an optical interface where the refractive index gradually varies along the interface, instead of an abrupt alteration. GRIN provides a smooth transition of electromagnetic waves and increased emissivity by avoiding the reflection induced by abrupt refractive index interfaces. The GRIN interface induces a curved or helical path for light propagation (Figure 3c-vi), thereby increasing the optical path length and impedance and allowing more absorption of incident radiation. The GRIN interface can be constructed with a wide range of structures such as porous structure, grating patterns, pyramid morphology, moth-eye, and butterfly-wing, and fabricated with various materials such as silicon, SiO₂, metals, metal oxides, and even polymers.^[4c,40b,48]

In the following Sections 3, 4, 5, and 6, homogeneous, composite, dynamic, and multifunctional RC materials will be elaborated with a specific focus on the material/property/process relationship of various categories (Figure 1).

3. Static-Homogeneous Radiative Cooling materials

Static-homogeneous RC materials stand out for their simplicity and straightforward manufacturing. The easy fabrication and application not only streamline the production but also enhance versatility, making them a practical choice for diverse cooling solutions. Considering the application scenarios, these static-homogeneous materials can be categorized into nighttime RC materials and daytime RC materials. We will focus on introducing the characteristics of RC cooling materials.

3.1. Nighttime Radiative Cooling Materials

Nighttime RC technology has been employed for centuries in tropical and subtropical regions for building cooling and water desalination through freezing.^[22m] Unlike PDRC technology,

which requires materials with specific solar reflectance properties to minimize sunlight absorption, materials for nighttime RC are not restricted by such constraints.^[49] Consequently, a diverse array of materials, including polymers, inorganic solids, composites, and gases, have been explored for potential capability of thermal emission to achieve effective RC.^[49] Extensive research has been conducted on polymers such as polyvinyl fluoride (PVF),^[3a] polyvinylchloride (PVC),^[50] and poly(4-methylpentene) (TPX),^[51] which demonstrate abundant vibrational modes within the atmospheric window region. Inorganic solids, despite having fewer infrared vibration modes compared to polymers, play a crucial role in nighttime RC strategies. TiO₂, BaSO₄, and SiO₂ are notable examples of inorganic solids investigated for RC technology.^[3b,52] Beyond solids, gases have been proposed as encapsulated entities for RC. Ethylene gas, ammonia, and ethylene oxide have garnered attention for their potential contributions to RC.^[53] These gases with highly selective ϵ_{LWIR} need to be encapsulated using infrared-transparent materials (e.g., polyethylene), and integrated with solar reflectors to realize RC devices. Although gases-based RC devices are more complicated than solid RC materials, they introduce a novel dimension to materials selection, paving the way for innovative RC applications.

3.2. Daytime Radiative Cooling Materials

Materials designed for PDRC demand both high ϵ_{LWIR} and high R_{sol} .^[22u] Various nano-micro structures including hierarchically air pores,^[4c,54] inorganic particles,^[5a] and nanofibers,^[4d] have been proposed to effectively scatter sunlight.

3.2.1. Porous Polymers-Based Radiative Cooling Materials

Polymers generally possess abundant IR-absorbing bonds, leading to high IR emissivity. The incorporation of a hierarchical porous structure is expected to strongly reflect solar radiation. Yang et al. introduced a straightforward, cost-effective, and scalable phase-inversion-based approach for fabricating hierarchically porous P(VdF-HFP) coatings with excellent PDRC capability (Figure 4a).^[4c] The P(VdF-HFP) coatings with rich hierarchical pores, featuring $\approx 50\%$ porosity and thickness exceeding 300 μm , exhibit an exceptional, substrate-independent hemispherical R_{sol} of 96% and a ϵ_{LWIR} of 0.97 (Figure 4b). The high R_{sol} ensures high reflection of sunlight from all angles, obviating the necessity for silver reflectors employed in previous designs. In pursuit of sustainability and practicality, Yang et al. further developed water-based paint coatings, that free of organic solvents.^[54b] Through a direct “bottom-up” ball milling approach, they successfully created a uniform micro-assembly of P(VdF-HFP) nanoparticles (NPs). The incorporation of micro- and nanopores among secondary and primary particles significantly enhances light scattering, resulting in excellent PDRC performance. This achievement marks a critical step forward in terms of sustainability and practical implementation.^[54b]

Wood is a significant and sustainable building material that offers both economic and environmental advantages,^[55] with an intrinsic interior porous structure.^[56] Building upon this characteristic, Hu et al. presented a structural PDRC wood achieved

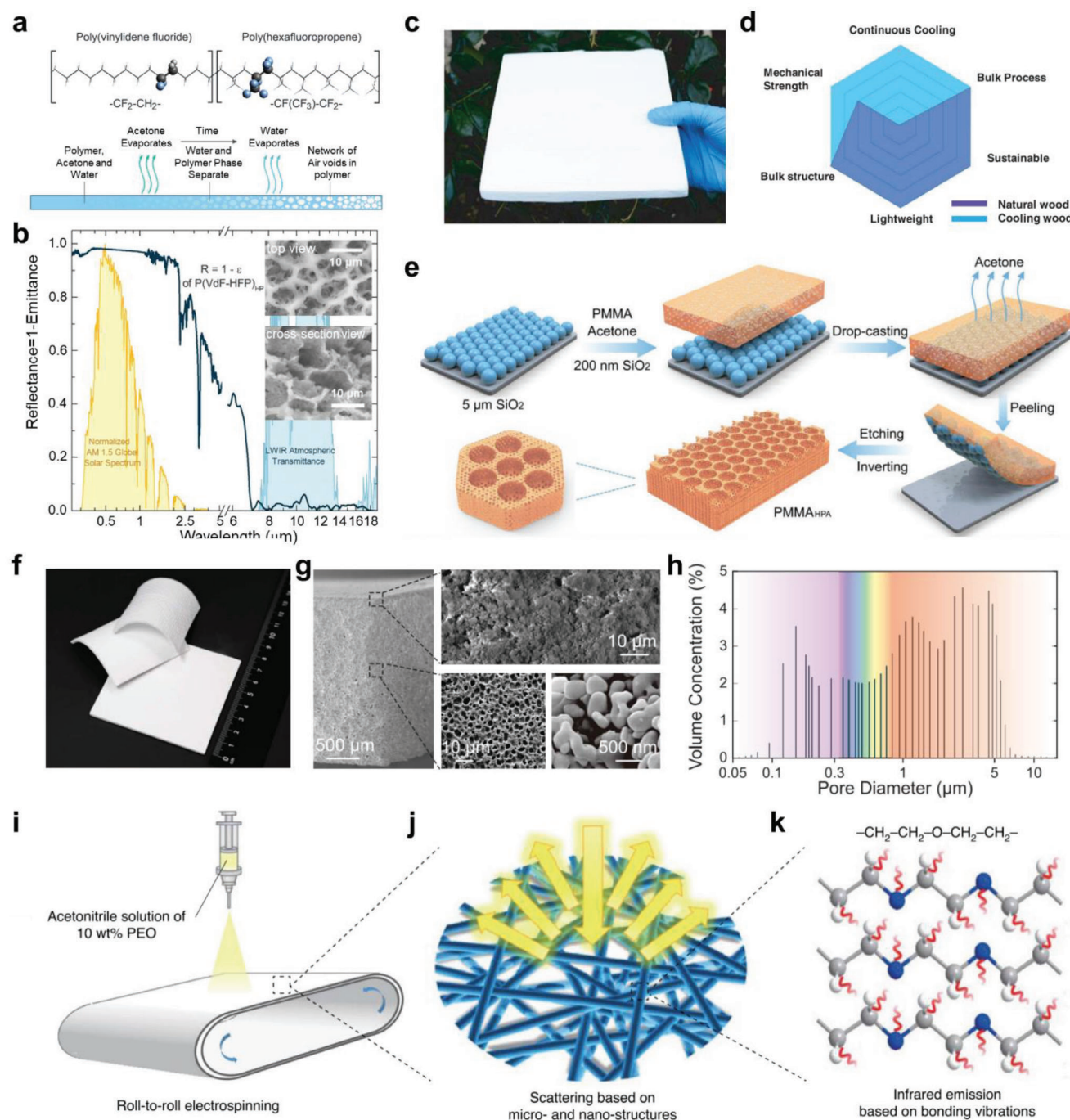


Figure 4. Static-homogeneous RC materials. a) Schematic of the P(VdF-HFP) structure with the VdF and HFP repeating units (top) and the phase inversion process of the formation of a P(VdF-HFP) film (bottom). b) Spectral reflectance of a P(VdF-HFP) film. The inset shows micrographs depicting top and cross-section views of P(VdF-HFP). Reproduced with permission.^[4c] Copyright 2018, AAAS. c) Image of the cooling wood. d) Performance comparison of cooling wood and natural wood. Reproduced with permission.^[4a] Copyright 2019, AAAS. e) Schematic of the fabrication of hierarchically porous array PMMA. Reproduced with permission.^[9] Copyright 2021, Springer Nature. f) Photograph of the cooling ceramics. g) SEM images of a fabricated cooling ceramic sample showing the hierarchically porous structure. h) Volume concentration of pores. Reproduced with permission.^[5a] Copyright 2023, AAAS. i) Schematic of the fabrication process. j) Schematic of the solar scattering of the es-PEO thin film. k) Schematic of the infrared emission based on bonding vibrations. Reproduced with permission.^[4d] Copyright 2021, Springer Nature.

through delignification and densification processes.^[4a] The resulting cooling wood (Figure 4c), characterized by largely disordered mesoporous cellulose structures, features a reflective and hazy surface that effectively scatters solar radiation with angular independent high emission. The high ϵ_{LWIR} is primarily attributed to the infrared absorption of cellulose, indicating significant cooling potential. The distinctive features of this cooling wood, including continuous subambient cooling, robust mechanical strength, bulk structure, low density, sustainability, and bulk fabrication process, position it as an appealing structural alternative to natural wood, contributing to energy-efficient and sustainable building applications (Figure 4d).

In addition to the previously discussed random porous structures, ordered porous structures can also be formed using the chemical etching method. Wu et al. presented a hierarchically structured PMMA film with a combination of micropore array and randomly distributed nanopores for highly efficient day- and nighttime passive RC.^[9] The hierarchical structure is formed using hydrofluoric acid aqueous solution to remove the ordered SiO_2 array and randomized SiO_2 nanospheres (Figure 4e). The fabricated hierarchical PMMA film features ordered symmetrical micropores (diameter $\approx 4.6 \mu\text{m}$) and randomized nanopores (average diameter $\approx 250 \text{ nm}$), achieving a high R_{sol} of 95% and superior ϵ_{LWIR} of 0.98. Meanwhile, a structural PDMS film incorporating built-in three-dimensional (3D) microvoid arrays through a facile sacrificial template method, was proposed to achieve PDRC with a R_{sol} of $\approx 93.4\%$ and a ϵ_{LWIR} of ≈ 0.95 .^[57]

3.2.2. Inorganic Particles-Based Radiative Cooling Materials

Within the realm of inorganic particles-based PDRC materials, the presence of high-index particles, such as TiO_2 particles, presents a challenge that the notable UV absorption (A_{UV}) poses a large unwanted solar absorption, despite their impressive ϵ_{LWIR} . To address this challenge, the adoption of low-index inorganic particles with highly efficient optical scattering emerges as a crucial strategy for achieving robust R_{sol} . An exemplary demonstration of this strategy was seen in a white coating that incorporates randomly packed SiO_2 microspheres, each with a diameter of $0.9 \mu\text{m}$.^[38b] This microsphere arrangement scatters solar radiation without absorption, allowing for effective heat dissipation through high ϵ_{LWIR} (>0.94). Another notable example in this category is Al_2O_3 which is widely used in the production of ceramics. Tso et al. introduced a RC ceramic (Figure 4f) distinguished by highly efficient light scattering and near-perfect R_{sol} of 99.6%.^[5a] This exceptional performance is attributed to a meticulously designed structure featuring a densely packed outer layer and numerous internal voids (Figure 4g), similar to *Cyphochilus*, the whitest known insect on Earth. The distinctive structure is created through a combination of phase inversion and sintering processes. During the phase inversion, a polymer-rich membrane is created, forming an anisotropic porous network that accommodates alumina particles. Subsequent high-temperature sintering facilitates the bonding of alumina particles, resulting in a precisely preserved porous structure characterized by a broader distribution of structural dimensions (Figure 4h). The produced cooling ceramic is exclusively composed of porous alpha-alumina ($\alpha\text{-Al}_2\text{O}_3$), ensuring a low extinction coefficient across the solar

wavelength range, thereby achieving high R_{sol} . Moreover, the vibrational modes of the Al–O chemical bonds in Al_2O_3 contribute to a high extinction peak within the atmospheric window range, resulting in a remarkable ϵ_{LWIR} of 0.97. With a noon-time cooling power exceeding 130 W m^{-2} in an outdoor setting, this cooling ceramic demonstrates its energy-saving potential on a global scale. Key features of the cooling ceramics such as color, weather resistance, mechanical robustness, and the ability to depress the Leidenfrost effect (a phenomenon where liquids form a vapor layer on a hot surface to reduce direct contact and allow the liquid to slide rather than evaporate immediately) solidify its durability and versatility, positioning them for commercial success across various applications, particularly in building construction.

3.2.3. Nano Fibers-Based Radiative Cooling Materials

Fiber-based RC materials, characterized by unique multiscale morphological structures, offer distinct advantages in flexibility and scalability.^[58] Zhu et al. demonstrated the capabilities of a hierarchically designed polymer nanofiber-based film produced through a scalable electrostatic spinning process (Figure 4i).^[4d] This film enables selective LWIR emission and effective sunlight reflection, thereby delivering excellent all-day RC performance. The high R_{sol} (96.3%) is ascribed to the multilayer structure of randomly stacked nanofibers with diameters broadly distributed (centered at $\approx 800 \text{ nm}$) (Figure 4j). The chosen polyethylene oxide (PEO), with only C–C, C–O, and C–H bonds, exhibits a desirable selective absorption band overlapping with the atmospheric transparency window, imparting a ϵ_{LWIR} of 0.78 (Figure 4k). This hierarchically designed selective thermal emitter demonstrates a cooling improvement of -3°C compared to a non-selective emitter at night and achieves 5°C sub-ambient cooling under sunlight. Meanwhile, a SiO_2 fiber network was demonstrated to reflect sunlight efficiently via backscattering and emit infrared thermal radiation.^[59] Zhang et al. utilized SiO_2 fibers and SiO_2 fumed as emitters, resulting in a high R_{sol} of 97% and a ϵ_{LWIR} of 0.91. This material is envisioned to be used in RC hats, effectively lowering head temperatures by 12.9°C .^[60]

In a nutshell, we summarized and discussed the static homogeneous RC materials reported in recent years that employ a variety of nanostructures, such as hierarchically arranged air pores, inorganic particles, and nanofibers. The diverse features observed in these RC materials offer valuable insights into the design of PDRC materials, emphasizing the importance of different structural elements in improving RC performance.

4. Static-Composite Radiative Cooling Materials

Apart from homogeneous RC materials, composites are another category of static RC materials which consist of two or more components with different properties. According to the composition of matrix and additive, static-composite RC materials can be categorized into inorganic–inorganic, organic–organic, and organic–inorganic hybrid composites.

4.1. Inorganic–Inorganic Composites

Inorganic materials are traditionally preferred for outdoor applications of RC technology due to their exceptional physical and

optical properties. These include high hardness, strength, and melting point, robust resistance against abrasion, corrosion, and oxidation, as well as unique optical properties such as high refractive index and wide band gap.

4.1.1. Metal-Based Inorganic Composites

Metallic materials with high reflectivity, especially smooth metallic films, are competitive candidates for reflecting solar irradiation. In 2014, Fan et al. first introduced a selective radiative emitter consisting of several alternating layers of SiO_2 and hafnium dioxide (HfO_2) deposited on a silver mirror (Figure 5a).^[2] The combination of SiO_2 and HfO_2 layers imparts the film with sufficient ϵ_{LWIR} , as the phonon polariton of SiO_2 produces strong thermal emission near 9 μm and HfO_2 presents selective emission in the wavelength range of 8–13 μm . Experiments demonstrate that the multilayer photonic film could reflect 97% of solar irradiation and achieve a sub-ambient temperature difference of 4.9 °C when exposed to direct sunlight. Subsequently, some similar material combinations, such as HfO_2 , SiO_2 and Si_3N_4 multilayer, were employed to develop emitters.^[6,61] For example, Yuan et al. used HfO_2 , SiO_2 , and Ag to design a pyramidal multilayer metamaterial.^[61c] Due to the attenuation effect of moth-eye structure and electromagnetic resonant absorption in the metamaterial, the designed cooler avoids the high A_{sol} of metal materials and provides a strong thermal emission. Nevertheless, the fabrication of metal-based inorganic RC coolers with alternatively stacked layers is a challenging task in most instances. Based on this situation, Huang et al. presented a solution-based approach for constructing a selective emitter with a three-layer architecture (Figure 5b-i), including a self-assembly monolayer of SiO_2 microspheres, a solution-derived silicon oxynitride (SiO_xN_y) layer, and a silver layer on a substrate (Figure 5b-ii,iii).^[38a] The emitter has a ϵ_{LWIR} of ≈ 0.95 , which results from the combined effects of selective emission by SiO_xN_y and polariton resonance by SiO_2 microspheres, and a R_{sol} of 96.4% due to the reflectance of Ag layer and scattering of microspheres. The scalable-solution-based fabrication offers inspiration for tackling engineering challenges in scaling up all-inorganic RC materials.

Meanwhile, other materials such as zinc oxide (ZnO), Al_2O_3 , and TiO_2 were also explored to develop composite RC materials.^[62] For instance, a square periodic photonic structure consisting of an Al-doped ZnO (AZO) square surface at the top, a SiO_2 spacer in the middle, and an Al reflector at the bottom was fabricated.^[62c] The patterned surface produces broad plasmonic resonances with an enhanced infrared absorptivity, while the transparent conducting oxide and multilayer reflecting coating provide a A_{sol} of 16%. Additionally, the combination of porous Al_2O_3 (AAO) or silica-coated porous AAO and metal substrate optimized by the effective medium theory also exhibited high R_{sol} (94.6% for porous AAO or 86% for silica-coated AAO) and high ϵ_{LWIR} (0.98 for porous AAO or 0.96 for silica-coated AAO) for PDRC.^[63]

On the other hand, research efforts have also focused on coolers with simple components. Al and germanium (Ge) pyramid-patterned structures were reported to achieve a temperature drop of 12.2 °C with the optimized pattern diameter.^[64] Also, Fumeaux et al. proposed a square structure comprising only

two materials, namely phosphorous-doped n-type silicon and silver (Figure 5c-i).^[65] The doped silicon layer provides high thermal emission, while the silver layer enhances the R_{sol} . In particular, the combination of doped silicon and the silver layer forms a dielectric resonator metasurface, which achieves wide-angle thermal emission by utilizing its magnetic dipole resonance (Figure 5c-ii), thus lowering the temperature by 10 °C below ambient temperature. A bulk radiative cooler composed of 1-mm-thick lithium fluoride (LiF) crystal coated with Ag backing was proposed for PDRC.^[66] This bulk cooler has a low A_{sol} of 4.7% and a ϵ_{LWIR} of 0.89, owing to the spectral selectivity of LiF.

4.1.2. Metal-Free Inorganic Composites

Despite metal-based inorganic RC materials can provide high R_{sol} by the metal reflective layer, it has several drawbacks in practical applications, including inherent A_{UV} properties, low resistance to oxidation, and complex fabrication process. Therefore, considerable attention has been given to exploring metal-free inorganic RC composite materials.

Ceramics, renowned for their good chemical stability and oxidation resistance, have emerged as promising candidates for RC technologies.^[67] As displayed in Figure 5d, Hu et al. developed a glass-based ceramic cooling coating consisting of a mesoporous glass framework embedded with Al_2O_3 solar reflective particles.^[5b] Low sintering temperature (≈ 600 °C) is used to ensure that the glass is sufficiently softened to form a larger glass cluster framework. Taking into consideration the pores and structural strength of the glass skeleton, the size and content of the Al_2O_3 particles were optimized to enhance R_{sol} (Figure 5e), and the size of the glass particles was adjusted to facilitate infrared emission. The optimized ceramic-based composite (thickness ≈ 550 μm) gives a R_{sol} of $>96\%$ and a ϵ_{LWIR} of ≈ 0.95 . In addition to its optical properties, the materials' intrinsic properties, such as the close-packed crystal structure of SiO_2 and Al_2O_3 ,^[67] ensure their environmental stability against water, UV radiation, and high temperatures. Its long-term stability is further enhanced by applying a thin (<10 μm) and dense transparent glass layer on the surface to improve its resistance to liquids and dust. Sun et al. prepared a novel all-ceramic nano-fiber aerogel composed of SiO_2 and Al_2O_3 by electrospinning, as shown in Figure 5f-i.^[68] The addition of Al_2O_3 enables the aerogel to form a multilayer oxide fiber network structure (Figure 5f-ii) with mechanical integrity, overcoming the brittleness commonly observed in silica aerogels. The multilayer fiber network structure combined with the intrinsic properties of SiO_2 and Al_2O_3 provides the aerogel with a R_{sol} of 95% and a ϵ_{LWIR} of 0.93, enabling large-scale PDRC (Figure 5f-iii). In addition, the all-ceramic aerogel exhibits thermal insulation properties with a thermal conductivity of 30 $\text{mW m}^{-1} \text{K}^{-1}$ for stable operation in high-temperature environments. A structural porous Si_3N_4 -BN ceramic with a R_{sol} of $\approx 95\%$ and a ϵ_{LWIR} of ≈ 0.95 was also recently reported for subambient cooling.^[69] Commonly, ceramics are also used as additives to further enhance the cooling performance.^[70] SiO_2 and BaSO_4 particles were used as functional fillers for modified geopolymers to achieve a high ϵ_{LWIR} of 0.95 through phonon resonance of SiO_2 particles and an improved R_{sol} (98%) by the wide band gap of BaSO_4 .^[70a]

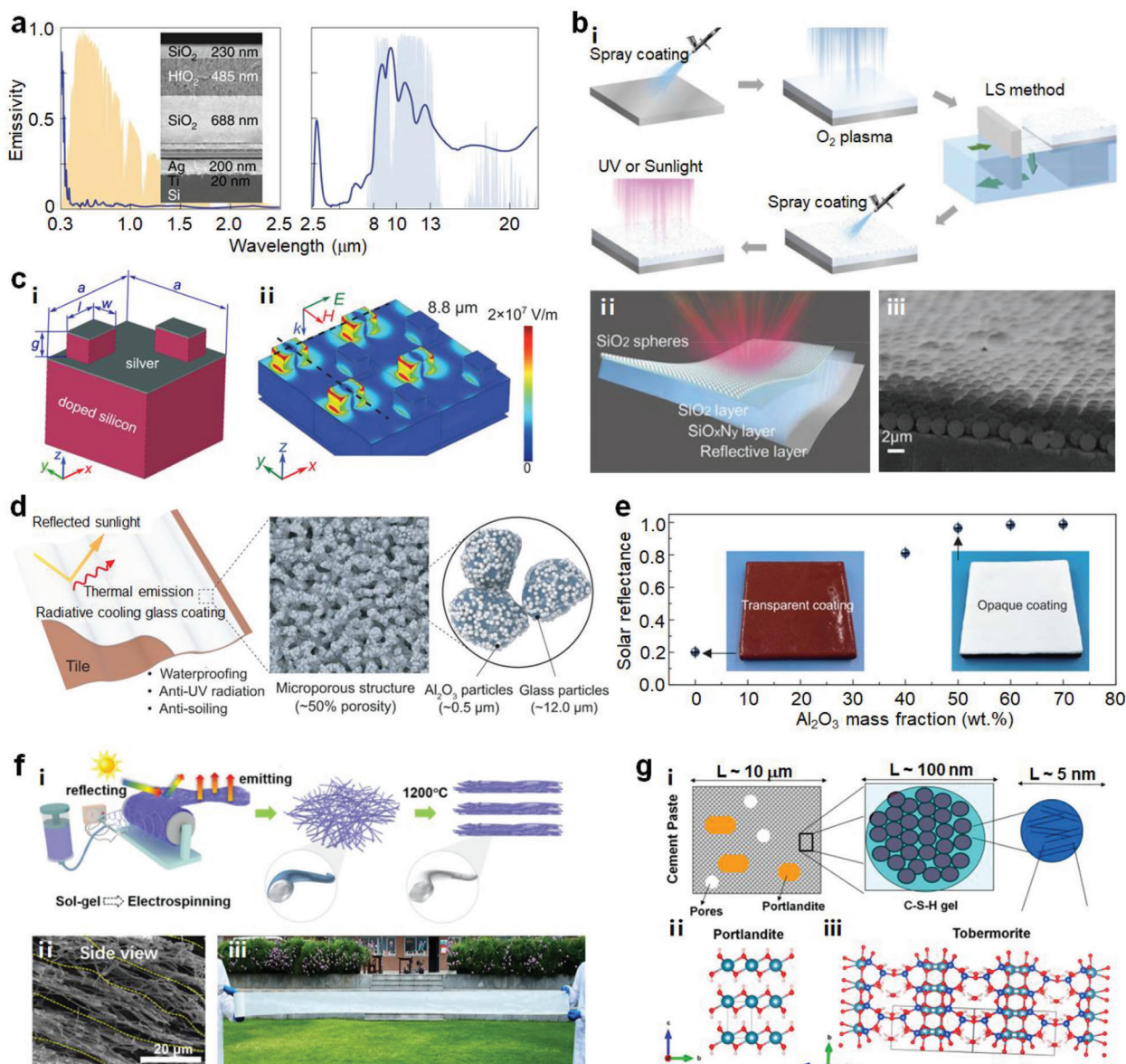


Figure 5. Inorganic-inorganic composite RC materials. a) Emissivity spectrum in the solar (left) and infrared (right) wavelengths for cooler with multilayers of SiO₂ and HfO₂ deposited on a silver mirror, along with an image of the multilayer structure in the inset. Reproduced with permission.^[2] Copyright 2014, Springer Nature. b) The solution-processed inorganic emitter. i) Schematic of the fabrication process and ii) cooling mechanism of the emitter. iii) SEM image of the emitter. Reproduced with permission.^[38a] Copyright 2022, John Wiley and Sons. c) Dielectric resonator metasurface. i) Schematic of the doped-silicon and silver square structure. ii) The simulated electric field distributions at resonance peaks of 8.8 μm . Reproduced with permission.^[65] Copyright 2017, John Wiley and Sons. d) Schematic illustration of the glass-based ceramic cooling coating on a ceramic roofing tile (left) and porous glass framework embedded with Al₂O₃ particles (right). Reproduced with permission.^[5b] Copyright 2023, AAAS. e) Solar reflectance of the glass coating (thickness of $\approx 550 \mu\text{m}$) with different Al₂O₃ particle mass fractions. The inset shows optical photographs of the glass coatings with 0 and 50 wt% Al₂O₃ particles. Reproduced with permission.^[5b] Copyright 2023, AAAS. f) All-ceramic Al₂O₃/SiO₂ nanofiber. i) Schematic of the fabrication process, ii) side-view SEM image and iii) optical image of the all-ceramic nanofiber aerogel composed of SiO₂ and Al₂O₃. Reproduced with permission.^[68] Copyright 2023, Elsevier. g) Cementitious materials: i) Schematic of the cement paste and ii,iii) crystalline structure of Portlandite (ii) and Tobermorite (iii). Atomic notations: light blue is calcium, dark blue is silicon, red is oxygen, and white is hydrogen. Reproduced with permission.^[71] Copyright 2023, American Chemical Society.

Cementitious materials, as the most commonly used and cost-effective building materials, were proposed to achieve PDRC by optimizing their optical properties. The RC properties of two cementitious minerals that are of great relevance in concrete science and technology, namely Portlandite (Ca(OH)₂) and Tobermorite (Ca₅Si₆O₁₆(OH)₂·4H₂O), were investigated (Figure 5g).^[71]

Figure 5g-i shows the cement paste, a complex hierarchical porous glassy composite. Combining spectra measurements with theoretical simulations, the study indicates that both Portlandite (Figure 5g-ii) and Tobermorite (Figure 5g-iii) have great potential for PDRC applications. Additionally, by modifying its mixing and curing conditions to alter the phase composition

and microstructure, cementitious composites can attain a R_{sol} of 90% and a ϵ_{LWIR} of 0.88, comparable to state-of-the-art coolers.^[72] Moreover, by applying Portland cement (a commonly used type of cement) with optimized RC properties in solar cells, it is possible to reduce the operating temperature of solar cells by up to 20 K, leading to improvements in efficiency and lifetime.^[73]

4.2. Organic–Organic Composites

Organic materials are promising RC materials in various practical applications due to their distinctive properties such as flexibility and ease of engineering.^[22] They can be mass-produced at low cost and easily processed into various forms such as films, coatings, and composites. This makes them compatible with different systems and devices like solar cells, electronic devices, and buildings. Moreover, their abundance of molecular vibrations allows for high infrared emission, making them efficient in emitting thermal radiation within the atmospheric window, which is a crucial aspect for achieving sub-ambient cooling. We have to bear in mind that RC materials need to have sufficient weatherability, which poses an inevitable challenge for polymeric-based RC materials.

Cellulose and silk, with unique properties such as biodegradability and non-toxicity, have recently gained attention as promising RC materials. Generally, the cellulose-fiber-based paper, with diameters ranging from 20 to 50 μm , can reflect up to 89% of solar irradiation and emit infrared radiation with a ϵ_{LWIR} of 0.92 due to the molecular vibrations of C—O—C and C—O bonds in cellulose.^[74] Based on this, Zheng et al. designed a dual-layered cellulose-fiber-based composite material with ethanolic poly(tetrafluoroethylene) (PTFE) microparticle suspensions that were partially embedded within the micro-sized pores of the cellulose fiber (Figure 6a).^[74] The PTFE particles on the surface of the paper not only make it hydrophobic and self-cleaning but also increase its R_{sol} to 93% by scattering sunlight. Moreover, PDMS and PMMA were used to improve the ϵ_{LWIR} and R_{sol} of cellulose-based materials.^[75] In addition to cellulose, natural silk also exhibits RC performance. As shown in Figure 6b, silk features broadband light scattering due to its unique self-assembled hierarchical structure and its molecular composition gives a high ϵ_{LWIR} .^[76] Zhang et al. fabricated a durable multilayer silk textile by sandwiching a silk nanotextile between an ultrathin expanded PTFE (e-PTFE, 35 μm) textile (top layer) and a commercial silk textile (bottom layer) (Figure 6c).^[76] This design enhances the mechanical properties of silk nanotextiles and improves their durability in outdoor applications.

Poly(lactic acid) (PLA), derived from biomass cellulose (Figure 6d-i),^[77] exhibits biodegradability and mechanical flexibility, making it a popular choice for flexible RC films. For instance, Zhang et al. employed silk fibroin (SF) and PLA to create an environmentally friendly RC film with a R_{sol} of 96% and a ϵ_{LWIR} of 0.95.^[78] The film could degrade rapidly in outdoor soil within a month, providing a new strategy that combines biodegradation with zero-energy cooling. Furthermore, a PLA film with a mixed micro- and nanofibrous structure was produced by Zeng et al. through a combination of meltblowing and electrospinning techniques (Figure 6d-ii).^[79] The fiber sizes are comparable to the wavelength of sunlight, which enabled the PLA film to re-

flect sunlight effectively. Zeng et al. incorporated this PLA film as a filter for a mask and a silk fabric as an inner support layer (Figure 6d-iii) to make RC wearable and improve human thermal comfort. Other biomass-based materials and their derivatives, such as chitosan (CS),^[80] cellulose acetate (CA),^[4b] and ethyl cellulose,^[35] are also potential candidates for RC applications.

Hydrophobic fluorinated polymers like polyvinylidene difluoride (PVDF), P(VdF-HFP), and PVF are commonly employed for RC. Wang et al. utilized PVDF and PVA to fabricate hierarchical hollow microfibers with interconnected micropores and secondary nanostructures (e.g., nanobumps and nanopores) using the coaxial electrospinning method.^[58] The PVDF/PVA microfibers exhibit a R_{sol} of 94% and a ϵ_{LWIR} of 0.94, enabling a temperature drop of about 9 °C when exposed to sunlight of 900 W m^{-2} . Importantly, the hollow structures and abundant nanopores not only promote solar light scattering but also provide thermal insulation to suppress environmental heat gain. In addition, a porous film of PVDF/polyurethane acrylate (PUA) with a stable cross-linking structure possesses a ϵ_{LWIR} of ≈ 0.93 , which is attributed to the C—O and C=C vibrations of PUA compensating for the emissivity deficiency of PVDF (Figure 6e-i).^[81] PUA also acts as a binder to form PVDF/PUA film by bonding the PVDF particles together. Figure 6e-ii shows the PVDF particles form clusters of different sizes, which facilitates light scattering. The PVDF/PUA film exhibits a R_{sol} exceeding 93% due to the strong interaction of various-sized clusters and multiple surfaces in the structure with sunlight (Figure 6e-iii).

PDMS with low A_{sol} and high ϵ_{LWIR} is another option for organic RC material. For example, Wang et al. fabricated the PDMS/P(VdF-HFP) coatings with a 3D porous structure and a micro-nano rough surface by using the liquid template method.^[82] The synergistic effect of the hierarchical structures in the bulk and the surface enhances the R_{sol} (96.5%) and ϵ_{LWIR} (0.94) capabilities of the coating. Additionally, Zhou et al. designed a flexible composite film consisting of PDMS and polystyrene (PS) microsphere photonic crystals by a facile self-assembling method (Figure 6f-i).^[83] The R_{sol} of the PDMS/PS composite film reached 92.7% by optimizing and engineering the diameter of the PS microspheres (diameter $\approx 1 \mu\text{m}$, Figure 6f-ii). The film also exhibits a ϵ_{LWIR} of 0.94 because of the molecular vibrations of PDMS and PS and the sphere-to-sphere, sphere-to-layer, and layer-to-layer optical resonances induced by the built-in PS microsphere arrays (Figure 6f-iii). Moreover, Guo et al. utilized poly(ethyl cyanoacrylate) (PECA) with C—O—C bonds to prepare PS/PDMS/PECA polymer composite coatings with a porous texture and hierarchical micro/nanoscale structure.^[84] The hierarchical porous structure enhances Mie scattering across the UV–vis–NIR spectrum (R_{sol} of 96%), whilst also boosting emission due to multiple absorption and scattering within the porous structure. Combining the molecular vibrations of PDMS and PECA, the porous coating achieved a ϵ_{LWIR} of 0.95.

PMMA is another common option for RC. He et al. employed polar polyethylene terephthalate (PET) and non-polar polyolefin as incompatible materials to manufacture a large-area coating with hierarchical eye-like air pores by the film-stretching method (Figure 6g-i).^[85] PMMA microspheres are further coated on the hierarchical porous film (Figure 6g-ii,iii) to increase ϵ_{LWIR} and reduce the specular light pollution, a phenomenon in which smooth surfaces reflect light in parallel and cause harsh glare.

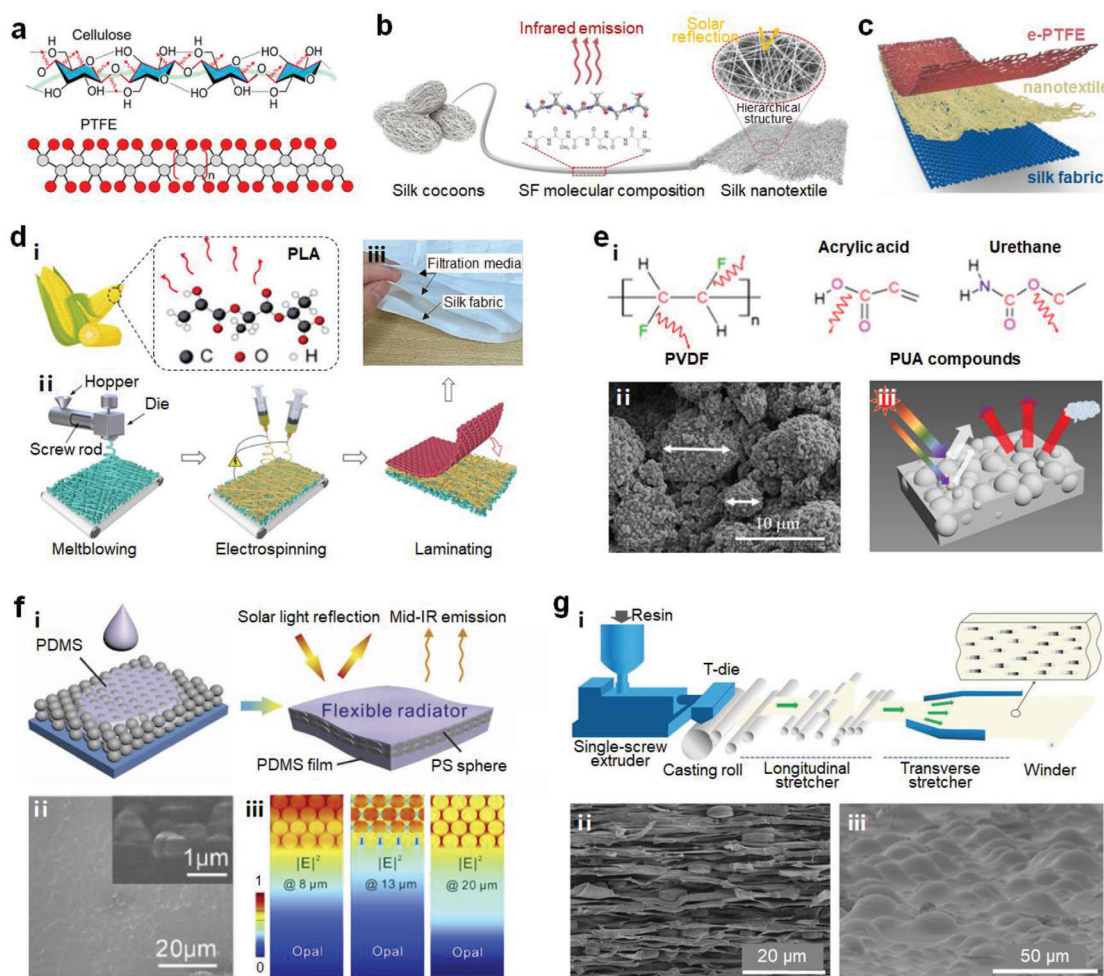


Figure 6. Organic-organic composite RC materials. a) Schematic molecular structure of cellulose and PTFE in the paper composites. Reproduced with permission.^[74] Copyright 2021, American Chemical Society. b) Schematic diagram of the fabrication process of the silk nanotextile, with illustrations showing its optical properties. c) Schematic structure of the multilayer silk textile. Reproduced with permission.^[76] Copyright 2023, John Wiley and Sons. d) Biodegradable PLA polymer film. i) Schematic of PLA molecular chain vibrations and derived from natural corns. Reproduced with permission.^[77] Copyright 2023, American Chemical Society. ii) Schematic of fabrication processes and iii) optical photo of the PLA/silk mask. Reproduced with permission.^[79] Copyright 2024, Elsevier. e) PVDF/PUA cooling film. i) Intrinsic molecular vibrations of PVDF and PUA functional groups. ii) SEM image and iii) schematic cooling mechanism of the PVDF/PUA film. Reproduced with permission.^[81] Copyright 2020, American Chemical Society. f) PDMS/PS composite film. i) Schematic of fabrication process of the PDMS/PS film. ii) SEM image of the surface morphology of PDMS/PS film with 1 μm PS microspheres. The inset shows a cross-sectional view of the film. iii) Simulated electric field distributions of the PDMS/PS film at wavelengths of 8, 13, and 20 μm . Reproduced with permission.^[83] Copyright 2022, The Optical Society. g) PMMA/PET hierarchical porous film. i) Schematic diagram of the film-stretching process for fabricating PMMA/PET hierarchical porous film. ii) Cross-section and iii) top-view SEM images of the PMMA/PET film. Reproduced with permission.^[85] Copyright 2022, John Wiley and Sons.

Benefiting from the scattering effect of the multiple-sized pores and the molecular vibration of PMMA and PET, the film yielded a R_{sol} of 97.6% and a ϵ_{LWIR} of 0.9. In addition to the polymers mentioned above, many other polymers have also emerged for RC recently, including polypropylene (PP),^[86] polyethylene (PE),^[87] and polyamide (PA).^[88]

4.3. Inorganic-Organic Hybrid Composites

Inorganic and organic materials have their respective pros and cons. To combine the benefits of both materials, researchers have proposed hybridizing inorganic and organic compositions to fur-

ther improve the cooling efficiency and comprehensive functionality of RC materials.

4.3.1. Metal-Polymer Composites

Generally, metallic materials feature high R_{sol} due to their free electrons, while polymers containing numerous chemical bonds or functional groups possess high ϵ_{LWIR} . Therefore, a combination of metals and polymers generally yields effective RC systems.

Gan et al. developed a planar PDMS/metal (Al or Ag) film structure for large-scale deep subambient cooling using a fast solution-based roll-to-roll manufacturing technique.^[31] The film,

with a thickness of 150 μm , exhibits a satisfied ϵ_{LWIR} of 0.95 while displaying relatively weak absorption in the vis–NIR spectral range. In addition to PDMS, metallic films incorporating silicon oxycarbonitride (SiCNO), derived from polysilazane, also demonstrated effective RC performance. A thin SiCNO coating (5 μm) on a 230 nm thick silver-coated Al substrate has a ϵ_{LWIR} of 0.86 and a R_{sol} of 97%.^[89] There are many other polymer options, such as P(VdF-HFP),^[90] PVF,^[91] acrylic polymer (3 M Scotch tape),^[92] PVDF and PMMA,^[93] which provide RC performance in combinations with metals. Additionally, Jia et al. designed a periodic trench-like metasurface structure based on a polymer film using a roll-to-roll printing technique (Figure 7a-i) to enable precise spectral control.^[94] The film consists of a 3D trench-like polymer (e.g., PDMS and polycarbonate (PC)) metasurface structure printed on a 50- μm -thick PET film with a 100 nm thick Ag layer deposited on the backside of the film (Figure 7a-ii,iii). The polymer metasurface RC film with optimized dimensions of trench structure provides a ϵ_{LWIR} of 0.96 and a R_{sol} of 95% and is insensitive to both emission angle and radiation polarization.

Song et al. designed a *Janus* emitter, as illustrated in Figure 7b-i, comprising a top selective emitter and a bottom broadband emitter, for enclosed spaces cooling.^[17] The *Janus* emitter consists of a 4 μm -thick PDMS layer, a 100 nm-thick Ag layer, and a 500- μm -thick micropatterned quartz coated with a 10 μm -thick PDMS. The top surface is designed to reflect sunlight and emit thermal radiation selectively (Figure 7b-ii), while the bottom surface spontaneously absorbs and emits heat from objects within the enclosed space (Figure 7b-iii). To enhance energy efficiency under certain scenarios, angle-selective emitters are desirable, as they can change their ϵ_{LWIR} depending on the angle. Recently, Cui et al. fabricated a tilted wedge-shaped microstructure of ≈ 150 μm thickness from PDMS with a 150 nm-thick Al deposited on the surface (Figure 7c).^[95] Typically, the intrinsic emissivity of PDMS is high, ≈ 0.9 near 10 μm , while the $\epsilon_{\text{Broadband}}$ of metallic Al is low (< 0.1). This design enables broadband tuning of the directional emissivity, which varies from 0.9 to 0.1 at different angles (Figure 7d).

4.3.2. Metal-Polymer-Dielectric Composites

Dielectric materials with long-term environmental stability also exhibit strong emission in the broadband IR region due to phonon resonance. The introduction of dielectrics therefore substantially improved the applicability of metal-polymer composite RC materials.

As demonstrated in Figure 7e, Yin et al. produced a scalable-manufactured flexible glass-polymer hybrid metamaterial by randomly embedding resonant SiO₂ microspheres in a polymeric matrix of TPX and backing with a thin silver layer.^[8] Appropriately sized SiO₂ microspheres can collectively excite surface phonon polariton resonances, resulting in a ϵ_{LWIR} greater than 0.93. A 200 nm-thick silver layer effectively reflected most of the solar energy, enabling sub-ambient cooling of over 8 $^{\circ}\text{C}$ at noon when the solar irradiance exceeds 800 W m^{-2} . Based on this metamaterial, Yang et al. developed a RC-based cooling system that can cool water to 10.6 $^{\circ}\text{C}$ below ambient at noon under stationary conditions, demonstrating the potential of RC technology for building applications.^[96] Additionally, Hu et al. in-

troduced a paper-based radiative cooler with a multiphase interface that employs cellulose nanofibers (filter paper) with SiO₂ microspheres and PDMS components.^[97] The designed microsphere/polymer/air three-phase interface enhances light scattering, and combined with the reflection from the Ag layer on the back of the paper, yields a R_{sol} of 91% for the cooler. In addition, the multiscale networks and multiphase interfaces enhance the light scattering and the photon propagation paths in the broadband IR region, resulting in an improved ϵ_{LWIR} of 0.9. On the other side, the oxidation resistance of the metal layer and its adhesion to the substrate are crucial for practical applications. Concerning this issue, Poulikakos et al. enhanced the adhesion between the silver and the SiO₂ substrate by depositing a thin chromium (Cr) layer, while the Cr layer on the other side acted as a protective barrier to prevent oxidation of the silver layer.^[21] The prepared cooler exhibits a R_{sol} of 96% and a ϵ_{LWIR} of 0.93. Besides SiO₂, other dielectric materials such as Al₂O₃,^[98] SiC,^[99] and calcium molybdate (CaMoO₄)^[100] are also commonly used in metal-polymer-dielectric composite RC materials.

4.3.3. Dielectric-Polymer Composites

Dielectric-polymers are a type of RC material with the integrated durability of dielectrics and the flexibility/easy processing of polymers. We will elaborate the different types of dielectric-polymers based on the dielectric material added, i.e., TiO₂, Al₂O₃, SiO₂, BaSO₄, salt, and other emerging dielectrics.

TiO₂ is a widely used dielectric material for RC applications. As depicted in Figure 7f,g, Tao et al. developed a large-area bilayer RC metafabric by laminating a TiO₂-PLA composite with a PTFE layer through a scalable industrial textile manufacturing route.^[16] The PTFE layer provided UV reflectivity (R_{UV}), multi-dispersed TiO₂ particles contributed to vis–NIR reflectivity and PLA fibers reflected solar radiation while exhibiting high ϵ_{LWIR} (Figure 7f), giving a R_{sol} of 92.4% and a ϵ_{LWIR} of 0.95, consistently maintaining a temperature over 2 $^{\circ}\text{C}$ below the ambient temperature. However, TiO₂ has a bandgap of 3.0 eV (0.413 μm), leading to intrinsic absorption of some solar irradiation (0.3–0.41 μm). To mitigate the intrinsic A_{UV} of TiO₂, Fan et al. added fluorescent pigments to convert the absorbed solar energy into fluorescent emission.^[101] In another work, Yu et al. employed a polystyrene-acrylate matrix with fluorescent pigments, TiO₂ NPs, and hollow SiO₂ microspheres to fabricate a flexible RC film.^[102] The optimized film exhibits a $\epsilon_{\text{Broadband}}$ of 0.97 and a R_{sol} of over 91%, showcasing its potential for wearable devices. Besides, mechanical strength is a crucial factor that affects the applicability of RC materials. To enhance this aspect, Deng et al. designed a layered structure with core-shell TiO₂-coated Mica microplatelets and aramid nanofibers (Figure 7h).^[103] This interfacial network design enhances the film's strength and durability without compromising its optical properties. The thin film holds a mechanical strength of ≈ 112 MPa and a Young's modulus of ≈ 4 GPa while maintaining a R_{sol} of 92% and a ϵ_{LWIR} of 0.87.

Al₂O₃ is another commonly used dielectric for RC technology. As shown in Figure 7i, Fan et al. prepared a flexible RC film using a PDMS substrate with micropyramid textures and randomly distributed Al₂O₃ microspheres.^[40b] The film demonstrated a R_{sol} of $\approx 95\%$, benefiting from the synergistic effect of total internal

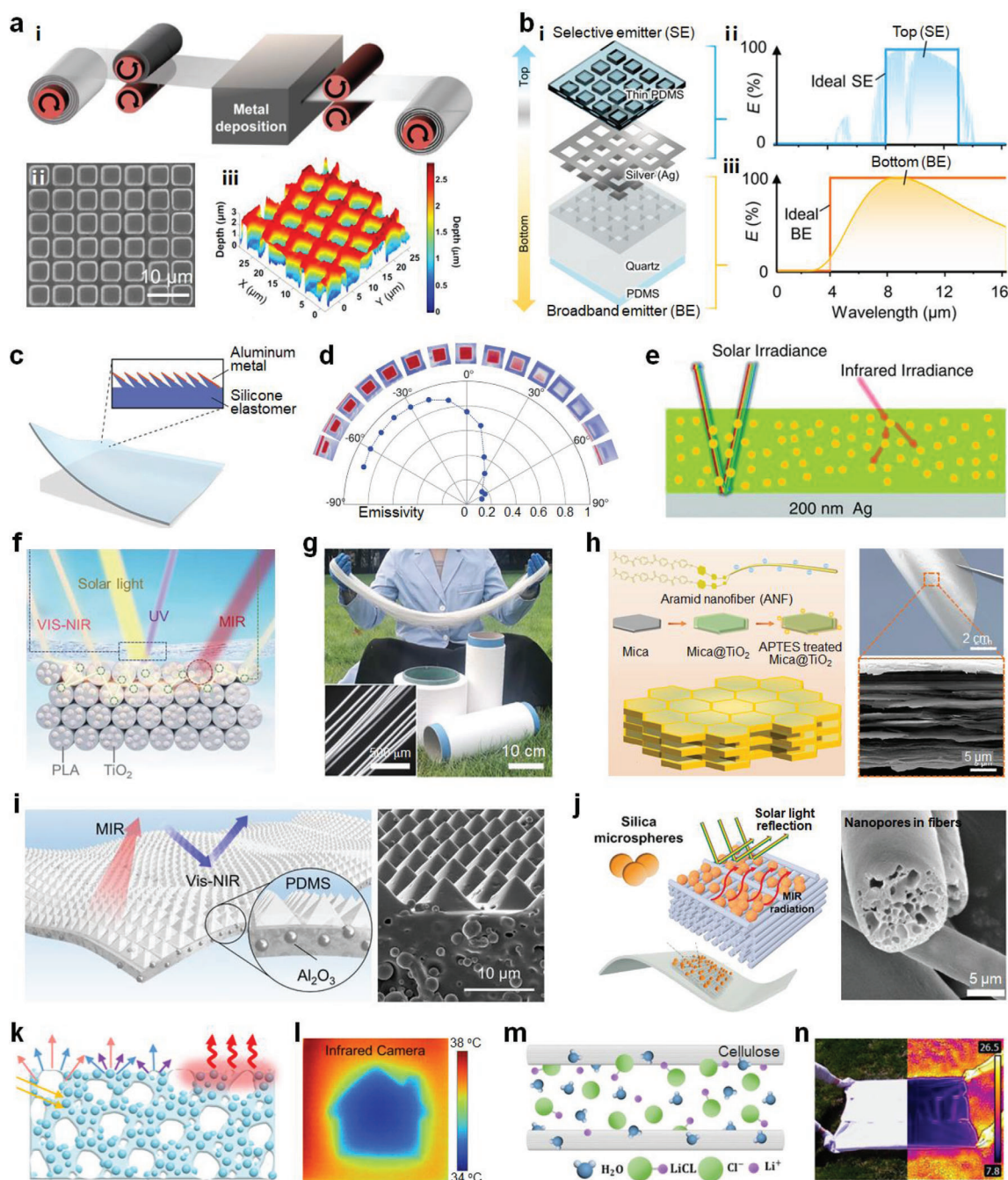


Figure 7. Inorganic–organic hybrid composite RC materials. a) The 3D trench-like metasurface RC film. i) Schematic of the fabrication processes, ii) top-view SEM image, and iii) optical profilometer 3D image of the film. Reproduced with permission.^[94] Copyright 2023, Springer Nature. b) *Janus* emitter. i) Structure schematic of the *Janus* emitter. ii,iii) Ideal emission spectra of the selective emitter (SE) on the top (ii) and the broadband emitter (BE) on the bottom (iii). Reproduced with permission.^[17] Copyright 2020, AAAS. c) Structure schematic of the angle-selective emitter with tilted wedge-shaped microstructure. d) Emissivity and correspondingly infrared images taken at different angles for samples. Reproduced with permission.^[95] Copyright 2023, Elsevier. e) Schematic diagram of the SiO₂/TPX metamaterial film. Reproduced with permission.^[8] Copyright 2018, AAAS. f) Schematic cooling mechanism of the fabricated TiO₂/PLA/PTFE metafabric. g) Photo of the fabricated metafibers. Reproduced with permission.^[16] Copyright 2021, AAAS. h) The design of Mica@TiO₂ thin lamellar film (left) with optical and SEM photographs (right). Reproduced with permission.^[103] Copyright 2023, Springer Nature. i) Schematic diagram (left) and SEM image (right) of Al₂O₃/PDMS micropyramid-structured photonic film. Reproduced with permission.^[40b] Copyright 2020, PNAS. j) Schematic (left) and cross-section morphology (right) of PVDF/TEOS fibers distributed with SiO₂ microspheres. Reproduced with permission.^[105] Copyright 2020, John Wiley and Sons. k) Schematic working principle of the porous ethyl cellulose/BaSO₄ coating. l) IR image of the ethyl cellulose/BaSO₄ coating under direct sunlight. Reproduced with permission.^[112] Copyright 2022, American Chemical Society. m) Schematic of the LiCl/cellulose composite. n) Optical and IR photographs of a large-scale LiCl/cellulose composite. Reproduced with permission.^[123] Copyright 2023, Springer Nature.

reflection from the pyramidal structure and Mie scattering from the Al_2O_3 particles. Additionally, the gradual effective refractive index transition of the micropillar array and the combined effect of phonon polariton resonance of Al_2O_3 particles lead to a ϵ_{LWIR} of exceeding 0.96 for the film. Similarly, Fan et al. embedded Al_2O_3 randomly into a porous polyurethane (TPU) matrix with a micro-humped profile, achieving a R_{sol} of 97.6% and a ϵ_{LWIR} of 0.96.^[104] Also, Al_2O_3 is utilized to mitigate the high A_{UV} of natural silk protein, thus enabling effective daytime cooling of silk textiles. Al_2O_3 NPs with a size of ≈ 300 nm were bonded to silk fibers to enhance their R_{UV} , increasing the R_{sol} of silk from 86% to $\approx 95\%$.^[19]

SiO_2 is a promising dielectric material for RC. Fan et al. randomly distributed SiO_2 microspheres with diameters of 4–16 μm on the surface of PVDF/tetraethoxysilane (PVDF/TEOS) fibers, which contained internal nanopores with diameters of 100–200 nm (Figure 7j).^[105] The fabricated PVDF/TEOS/ SiO_2 composite yielded a R_{sol} of $\approx 97\%$, attributed to light scattering induced by the stacked interfaces and abundant internal nanopores. It demonstrated a ϵ_{LWIR} of >0.96 owing to the molecular vibrations of the C–F bond in PVDF and the Si–O bond in TEOS, along with the phonon-polariton resonance of the SiO_2 microspheres. Other polymers such as PE,^[106] P(VdF-HFP),^[107] CA,^[108] CS,^[109] ethylene-propylene-diene copolymer (EPDM),^[110] have also been reported to hybridize with SiO_2 particles for RC. Moreover, Yoo et al. synthesized raspberry-like hollow SiO_2 spheres, which have stronger light scattering compared to solid SiO_2 spheres due to the high refractive index difference between the inner air and the solid shell, thereby improving the cooling performance of RC coatings.^[111]

BaSO_4 is suitable for the fabrication of dielectric-polymer RC materials. For instance, Ruan et al. adopted a broad particle size distribution and high filler concentration strategy to produce a BaSO_4 /acrylic paint with a R_{sol} of 98.1% and a ϵ_{LWIR} of 0.95.^[26] Further, Raman et al. used P(VdF-HFP) with fewer C–H and O–H bonds as a polymer binder alternative to acrylics for reducing NIR absorption.^[14] The resulting BaSO_4 /porous P(VdF-HFP) coating achieved a R_{sol} of $\approx 98\%$. However, constructing porous P(VdF-HFP) involves using acetone, which is harmful to the environment in practical applications. To address this issue, Duan et al. utilized a green solvent (i.e., ethanol/water) to fabricate coatings consisting of porous ethyl cellulose matrix-random BaSO_4 NPs (Figure 7k), avoiding the usage of toxic volatile organic solvents like acetone.^[112] The ethyl cellulose/ BaSO_4 coating has a R_{sol} of 98.6% and a ϵ_{LWIR} of 0.98, maintaining a low temperature under direct sunlight (Figure 7l). Furthermore, Shuai et al. proposed to optimize the optical properties of RC coatings using biomimetic human skin wrinkle structure.^[113] In a PVDF matrix, BaSO_4 particles (0.4 μm) were added to scatter sunlight and SiO_2 particles (4 μm) to emit thermal radiation. The PVDF/ BaSO_4 / SiO_2 coating with optimized wrinkle structures achieved a R_{sol} of $\approx 95\%$ and a ϵ_{LWIR} of ≈ 0.96 .

Salt-polymer composites, besides the above-mentioned BaSO_4 , have also received considerable attention as an emerging type of RC material. In the broadband IR region, salts also exhibit abundant bond vibrations besides polymers. For example, the vibrational absorption peaks of P=O (7–9 μm),^[114] P–O (9–9.5 μm),^[115] P–O–H (9–11 μm),^[116] P–H (12–13 μm),^[117] and Mo–O (11–14 μm),^[100] bonds drop in the atmospheric

window. Notably, the absorption peaks of the salt and the hybrid polymer in the broadband IR region are complementary, and their synergistic effect enhances the thermal emission of the RC material. In the solar region, salts with narrow band gaps possess the intrinsic ability to absorb UV light and convert it into less harmful heat, thereby preventing the polymer from being damaged. For instance, Zhu et al. doped PEO with potassium titanate ($\text{K}_2\text{Ti}_6\text{O}_{13}$ (PT), 3.0 eV)^[118] nanofibers to improve the UV stabilities of polymer. The resulting material shows enhanced durability while having a R_{sol} of $\approx 94\%$ and a ϵ_{LWIR} of 0.91.^[119] On the other hand, salts with wide bandgaps and with appropriate light scattering particle sizes can be used to enhance R_{sol} , as in the case of a coating consisting of nanoporous $\text{MgHPO}_4 \cdot 1.2\text{H}_2\text{O}$ (5.2 eV) with a size of ≈ 200 nm and PVDF has a R_{sol} of 92%.^[116] Researchers have explored various salts, such as aluminum phosphate (AlPO_4),^[114] barium titanate (BaTiO_3),^[120] and CaCO_3 ,^[121] to develop RC materials. Another special salt, lithium chloride (LiCl), with its high hygroscopicity, bridges the fields of RC and ambient water harvesting.^[122] For example, a novel RC sorbent was developed for atmospheric water harvesting by intercalating LiCl into a cellulose matrix (Figure 7m).^[123] The multiscale structure of cellulose and the multimode vibration of its molecular chains endow the LiCl/cellulose composite with RC performance (Figure 7n). On one side, RC promotes adsorption, on the other side, adsorbed water molecules contribute to the cooling power due to their high ϵ_{LWIR} . Therefore, the LiCl/cellulose composite that synergistically integrates the thermal adsorption effect, RC, and multiscale cellulose–water interactions yielded a water production rate of 5.97 L kg^{−1} day^{−1} under 0.9–1.1 kW m^{−2}.

Other dielectric materials, including but not limited to MgO ,^[29] ZnO ,^[124] Si_3N_4 ,^[125] h-BN,^[126] chromium oxide (Cr_2O_3),^[127] yttrium oxide (Y_2O_3),^[33] and zirconia (ZrO_2),^[128] are also applicable for the ongoing development of dielectric-polymer RC materials. With the continuous evolution of RC materials, we will be better equipped to develop a broader range of dielectric-polymer materials of interest in the future.

We summarized the static-composite RC materials reviewed in this chapter in Table 1. As we discussed, different composites have advantages and disadvantages and can be tailored to meet specific requirements. For instance, inorganic-inorganic composites can withstand harsh environments, while organic-organic composites can offer flexible and easy-to-manufacture solutions. Organic-inorganic hybrid materials can harvest the advantages of both options and achieve superior performance in terms of durability, adaptability, and spectral properties but the weatherability could remain as a challenge. Therefore, we expect that the development of novel static-composite RC materials will continue to advance the field of RC and contribute to energy-efficient and sustainable technologies.

5. Dynamic Radiative Cooling Materials

Static RC material in principle is only preferred to be used in the regions where cooling is dominant. Dynamic regulative materials for RC represent a significant advancement in thermal management technologies as they offer extensive application versatility and enhanced alignment with diverse thermal demands in different climates. In this review, we systematically categorize the burgeoning research in this field according to the stimuli these

Table 1. Summary of various static-composite RC materials discussed above, their design strategies, and optical properties.

Type	Materials	Design strategy	Optical properties		Ref.
			R_{sol}	ϵ_{LWIR}	
Inorganic–inorganic	SiO ₂ /HfO ₂ /Ag	Metal reflection/PhPs resonance	97%	0.65	[2]
	AAO/Al		94.6%	0.98	[63a]
	AAO/SiO ₂ /Ag		86%	0.96	[63b]
	AZO/SiO ₂ /Al	Metal reflection/plasmonic resonances	84%	0.79 ^{a)}	[62c]
	SiO _x N _y /SiO ₂ /Ag	Metal reflection, Mie scattering/molecular vibration	96.4%	0.95	[38a]
	SiO ₂ /HfO ₂ /Ag	GRIN, electromagnetic resonant absorptivity	95%	/	[61c]
	Ge/Al	GRIN	/	/	[64]
	Doped-Si/Ag	Magnetic dipole resonance	/	/	[65]
	LiF/Ag	Metal reflection	95.3%	0.89	[66]
	Al ₂ O ₃ /glass	Light scattering/PhPs resonance	>96%	≈0.95	[5b]
	Si ₃ N ₄ /BN		≈95%	≈0.95	[69]
	BaSO ₄ /SiO ₂ /geopolymer		97.6%	0.95	[70a]
	SiO ₂ /Al ₂ O ₃	Backscattering reflection/molecular vibration, GRIN	≈95%	≈0.93	[68]
	Portlandite and tobermorite cement	Light scattering	93% (Portlandite) 75% (Tobermorite)	0.84 (Portlandite) 0.87 (Tobermorite)	[71]
	Cementitious composites		90%	0.88	[72]
Organic–organic	PVDF/PVA	Light scattering/molecular vibrations	94%	0.94	[58]
	PTFE/paper		93%	0.92	[74]
	PDMS/PMMA/cotton		94%	0.79	[75]
	e-PTFE/silk nanotextile/commercial silk textile		96.5%	0.97	[76]
	SF/PLA		96%	0.95 ^{a)}	[78]
	PLA/silk		>85%	>0.96	[79]
	PVDF/PUA		93.4%	0.93	[81]
	P(VdF-HFP)/PDMS		96.5%	0.94	[82]
	PDMS/PS		92.7%	0.94	[83]
	PS/PDMS/PECA		96%	0.95	[84]
	PET/polyolefin/PMMA		98%	0.90	[85]
	TiO ₂ /PLA/PTFE	Light scattering/molecular vibrations	92.4%	0.95	[16]
	Al ₂ O ₃ /silk		95%	≈0.90	[19]
	Mica@TiO ₂ /aramid nanofibers		≈92%	0.87	[103]
	MgHPO ₄ ·1.2H ₂ O/PVDF		92.2%	0.94	[116]
Inorganic–organic	K ₂ Ti ₆ O ₁₃ /PEO		94%	0.91	[119]
	PDMS/metal	Metal reflection/molecular vibrations	>90%	0.95	[31]
	SiCNO/Ag/Al		97%	0.86	[89]
	Polymer/PET/Ag		95%	0.96	[94]
	PDMS/Ag/quartz	Metal reflection/plasmonic resonances	/	/	[17]
	PDMS/Al	Angle-selective emitter	/	0.9–0.1 ^{a)}	[95]
	SiO ₂ /TPX/Ag	Metal reflection/PhPs resonance	96%	>0.93	[8]
	SiO ₂ /PDMS/paper/Ag	Metal reflection, light scattering/molecular vibrations, PhPs resonance	91%	0.9	[97]

(Continued)

Table 1. (Continued)

Type	Materials	Design strategy	Optical properties		Ref.
			R_{sol}	ϵ_{LWIR}	
	SiO ₂ /PDMS/Cr/Ag/Cr	Metal reflection/molecular vibrations, PhPs resonance	96%	0.93	[21]
	TiO ₂ /SiO ₂ /fluorescent pigment/polystyrene-acrylates	Sunlight-induced fluorescence, light scattering/molecular vibrations, PhPs resonance	≈90%	≈0.96	[101]
	TiO ₂ /SiO ₂ /fluorescent pigment/polystyrene-acrylic		>91%	0.97 ^{a)}	[102]
	SiO ₂ /PVDF/TEOS	Light scattering/molecular vibrations, PhPs resonance	≈97%	>0.96	[105]
	Al ₂ O ₃ /TPU	Light scattering, high band gap/molecular vibrations, PhPs resonance	97.6%	≈0.96	[104]
	PVDF/BaSO ₄ /SiO ₂		95%	0.96	[113]
	BaSO ₄ /ethyl cellulose		≈98.6%	0.98	[112]
	BaSO ₄ /P(VdF-HFP)		≈98%	/	[14]
	BaSO ₄ /acrylic	Light scattering, high band gap/PhPs resonance	98.1%	0.95	[26]

^{a)} The emissivity is $\epsilon_{\text{Broadband}}$.

materials respond to, encompassing thermal, electrical, mechanical, and other stimuli-responsive variations. This classification aims to provide a coherent and comprehensive understanding of the current state and future potential of dynamic RC materials.

5.1. Thermal-Responsive Materials

5.1.1. Vanadium Dioxide

Vanadium dioxide (VO₂) exhibits remarkable thermal-responsive characteristics as it changes its optical properties in response to ambient temperature variations. This unique behavior is depicted in **Figure 8a**. At low temperatures, VO₂ has a high optical conductivity for infrared within the wavelength of 0.8–25 μm . Its optical conductivity transitions to significantly reduced above its transition temperature.^[129] In practical applications, VO₂ has been innovatively used in devices that operate distinctively in daytime photothermal and nighttime RC modes. The working principles of these devices are illustrated in **Figure 8b**.^[130] During the day, these devices are characterized by high A_{sol} and low ϵ_{LWIR} to promote photothermal. Conversely, at night, they exhibit high emissivity within the atmospheric window to facilitate effective RC. Their structural composition, detailed in **Figure 8c** and designed by Zou et al.,^[131] typically involves a multi-layered arrangement comprising an Al₂O₃ protective overcoat, a VO₂ coating, and an Al backside reflective layer on a silica substrate. As shown in **Figure 8d**, photothermal/RC dual-functional devices show a high ϵ_{LWIR} of 0.75 at 30 °C and a low ϵ_{LWIR} of 0.21 at 90 °C, while maintaining a consistently high A_{sol} of 88%.

An alternate application of VO₂ is in the creation of F–P resonators, which function in a manner contrary to the aforementioned devices. This concept, as elucidated by Wang et al.^[43] and illustrated in **Figure 8e**, involves the formation of a resonator with metallic VO₂ as the top mirror at elevated temperatures, result-

ing in high emissivity. At low temperatures, the transparency of VO₂ transforms the device into a highly reflective surface with low emissivity. Utilizing this principle, Long et al.^[11a] and Wu et al.^[11b] developed VO₂-based dynamic RC materials for energy-efficient smart windows and wall coatings in building envelopes. Long's innovation, depicted in **Figure 8f**, is a passive RC regulating thermochromic (RCRT) smart window made of a layered structure of VO₂, PMMA transparent spacer, and indium-doped tin oxide (ITO) low-emissivity (low-E) coating. It gives a solar modulation ability (ΔT_{sol}) of 9.3% and a broadband emissivity modulation ability ($\Delta \epsilon_{\text{Broadband}}$) of 0.40 (**Figure 8g**). Enhancement of this design by Muskens et al.^[132] involved replacing PMMA with transparent silica and forming a periodic VO₂ array. Their design improved the window's visible transmittance (T_{is}) from 27.8% to 62%. Moreover, through tuning the porosity of VO₂ layer, the $\epsilon_{\text{Broadband}}$ of F–P resonator at different temperatures can be tuned to meet the different energy saving demands in regions with different climates.^[133] Wu's design, on the other hand, created a flexible dynamic RC wallpaper using VO₂ metamaterials, which is capable of adjusting its emissivity in response to temperature changes (**Figure 8h**). Its capabilities of tuneable ϵ_{LWIR} and dynamic RC are illustrated under an IR camera, as seen in **Figure 8i**. At low temperatures of 20 °C, the temperature-adaptive radiative coating (TARC) has comparable ϵ_{LWIR} to the low-E control group. When the temperature exceeds its transition temperature, the TARC's ϵ_{LWIR} increases drastically to a similar value to the high-emissivity (high-E) control group.

5.1.2. GST and Other Chalcogenide Phase Change Materials

Germanium-antimony-telluride (GST) and other chalcogenide phase change materials such as indium-antimony-telluride (IST) exhibit significant temperature-responsive characteristics, making them ideal for dynamic RC applications. As illustrated in

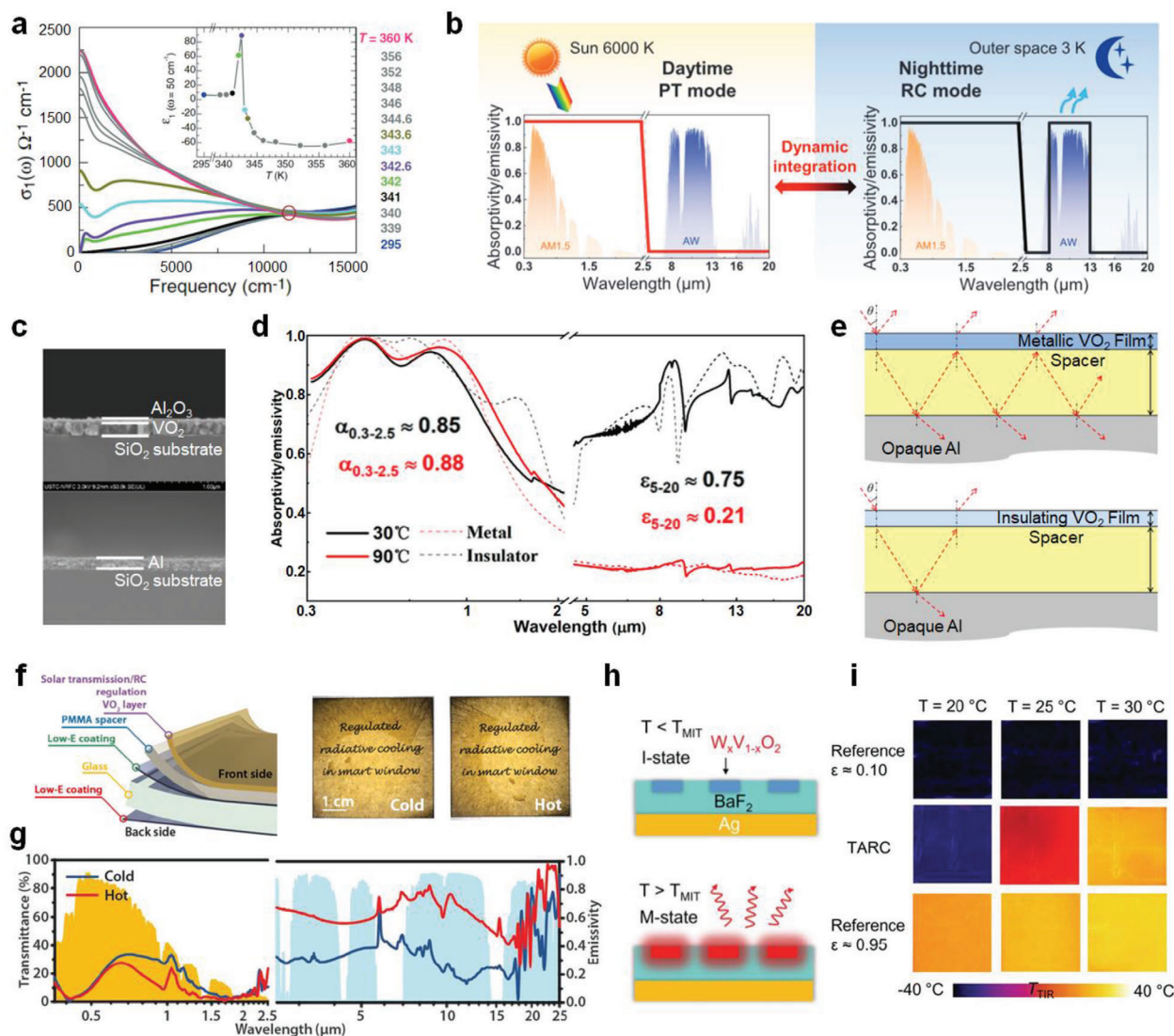


Figure 8. Examples of thermochromic-responsive dynamic RC materials based on VO_2 . a) The real part of the optical conductivity of VO_2 is plotted as a function of frequency for various representative temperatures. The open circle denotes the isosbestic (equal conductivity) point for all spectra. Inset: The temperature dependence of the real part of the dielectric function in the low-frequency limit ($\omega = 50 \text{ cm}^{-1}$). Reproduced with permission.^[129] Copyright 2017, AAAS. b) The ideal spectrum requirement of the material for the dynamic integration of PT and RC during the day and night. The orange shadows represent the normalized AM1.5G solar spectrum. The blue shadows represent the atmospheric window. Reproduced with permission.^[130] Copyright 2022, PNAS. c) SEM images for the cross-section of the photothermal/RC dual-regulation material. The $\text{Al}_2\text{O}_3/\text{VO}_2/\text{SiO}_2$ interface at the top-side and SiO_2/Al interface at the back-side were clearly observed. d) The absorption/emissivity of the photothermal/RC dual-regulation material at 30°C (black line) and 90°C (red line). Simulation results of the absorption/emissivity in the metal (red dotted line) and insulator states (black dotted line) were also performed for comparison. Reproduced with permission.^[131] Copyright 2023, American Chemical Society. e) Schematic of the F-P resonator and its wave propagation when the VO_2 top layer is metallic (top, $T > 345 \text{ K}$) and insulating (bottom, $T < 341 \text{ K}$). Reproduced with permission.^[43] Copyright 2017, Elsevier. f) Schematic structure of the RCRT window (left); and its photos at low and high temperature (right). g) Spectra of the RCRT smart window in the vis-NIR and LWIR range at 20°C (blue line) and 90°C (red line) against a normalized AM1.5 global solar spectrum (yellow shadow) and LWIR atmospheric transmittance window (blue shadow). Reproduced with permission.^[11a] Copyright 2021, AAAS. h) Schematics of materials composition and working mechanism of the TARC (top). Photograph (2 cm $\times$ 2 cm) the false-color SEM image of TARC showing high flexibility and structural consistency with the design (bottom). i) IR images of TARC compared with two conventional materials (references) with constantly low or high thermal emittance showing the temperature-adaptive switching in thermal emittance of TARC. Reproduced with permission.^[11b] Copyright 2021, AAAS.

Figure 9a, GST experiences crystallinity changes between amorphous and crystalline structures. This transformation occurs when GST is annealed above its glass-transition temperature of 160°C , changing from a low-E amorphous state to a high-E crystalline state. Conversely, Adibi et al. elucidated that heating crys-

talline GST above its melting temperature (around 600°C) and then quenching it reverts the material to its amorphous state.^[134]

Qiu et al. first reported the application of GST in dynamic RC.^[135] They demonstrated the use of GST for temperature-responsive tuning of emissivity by forming a gold nanodisk array

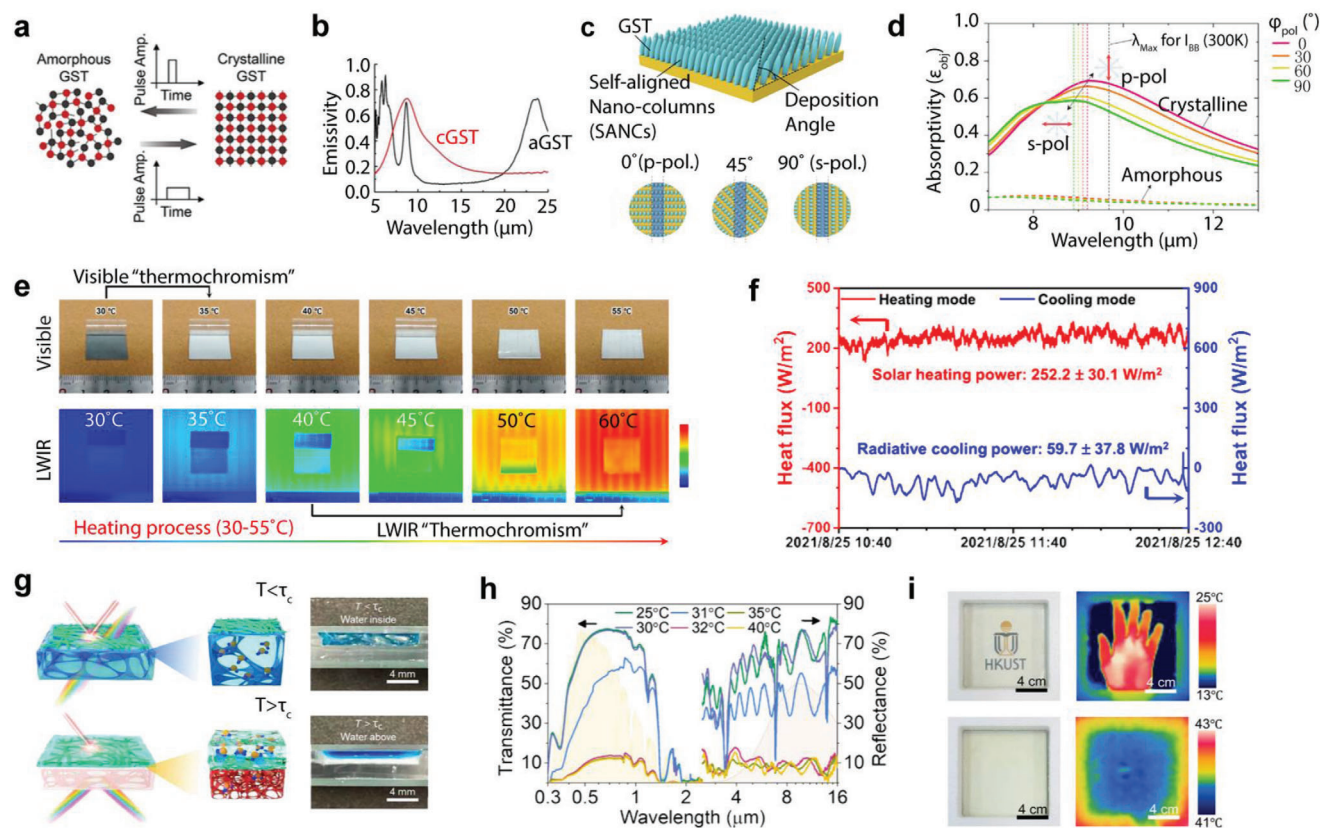


Figure 9. Thermochromic-responsive dynamic RC materials based on chalcogenide phase change material, polymer, and hydrogel. a) Schematic illustration of the reversible switching of the GST phase between amorphous and crystalline states. A long pulse with a moderate amplitude enables amorphous-to-crystalline conversion. A short pulse with a large amplitude enables crystalline-to-amorphous conversion via the melt-quenching process. Reproduced with permission.^[134] Copyright 2020, De Gruyter. b) Experimental thermal emissivity of the amorphous GST (aGST) thermal emitter (black line) and crystalline GST (cGST) phases thermal emitter (red line). Reproduced with permission.^[135] Copyright 2017, John Wiley and Sons. c) Illustration of self-aligned GST nanocolumns (SANCs) with top views corresponding to different polarization angles. d) Calculated emissivity spectra of optimized SANCs with different polarization angles and phases. The short dashed lines indicate the wavelengths of the emission peak. Reproduced with permission.^[136] Copyright 2023, Elsevier. e) Simultaneously captured optical and infrared images (set background emissivity of 1) of the dual-mode device during the heating process of the heating plate, which visually presents the visible and infrared thermochromism. f) Continuous time-resolved PT power (red line) and RC power (blue line) measured in an open-space outdoor environment. Reproduced with permission.^[139] Copyright 2022, PNAS. g) Schematic and cross-sectional images of the working mechanism of STR window in the cold condition (top) and hot condition (bottom). The blue-dyed water was used to indicate the water distribution in hydrogel at different temperatures. h) Spectra of the STR windows at different temperatures. The orange and pink shadows represent the spectra of AM1.5G solar radiation and blackbody radiation (300 K). i) Visible and infrared images of the STR window in cold (top) and hot (bottom) conditions. Reproduced with permission.^[140] Copyright 2022, AAAS.

on a GST substrate, as shown in Figure 9b. To enhance the emissivity modulation capability, Song et al. developed a polarization-driven ϵ_{LWIR} regulator.^[136] This device, as illustrated in Figure 9c, comprises self-aligned nano-GST columns deposited by angled vapor deposition on a gold substrate. Acting as a polarizer, the angled columnar structure enables the device to dynamically adjust its ϵ_{LWIR} through both phase change and polarization tuning. The device achieved an LWIR emissivity modulation ability ($\Delta\epsilon_{\text{LWIR}}$) of ≈ 0.4 through a phase change, which further improved to about 0.5 when polarization was adjusted (Figure 9d). Other chalcogenide phase change materials, like IST, have also been explored for dynamic RC applications. Li et al. reported the utilization of IST in a silicon-IST-silicon-metal multilayered structure to achieve a significant ϵ_{LWIR} modulation with a $\Delta\epsilon_{\text{LWIR}}$ of 0.83.^[137] This showcases the versatility and efficacy of chalcogenide materials as dynamic RC materials.

5.1.3. Thermal-Responsive Polymers and Hydrogels

Temperature-responsive polymers and hydrogels represent a novel approach in the field of dynamic RC materials and distinguish themselves from VO_2 and chalcogenide phase change materials through their unique mechanisms of RC adjustment. Unlike VO_2 and chalcogenides, which alter their optical properties to regulate ϵ_{LWIR} , temperature-responsive polymers maintain a near-unity ϵ_{LWIR} . Their RC tuning mechanism involves covering or uncovering an underlying low-E layer via reversible shape deformation, as detailed by Chen et al.^[138] This approach contrasts with the phase change method employed by other thermal-responsive materials and offers a different avenue for dynamic RC regulation.

An exemplary implementation of this technology is seen in the shape-memory polymer-based dynamic RC regulation device

designed by Ma et al.^[139] Drawing inspiration from the reactive nature of *Mimosa pudica*'s leaves and integrating a Himalayan rabbit-inspired thermochromic coating, this device exhibits significant responsiveness to temperature changes. As shown in Figure 9e, when the temperature rises from 40 to 55 °C, the shape-memory polymer flattens and covers the low-E Al substrate beneath it. This shape-morphing process results in an increase in the device's temperature under an IR camera, aligning it with the high-E background. Furthermore, the device incorporates an IR-transparent thermochromic coating applied to the low-E Al substrate, leading to a color change from grey to white as temperature increases. This dual-regulation system, illustrated in Figure 9f, effectively manages both A_{sol} and RC to achieve a photothermal power of 252.2 W m⁻² and a RC power of 59.7 W m⁻².

Similarly, temperature-responsive hydrogels offer another innovative approach to dynamic RC regulation. Huang et al. reported the development of a poly(*N*-isopropylacrylamide) (PNIPAm) hydrogel-based smart window (STR window) capable of synchronously regulating its solar and thermal radiation.^[140] This window, as shown in Figure 9g, leverages the temperature-dependent phase separation of the hydrogel to control its ϵ_{LWIR} through the absorption and release of water. At room temperature, the hydrogel retains water within its polymer network, allowing high solar transmittance while inhibiting RC by exposing the topmost low-E silver nanowire (AgNWs). In contrast, at the temperature above its lower critical solution temperature (LCST), the hydrogel expels water and agglomerates, reducing solar transmittance and prompting RC by covering the AgNWs with a high-E water layer. Figure 9h,i demonstrates that this smart window achieves a ΔT_{sol} of 58.4% and a $\Delta \epsilon_{\text{LWIR}}$ of 0.57.

5.2. Electrical-Responsive Materials

5.2.1. Reversible Electrodeposited Metals and Conductive Metal Oxides

Reversible electrodeposition represents an innovative approach in dynamic RC utilizing the deposition of metal particles to regulate its ϵ_{LWIR} . As illustrated in Figure 10a,^[141] this electrodeposition process can be exemplified with silver NPs (AgNPs). Initially, in the absence of voltage, silver ions are dispersed in the electrolyte and resulting in a device with high ϵ_{LWIR} due to the strong IR absorption of the gel electrolyte. Upon applying a deposition voltage, silver ions are electrodeposited onto nanoscopic Pt film electrode surfaces, transforming the device into a highly IR reflective state and thereby achieving low ϵ_{LWIR} . The performance of reversible electrodeposition is notably efficient. Figure 10b demonstrates the change in MWIR emissivity (ϵ_{MWIR}) and ϵ_{LWIR} of the device with increasing deposition time. A mere 10 s of electrodeposition can modulate its ϵ_{MWIR} and ϵ_{LWIR} by 0.77 and 0.71, respectively. This effect is further validated by the broadband IR reflectance spectra, as shown in Figure 10c, where a 15-s deposition leads to a significant increase in $\epsilon_{\text{Broadband}}$ from 0.08 to 0.83. To enhance the dynamic RC modulation performance of electrodeposition, Hsu et al. replaced the BaF₂ protective layer and nanoscopic Pt electrode with PE films and monolayer graphene to create a flexible RC regulator.^[142] This innovation has a high

T_{vis} of 87.9% and can switch its ϵ_{LWIR} between 0.12 and 0.94 through reversible silver electrodeposition.

Further advancements in this field are seen in copper-based reversible electrodeposition. Hsu et al. extended the lifespan of such devices by substituting the conventional silver-DMSO electrolyte system with an aqueous-based copper electrolyte.^[10a] This copper-based electrodeposition device exhibited enhanced cycle durability, sustaining over 2500 deposition/dissolve cycles while maintaining a $\Delta \epsilon_{\text{LWIR}}$ of 0.85, as shown in Figure 10d. Liu et al. introduced a high-transparency electrodeposition RC regulator using AZO NPs.^[10b] This novel device, unlike conventional electrodeposition systems, maintains a consistent T_{vis} of 84.7% in both high- and low-E states, as shown in Figure 10e. Additionally, it demonstrates promising emissivity modulation abilities (0.51 for MWIR, 0.41 for LWIR), marking a significant improvement over previous designs.

5.2.2. Conjugated Conductive Polymers

Conjugated conductive polymers (CPs) like poly(3,4-ethylenedioxythiophene) (PEDOT) and polyaniline (PANI) have emerged as pivotal materials in the development of electrochemical responsive dynamic RC materials. Xu et al. identified that the core mechanism underlying their emissivity regulation is the alteration of energy levels and bands.^[143] This process involves the intercalation or deintercalation of electrons into CPs upon voltage application, leading to the oxidation or reduction of these polymers. Such changes cause a transition from aromatic to quinoidal forms in the conjugated backbones of CPs, thus creating new energy levels and altering their optical properties and ϵ_{LWIR} .

Focusing on PEDOT-based CPs, Jonsson et al. developed an innovative electrochemical responsive RC material using poly(3,4-ethylenedioxythiophene):tosylate (PEDOT:Tos).^[144] This design features a sandwich structure with two PEDOT:Tos electrodes separated by an electrolyte-filled porous cellulose layer (Figure 10f). In the "ON" state, the top PEDOT:Tos electrode is reduced while the bottom one is oxidized to enhance the sky-toward ϵ_{LWIR} of the device. Conversely, in the "OFF" state, the oxidation states of the electrodes are reversed, resulting in a low-E state that inhibits the RC effect.

Polyaniline (PANI) is another commonly used electrochromic CPs. Li et al. created an emissivity-adjustable PANI-based thin film inspired by a chameleon.^[145] By combining PANI with an electrolyte to form pixels, they produced a flexible film capable of modulating ϵ_{LWIR} by 0.58. Hsu et al. further enhanced the flexibility of PANI-based RC regulators by employing a kirigami design (Figure 10g).^[146] Through cutting patterns that conform to the contours of the human body, this approach enables the PANI-based personal thermal management device to fit closely to the body while maintaining a $\Delta \epsilon_{\text{LWIR}}$ of 0.28.

5.2.3. Valance/State Change Metal Oxides

Valance and state change metal oxides, such as Li₄Ti₅O₁₂ (LTO) and WO₃, offer a method for regulating emissivity through voltage-induced changes in their oxidation/reduction states. With

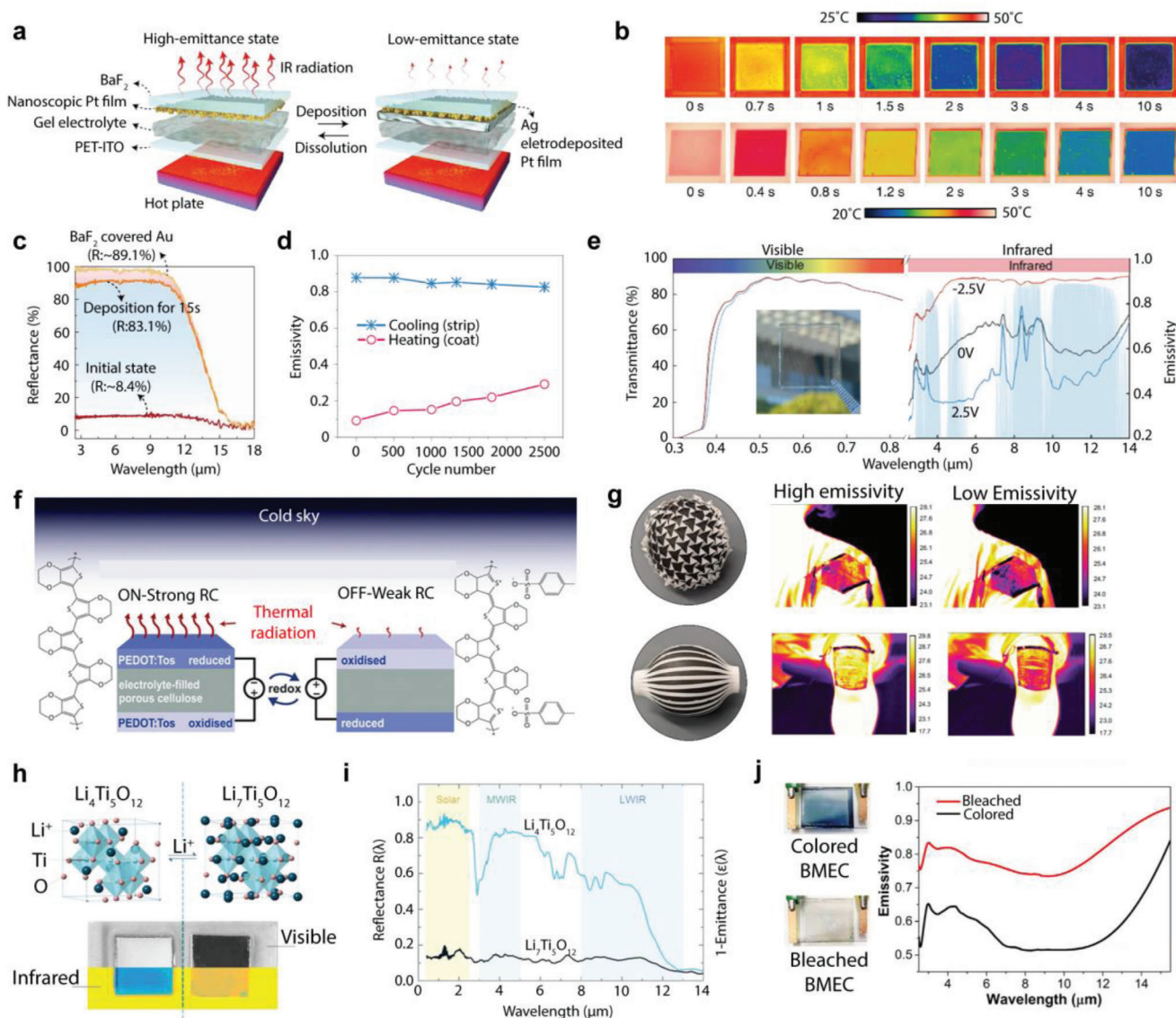


Figure 10. Electrical-responsive dynamic RC materials. a) Schematics of a nanoscopic Pt film-based reversible silver electrodeposition (RSE) device before (left) and after (right) electrodeposition. b) Real-time MWIR (top) and LWIR (bottom) images of the RSE device during the electrodeposition process. c) The broadband IR reflectance spectra of the RSE device before and after Ag electrodeposition (15 s). The broadband IR reflectance spectrum of the BaF_2 substrate covered standard gold (Au) film represents an ideal case. Reproduced with permission.^[141] Copyright 2022, AAAS. d) Cycle life test of the Cu-based electrochromic broadband IR system. Reproduced with permission.^[10a] Copyright 2023, Springer Nature. e) Visible transmittance and infrared emissivity spectra of transparent electrochromic RC regulator at various applied voltages. The blue shaded areas are infrared atmospheric windows. Inset: Photo of the transparent electrochromic RC regulator. Reproduced with permission.^[10b] Copyright 2023, Springer Nature. f) Schematic illustration depicting the concept of reversible electrically tunable RC based on the tunable thermal emissivity of PEDOT:Tos. The chemical structures of PEDOT:Tos are shown for the reduced (left) and oxidized (right) states. Reproduced with permission.^[144] Copyright 2023, Elsevier. g) Schematic of kirigami inspired personal thermal management device and their corresponding IR photos at high-E and low-E states. The kirigami cut allows the device to stretch and deform conformally along with the human body contour and movement. Reproduced with permission.^[146] Copyright 2023, Oxford University Press. h) Crystal structures of $\text{Li}_4\text{Ti}_5\text{O}_{12}$ and $\text{Li}_7\text{Ti}_5\text{O}_{12}$ and their corresponding visible and IR photos. i) Visible-to-infrared spectral reflectance $R(\lambda)$ of $\text{Li}_4\text{Ti}_5\text{O}_{12}$ and $\text{Li}_7\text{Ti}_5\text{O}_{12}$ NPs (mass loading 2 mg cm^{-2}) deposited on Al foil and covered by BaF_2 . Solar, MWIR, and LWIR ATWs are highlighted. Reproduced with permission.^[148] Copyright 2018, John Wiley and Sons. j) Digital photograph of bleached and colored WO_3 -based BMEC devices and their corresponding emissivity spectra. Reproduced with permission.^[150] Copyright 2021, John Wiley and Sons.

a working principle similar to conjugated polymers, these metal oxides undergo lithium-ion (Li^+) intercalation or deintercalation upon voltage application and lead to a shift in their energy bands.^[147] This process allows them to transit between insulating and metallic states, thereby altering their optical properties.

LTO was first introduced by Yang et al. as a zero-strain material with highly reversible lithium-ion intercalation and negligible volumetric changes during ion intercalation/deintercalation.^[148] Figure 10h illustrates the delithiated state ($\text{Li}_4\text{Ti}_5\text{O}_{12}$) of LTO NPs to showcase their transparency in the visible spectrum and

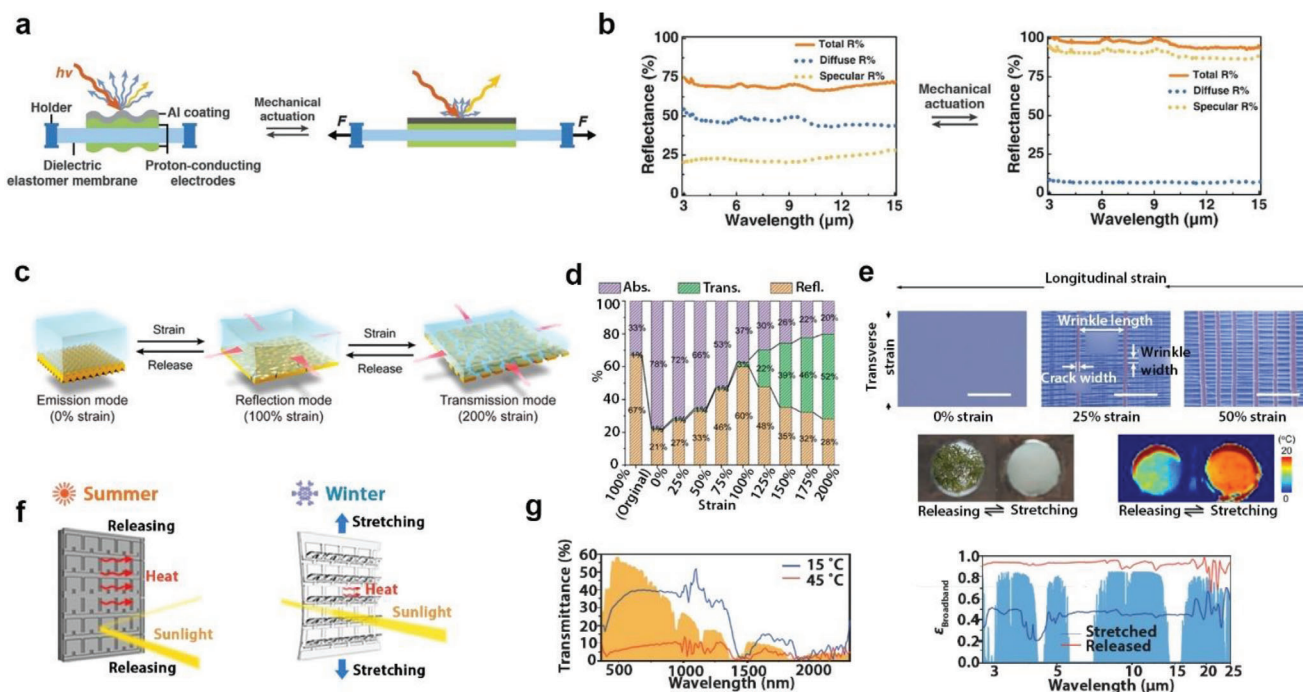


Figure 11. Mechanical-responsive dynamic RC material based on wrinkle, crack, and kirigami structure. a) Schematics of an Al-modified device before (left) and after (right) mechanical actuation in an adjustable holder, illustrating the change in the surface morphology and the reflection of infrared light. b) The broadband IR reflectance spectra of the Al-modified device before (left) and after (right) mechanical actuation. The broadband IR reflectance (solid orange traces) are shown along with their specular (dotted yellow traces) and diffuse (dotted blue traces) components. Reproduced with permission.^[151] Copyright 2018, AAAS. c) Schematic of mechanical actuation of the triple-mode modulator with biaxial strains of 0% (left), 100% (middle), and 200% (right). d) Weighted-average absorbance, transmittance, and reflectance of the film under the biaxial strain from original 100% to 0% and then stepwise to 200%. Reproduced with permission.^[153] Copyright 2021, American Chemical Society. e) SEM images of a bilayer film with different longitudinal strains. The tinted zone distinguished the cracks and wrinkles (top). Scale bars are 50 μm. Visible and IR images of the bilayer dynamic windows with released and stretched states (bottom). Reproduced with permission.^[154] Copyright 2021, Elsevier. f) Schematic of kirigami-inspired dual-regulation window in summer and winter. g) The UV-vis-NIR spectra of the durable solar/RC dual-control smart window at 15 and 45 °C (left). The orange shadow represents the AM1.5 global spectrum. Broadband emissivity spectra of the durable solar/RC dual-control smart window in released and stretched states (right). Reproduced with permission.^[158] Copyright 2023, Royal Society of Chemistry.

allow the underlying Al layer to be visible. In this state, LTO-Al exhibits high reflectivity in the broadband IR region. However, upon lithium intercalation, the lithiated $\text{Li}_7\text{Ti}_5\text{O}_{12}$ becomes highly absorptive in both visible and broadband IR regions. As demonstrated in Figure 10i, LTO-based electrically responsive devices have a significant broadband modulation capability, with a visible reflectance difference (ΔR_{vis}) of 74% and a $\Delta \epsilon_{\text{LWIR}}$ of 0.30. Utilizing LTO's remarkable capabilities in regulating T_{vis} , NIR transmission (T_{NIR}), and ϵ_{LWIR} , Wang et al. developed an LTO-based electrochromic smart window.^[149] This window transitions from transparent to dark blue upon lithium intercalation, indicating its potential in applications requiring variable visual appearances.

WO_3 , or tungsten oxide, is well-known for its electrochromic properties in modulating visible light. Fan et al. uncovered its ability to modulate ϵ_{LWIR} .^[150] By integrating both amorphous and crystalline WO_3 layers into a multilayered electrochemical device, it is possible to achieve dual-modulation of both T_{vis} and ϵ_{LWIR} . In its bleached state, the broadband modulation electrochromic (BMEC) window is transparent with a ϵ_{LWIR} of 0.8 (Figure 10j). Upon voltage application, the BMEC window adopts a dark blue coloration, and its ϵ_{LWIR} decreases to 0.6.

5.3. Mechanical- and Humidity-Responsive Materials, and Other Designs

5.3.1. Mechanical-Responsive Materials

Mechanical-Responsive Materials Based on Wrinkle and Crack: Mechanical-responsive dynamic RC materials leveraging the principles of wrinkles and cracks offer innovative methods for thermal regulation. Inspired by cephalopods, Gorodetsky et al. designed wrinkle-based devices to modulate the intensity of reflected incident infrared radiation by altering their surface structure.^[151] In Figure 11a, the device's wrinkled state is characterized by diffuse reflection, which accounts for about 48% of the total LWIR reflection. Conversely, upon actuation to a smooth state, the average total LWIR reflectance of the surface exceeds 90%. They further introduced a dynamic thermoregulatory device based on cracking mechanism.^[152] Initially, as depicted in Figure 11b, this device is covered with a continuous layer of low-E copper coating. Mechanical actuation causes this layer to spread apart, revealing portions of the stretched, underlying high emissivity polymer matrix. This structural alteration results in a significant change in the device's LWIR reflectivity (R_{LWIR}) of 47%.

Combining both wrinkling and cracking approaches, Hsu et al. developed a triple-mode LWIR modulator.^[153] This device, created by coating a low-E metal coating on a pre-stretched stretchable styrene-ethylene-butylene-styrene (SEBS) block copolymer, exhibits diverse modes of operation (Figure 11c). In the emission mode, the device is in its released state and forms wrinkles on the metal surface, leading to high ϵ_{LWIR} . In the reflection mode, stretching the device flattens the metal coating, thereby reducing its ϵ_{LWIR} and increasing its R_{LWIR} . The transmission mode, achieved by further stretching to induce cracks in the metal coating, allows visible light to pass through to render the device a transparent appearance. The performance of this multimodal functionality is numerically analyzed in Figure 11d, showing the dominance of absorptance (78%) at 0% strain (emission mode), a significant increase in reflectance from 21% at 0% strain to 67% at 100% strain (reflection mode), and a substantial transmittance of 52% at 200% strain (transmission mode), compared to emission (20%) and reflection (28%).

In a similar vein, Hu et al. fabricated a mechanical-responsive dynamic RC/solar regulation smart window by applying transparent conductive low-E poly(3,4-ethylenedioxythiophene) polystyrene sulfonate (PEDOT:PSS) polymer on a stretchable PDMS substrate.^[154] In its released state, the PEDOT:PSS overcoating is continuous and smooth, allowing the window to exhibit a high solar transmittance (T_{sol}) of $\approx 80\%$ and a R_{LWIR} of 41%. However, upon stretching, micro-cracking and wrinkling occur to scatter solar radiation and enhance IR emissivity (Figure 11e). This results in a ΔT_{sol} of 55% and a $\Delta \epsilon_{\text{LWIR}}$ of 0.25.

Mechanical-Responsive Materials Based on Kirigami Structures: Kirigami structures, derived from the ancient Asian art of cutting paper into various patterns, imbue materials with distinctive mechanical properties such as 2D to 3D transformation,^[155] predictable deformation,^[156] and enhanced extensibility with minimal impact on conductivity.^[157] These kirigami-inspired metamaterials can programmatically alter their physical properties by modifying configurations, thus enabling the creation of mechanical-responsive dynamic RC devices. By controlling the exposure of underlying low-E layers, kirigami patterns facilitate the regulation of ϵ_{LWIR} in response to environmental changes.

In the realm of dynamic RC regulation, Long et al. innovatively integrated PDMS-based kirigami structures with low-E AgNWs coatings and thermochromic PNIPAm hydrogel.^[158] This combination allows for dual-regulation of solar and RC: In summer, the high-E PDMS kirigami is in a relaxed state, covering the entire surface to enhance RC, while the hydrogel turns opaque to reduce solar transmission (Figure 11f). In winter, the kirigami structure undergoes an out-of-plane structural change, exposing the low-E AgNWs layer to reduce RC, while the hydrogel becomes transparent to facilitate solar transmission and sunlight heating. Figure 11g illustrates the window's capabilities, including a ΔT_{sol} of 21.7% and a $\Delta \epsilon_{\text{Broadband}}$ of 0.5.

Hu et al. further explored kirigami structures' potential by designing a device focusing on modulating IR emissivity through an angle-dependent thermal reflector.^[159] This device fabricated using a kirigami structure with transparent ITO-coated PET film, demonstrates significant temperature regulation in a building model during outdoor tests. Under cooling mode, it reduced the interior temperature by $\approx 3.3^\circ\text{C}$, while in heating mode, the temperature increased by around 3.9°C .

5.3.2. Humidity-Responsive Materials

Humidity-responsive materials are increasingly recognized for their potential in dynamic RC regulation, particularly in personal thermal management and smart window technologies. Wang et al. pioneered the use of such materials in textiles from triacetate-cellulose fused bimorph fibers coated with a thin layer of low-E carbon nanotubes to craft a dynamic RC fabric.^[160] As illustrated in Figure 12a, the yarn in this textile tightens under hot and wet conditions, causing neighboring fibers' meta-elements to come closer together and induce resonant electromagnetic coupling. This coupling shifts the textile's ϵ_{LWIR} to better match human body thermal radiation, thus enhancing heat exchange. Conversely, in cold and dry conditions, the yarn loosens to reduce heat dissipation and effectively gating infrared radiation through the textile. This humidity-responsive behavior of the novel textile is due to the difference in hydrophobicity between triacetate and cellulose. As visualized in Figure 12b, an increase in relative humidity from 10% to 75% at room temperature causes the meta-yarns to brighten under the IR camera due to increased ϵ_{LWIR} .

Yang et al. introduced a humidity-responsive materials-based smart window.^[161] This window leverages the optical scattering properties of porous polymer coatings (PPCs), which change upon wetting. In the "icehouse" mode, as shown in Figure 12c, the dry porous PE scatters sunlight while allowing LWIR emitted from the room to pass through. In the "greenhouse" mode, the porous polymer is wetted by alcohol and becomes transparent due to refractive matching between the polymer and liquid, and simultaneously blocks IR transmission. This design demonstrates a ΔT_{sol} of 33% and a ΔT_{LWIR} of 64% between the two states. Figure 12d visualizes the visible and IR regulation ability of the window. At dry state, the window is transmittance to LWIR while opaque to visible light. Conversely, the window becomes highly absorptive to LWIR while transparent to visible light when wet.

5.3.3. Dynamic Radiative Cooling Materials Based on Mechanical Actuation

Dynamic RC materials based on mechanical actuation stand out for their versatility and robustness compared to spontaneously responsive materials. They offer a broader range of material choices, simpler structures, and generally exhibit higher durability. Unlike materials that respond automatically to environmental changes, these mechanically actuated materials can be manually or electronically controlled for precise and adaptable thermal management.

Hsu et al. designed a notable application for energy-efficient building walls, which is the dual-mode electrostatically controlled thermal contact conductance regulator.^[12] This device includes rotary actuators or rollers and a thin-film polymer composite with solar heating and RC functions side-by-side (Figure 12e). The cooling segment, detailed in Figure 12f, maximizes RC by reflecting sunlight with a silver coating and utilizing the near-unity emissivity of PDMS. Conversely, the heating part, with a dark-colored copper/zinc coating, absorbs solar energy and minimizes radiation loss, thereby efficiently transferring heat to the building envelope (Figure 12f). Mode-switching is achieved by moving the materials along a track system with actuators and

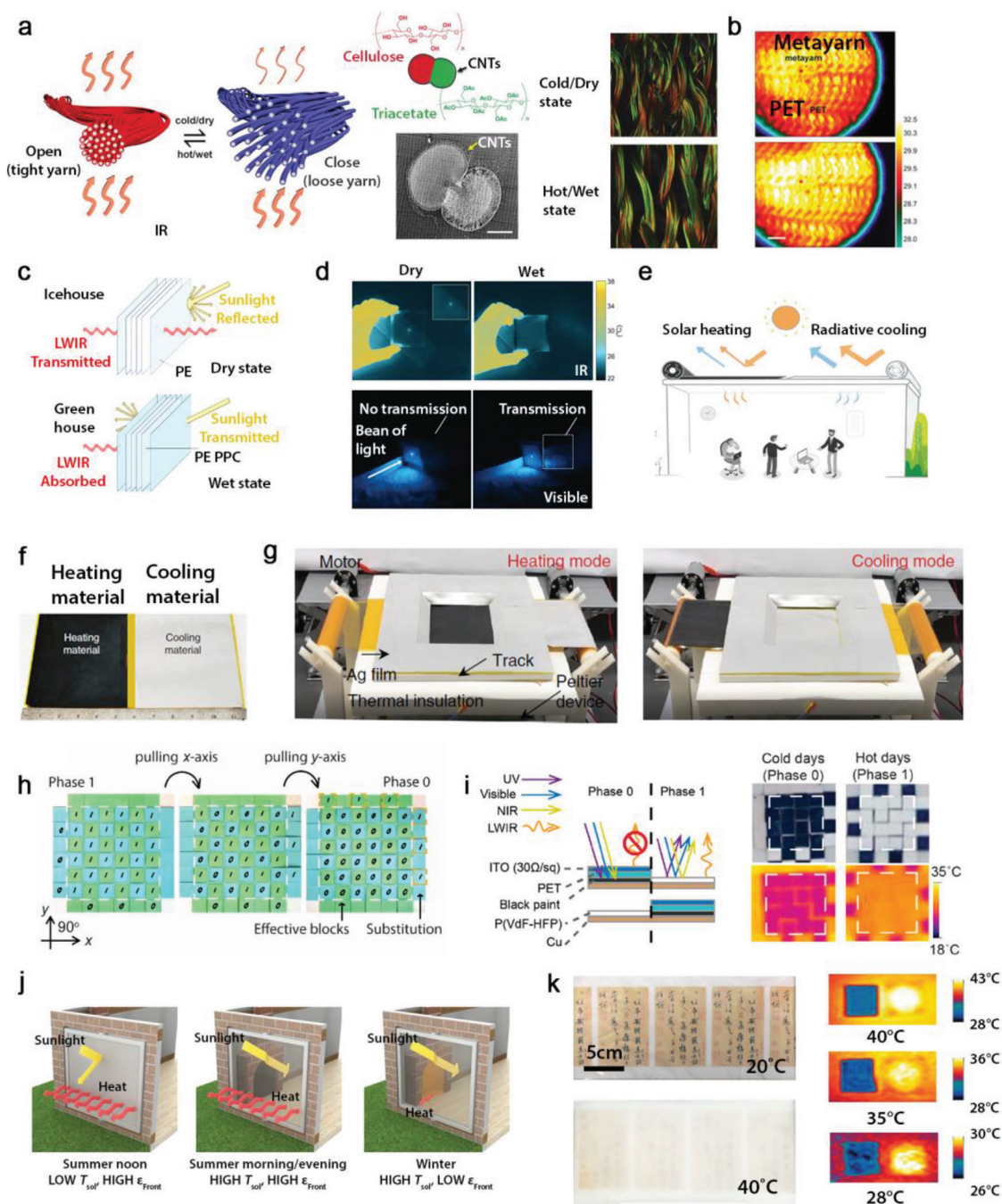


Figure 12. Dynamic RC materials response to humidity and mechanical and mechanical actuation. a) Design principles of the LWIR gating textile its structure and cross section SEM image, and its confocal fluorescent microscopy images of the knitted fabric in the closed state (low humidity) and the open state (high humidity). b) IR images from the same region of a fabric piece containing metayarns knitted into a PET matrix at 10% and 75% relative humidity. Reproduced with permission.^[160] Copyright 2019, AAAS. c) The icehouse-to-greenhouse switching of PPCs, exemplified by a PE-air/alcohol system. d) Dry icehouse and wet greenhouse states of the porous PE-air/alcohol system shown in the LWIR (top) and the visible wavelengths (bottom). Inset in LWIR photos shows the warm light-emitting diode without the switchable transmitter in front. Reproduced with permission.^[161] Copyright 2019, Elsevier. e) Schematic of the dual-mode device at heating (left) and cooling (right) mode. The switchable building envelope can utilize both renewable heating and cooling sources. f) The photo of the dual-mode material shows the different visible appearance of the heating/cooling parts. g) Photo of the switchable building envelope and its actuation device in heating (left) and cooling (right) mode. Reproduced with permission.^[12] Copyright 2020, Springer Nature. h) Photographs to demonstrate a complete phase 0–1 transition by pulling the x- (blue paper) and y- (green paper) belts. i) Left: Schematic of the Design 1. Phases 0 and 1 are for cold and hot days, respectively. Right: Photographs and corresponding thermal images of Design 1 sample on the two phases. Effective areas are marked by the dashed lines. Reproduced with permission.^[162] Copyright 2022, American Chemical Society. j) Working principle of the TET smart window panel in summer and winter. k) Left: Photo of TET smart window at 20 and 40 °C. Right: IR images for the low-E and high-E sides of the TET smart window at different temperatures. Reproduced with permission.^[163] Copyright 2021, Elsevier.

aligning the material to the desired mode as environmental conditions change (Figure 12g).

Switchable interwoven surfaces offer an innovative approach to dynamic RC regulation. Long et al. conceptualized a design where blocks with distinct R_{sol} and ϵ_{LWIR} are interlaced. These blocks can be alternately shifted by sliding the x-belts and y-belts, effectively altering the surface's absorptive and emissive characteristics (Figure 12h).^[162] Design 1 employs layers of ITO, black paint, and P(VdF-HFP) (Figure 12i). For cold days, solar energy penetrates the transparent ITO and is absorbed by the black paint layer for heating, while maintaining low ϵ_{LWIR} due to the top low-E ITO layer. On hot days, the top porous P(VdF-HFP) layer reflects solar energy and provides high ϵ_{LWIR} for RC. This design alternates between a high-absorbance black appearance (A_{sol} : 87%) and a high-reflectance white (R_{sol} : 95%), with corresponding ϵ_{LWIR} changes (0.38 for cold days and 0.91 for hot days). On the other hand, design 2 which is tailored for smart windows, consists of ITO, tungsten-doped VO_2 , and PVC plastic foil. Solar energy is transmitted through all layers in winter, with low ϵ_{LWIR} due to the top-most ITO layer. During summer, the NIR is blocked by the metallic VO_2 layer, while the high ϵ_{LWIR} top PVC layer is exposed for effective RC. This design efficiently modulates the thermal properties of the window and makes it cater to different environmental conditions.

Long et al. introduced a thermochromic smart window with a high emissivity modulation ability (TET smart window) based on mechanical flipping.^[163] This window, illustrated in Figure 12j, uses a dual-side design with high-E HPC hydrogel and low-E ITO coating on opposite sides of the glass. In summer, the high-E hydrogel side faces outward to promote RC, with varying solar transmission modes for energy saving and luminance. In winter, the window flips over, allowing solar transmission for heating while minimizing RC. Figure 12k highlights its thermochromic behavior and $\epsilon_{\text{Broadband}}$ difference between the two sides. The TET smart window has a $\Delta\epsilon_{\text{Broadband}}$ of 0.85 and a promising ΔT_{sol} of 50.3%. Wang et al. developed a dual-mode flappable cooler/heater, which utilizes MXene nanosheets on porous PVDF with coral-like hierarchical structures.^[164] This film offers high R_{sol} (96.7%) and high ϵ_{LWIR} (0.96) on the cooling side for effective subambient cooling, and low ϵ_{LWIR} (0.11) with high A_{sol} (75.7%) on the heating side to enhance its heating capability.

In summary, dynamic RC materials based on stimuli-responsive materials represent a versatile and effective approach to dynamic thermal management. Their ability to alter their emissivity and other optical properties opens new avenues in developing thermoregulatory devices such as dynamic RC coolers, smart windows, and personal thermal management textiles. Table 2 summarized the materials discussed in this chapter and their examples of application. It showcases the potential of stimuli-responsive materials in the field of dynamic RC.

6. Multifunctional Radiative Cooling Materials

The multifunctionality gives the RC materials a wider range of application scenarios. In this review, transparency, color, self-cleaning, durability, and thermal insulation are elaborated, aiming to demonstrate the versatility and practicality of RC materials.

6.1. Visibly Transparent Radiative Cooling Materials

In contrast to general PDRC materials designed for robust solar radiation reflection, which are commonly applied to building walls and roofs, visible transparency is a must in specific scenarios such as windows,^[24,165] electronic devices,^[166] and solar cells.^[167]

6.1.1. Visibly Transparent and Near-Infrared Reflective Radiative Cooling Materials

Zhou et al. proposed an innovative energy-saving strategy involving a transparent double-layer coating on windows (Figure 13a).^[24] This approach utilizes a NIR reflectance (R_{NIR}) underlayer to block undesired NIR in sunlight, while the top emitter, acting as a complementary layer, enhances ϵ_{LWIR} to cool the windows. PDMS functions as a transparent polymer emitter, and ITO acts as the NIR reflective layer (Figure 13b). Outdoor tests revealed that the addition of the top emitter reduced the temperature of the NIR reflective window by $\approx 7^\circ\text{C}$. Implementing this strategy results in additional cooling energy savings ranging from 20 to 49 $\text{MJ m}^{-2} \text{ year}^{-1}$ for a typical global building, constituting 3% to 8% of total annual cooling energy consumption. This research highlights the significant impact of window cooling on interior energy efficiency.

Building upon the window cooling strategy, various studies have explored multilayer structures for more energy-efficient enhancements.^[165a,168] Yan et al. presented a low-cost biomimetic RC (Bio-RC) glass, comprising three layers: PDMS, ITO, and SiO_2 (Figure 13c).^[168a] This Bio-RC glass demonstrates effective T_{vis} of 80.5%, T_{NIR} of 10.3%, and ϵ_{LWIR} of 0.95, to reduce temperatures during daylight while maintaining transparency and aesthetics (Figure 13d). Concurrently, Rho et al. proposed a transparent radiative cooler based on a dielectric multilayer consisting of five layers of hydrogenated amorphous silicon (a-Si:H) and SiO_2 stacked alternately, with a top PDMS layer on a glass substrate (Figure 13e).^[168b] The former transmits visible light while reflecting NIR, and the latter contributes to high emissivity in the atmospheric window. Despite exhibiting a yellow color, the radiative cooler transmits a large portion of visible wavelengths, preserving the color of objects underneath. Outdoor tests confirmed that the transparent radiative cooler can maintain the colors of objects underneath and cool itself by up to 10.1°C compared to a reference sprayed with the same paint, demonstrating high compatibility with commercially available paints. Furthermore, Li et al. fine-tuned the thickness of 30 layers of TiO_2 and SiO_2 (Figure 13f), achieving effective T_{vis} of 74.0%, R_{NIR} of 87.0%, and ϵ_{LWIR} of 0.88.^[168c] The high T_{vis} makes the $\text{SiO}_2/\text{TiO}_2$ stack film available to meet the high transparency requirement for some applications.

6.1.2. Visibly Transparent and Near-Infrared Transparent Radiative Cooling Materials

In the context of some solar cells where exceptionally high solar transmittance is needed to enhance power conversion efficiency, a high T_{NIR} is needed.^[167] Wan et al. presented an innovative, environmentally friendly solution—the transparent silk RC (TSRC)

Table 2. Summary of dynamic EC materials, their properties, and applications.

Type	Materials	Device structure	Application	Ref.
Thermal-responsive	VO ₂	Al ₂ O ₃ -VO ₂ -Quartz-Al multilayer	Photothermal/RC dual-functional heating/cooling source	[130,131]
		F-P resonator with VO ₂ /PMMA composite-PMMA spacer-ITO reflector	Solar transmittance/RC dual-regulation smart window	[11a]
		F-P resonator with VO ₂ -SiO ₂ spacer-AZO reflector-CaF ₂ substrate		[132]
		F-P resonator with W-doped VO ₂ -BaF ₂ spacer-Ag reflector	Flexible RC regulation wall coating	[11b]
	Chalcogenide phase change materials	Au nanodisk array-GST-Au substrate	Dynamic RC controller	[135]
		GST column-Au substrate		[136]
		Silicon-IST-silicon-Au multilayer		[137]
	Shape-memory polymers	Diocetyl phthalate-modified poly(4-methyl-1-pentene) polymer foil+Nano Cr coated Al plate	Photothermal/RC dual-functional heating/cooling source	[138]
		Two-way shape memory polymer foil-thermochromic coating-Al substrate	Solar absorption/RC dual-regulator	[139]
	LCST hydrogel	AgNWs coating+PNIPAm hydrogel	Solar transmittance/RC dual-regulation smart window	[140]
Electrical-responsive	Electrodeposited metals and metal oxides	Transparent electrodes (Pt/Graphene films)+Ag (particles and films)+Electrolyte	LWIR/MWIR emissivity regulator	[141,142]
		Cu films	Dynamic RC controller	[10a]
		ITO electrode+AZO NPs+Electrolyte	Transparent RC controller/smart window	[10b]
	Conjugated conductive polymers	PEDOT:Tos (oxidized)-Cellulose electrolyte-PEDOT:Tos (reduced) multilayer	Dynamic RC controller	[144]
		PANI foil-Au foil-Ce ⁴⁺ electrolyte multilayer	Dynamic RC controller	[145]
		PANI foil-Au foil-Gel electrolyte multilayer	Personal heat management device	[146]
	Valance/State change metal oxides	LTO NPs-electrolyte-Al counter electrode	Solar absorption/RC dual-regulator	[148]
		LTO nanocolumn on FTO substrate	Solar transmittance/RC dual-regulation smart window	[149]
		Amorphous WO ₃ -Ta ₂ O ₅ -Crystalline WO ₃ multilayer	Solar absorption/RC dual-regulator	[150]
		TiO ₂ -SiO ₂ alternate multilayer on acrylic stretchable film	Dynamic emissivity regulator, IR camouflage	[151]
Mechanical-responsive	Wrinkle and crack-based	Cu overcoating on SEBS stretchable film	Dynamic emissivity regulator, IR camouflage	[152]
		Ti/Au/reflective layer on SEBS stretchable film	Dynamic emissivity regulator, Smart window, IR camouflage	[153]
		PSDOT:PSS low-E layer on PDMS	Solar transmittance/RC dual-regulation smart window	[154]
	Kirigami-based	PDMS kirigami pattern-AgNWs-PDMS laminated PNIPAm hydrogel	Solar transmittance/RC dual-regulation smart window	[158]
		ITO coating on PET/PDMS kirigami pattern	Angle-dependent thermal reflector	[159]
Humidity-responsive	Humidity-responsive yarn	carbon nanotubes coated triacetate-cellulose bimorph fibers	Humidity and RC management textile	[160]
	Porous polymers	Porous PE+Ethanol	Solar transmittance/IR transmittance dual-regulation smart window	[161]
Mechanical actuation		High-E side: PDMS coated Ag film Low-E side: copper/zinc coating	Dual-mode electrostatically controlled thermal contact conductance regulator	[12]
		High-E block: P(VdF-HFP)/Low-E block: ITO+Black paint (Design 1)	Photothermal/RC dual-regulation wall (Design 1)	[162]
		High-E block: PVC/Low-E block: ITO+PET+VO ₂ (Design 2)	Solar transmittance/RC dual-regulation smart window (Design 2)	
		High-E side: PE laminated HPC hydrogel Low-E side: ITO	Solar transmittance/RC dual-regulation smart window	[163]
		High-E side: Porous PVDF Low-E side: MXene (Ti ₃ C ₂ T _x)	Photothermal/RC dual-functional heating/cooling source	[164]

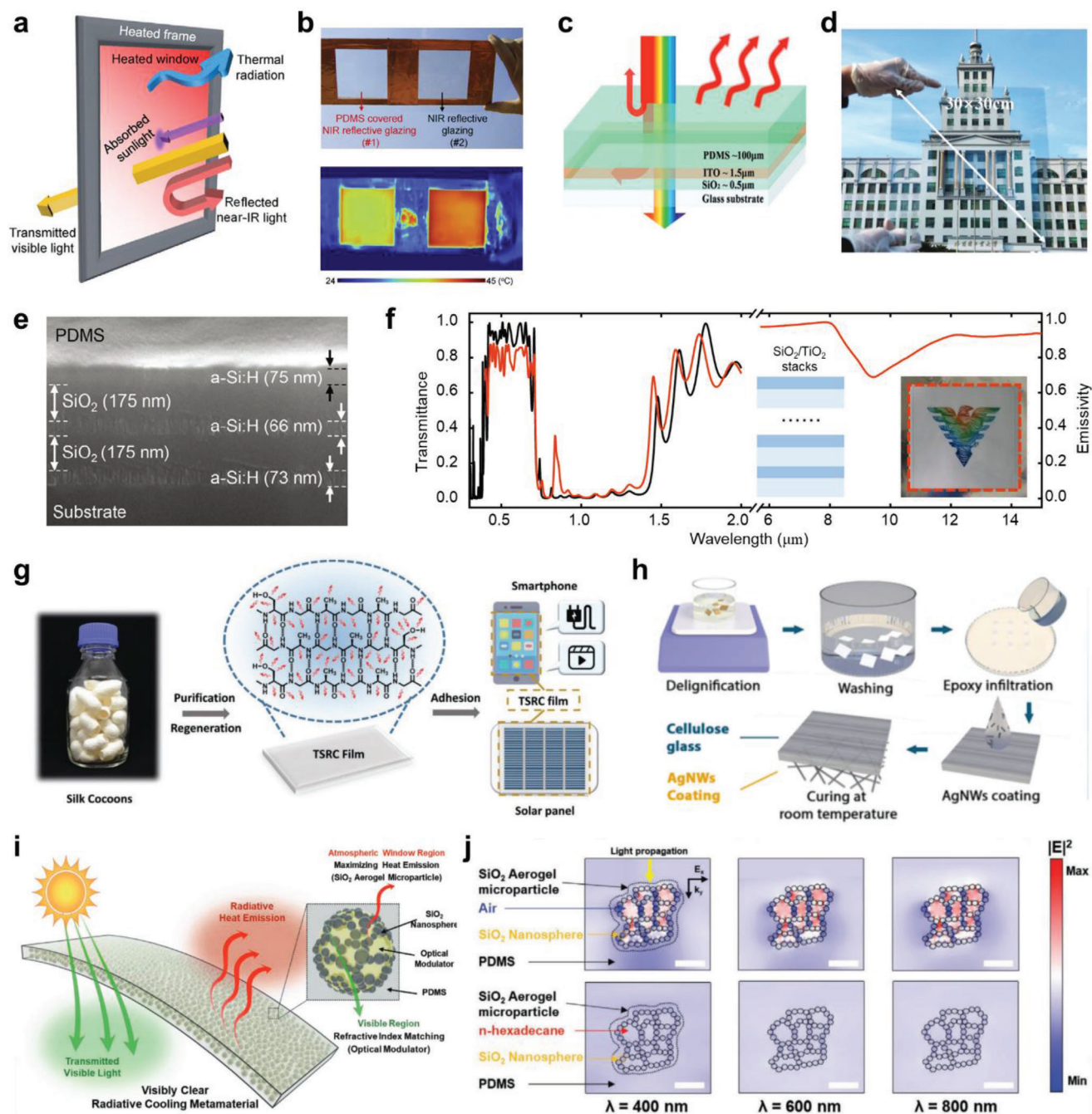


Figure 13. Visibly transparent RC materials. a) Energy diagram of window cooling, emphasizing NIR light reflection for minimizing solar irradiation and high thermal emittance to maximize heat loss under sunlight. b) Optical (top) and IR images (bottom) of NIR reflective glazing with PDMS (sample 1) and without (sample 2), highlighting identical appearance and significant temperature difference. Reproduced with permission.^[24] Copyright 2020, Elsevier. c) Schematic structure of the Bio-RC glass composed of only three films. d) Photograph of a building seen through Bio-RC glass. Reproduced with permission.^[168a] Copyright 2022, John Wiley and Sons. e) SEM image of the fabricated transparent radiative cooler. Reproduced with permission.^[56] Copyright 2021, John Wiley and Sons. f) Transmittance and emissivity spectra of the multilayer coating (SiO₂/TiO₂ stacks film) (red) and the simulation result (black). The inset shows the high transparency of the multilayer coating. Reproduced with permission.^[168c] Copyright 2022, Springer Nature. g) Schematic of the transformation of silk cocoons to a TSRC film and its application to the RC of electronic devices such as smartphones and solar panels. Reproduced with permission.^[166] Copyright 2023, John Wiley and Sons. h) Schematic manufacturing procedures. Reproduced with permission.^[169] Copyright 2021, American Chemical Society. i) Schematic illustration and working principle of the optical modulator infiltrated into SiO₂ aerogel microparticles. j) Simulated electric fields of simplified FDTD models without (top) and with n-hexadecane infiltration (bottom) at incident wavelengths of 400, 600, and 800 nm. Reproduced with permission.^[167] Copyright 2021, John Wiley and Sons.

film, derived from natural silkworm cocoons (Figure 13g).^[166] These TSRC films maintain transparency across the entire solar spectrum, giving a high T_{vis} of 91.0% and a notable ϵ_{LWIR} of 0.88. Consequently, the films effectively preserve the function and appearance of associated devices due to their high visible transparency. They induce a temperature drop of 5.1 °C for a smartphone during multitasking and charging, and 14 °C for a silicon solar panel, accompanied by an $\approx 7\%$ improvement in photoelectronic conversion efficiency. Thus, these films are expected to have promising applications in outdoor applications for portable and wearable electronic devices. Similarly, Long et al. reported a bamboo-derived composite with high optical transparency ($T_{\text{lum}} = 80\%$, $T_{\text{sol}} = 74\%$) and a high ϵ_{LWIR} of 0.95.^[169] The cellulose used in this composite is derived from rapidly growing bamboo, a member of the grass family, making it a more environmentally friendly alternative to wood. The fabrication involved dispersing AgNWs uniformly in ethanol and then spin coating them on one side of the cellulose, providing a low ϵ_{LWIR} (Figure 13h). Consequently, these cost-effective and environmentally friendly composites have great potential to significantly reduce cooling energy consumption.

Meanwhile, Kim et al. reported an innovative RC metamaterial characterized by both visual clarity and flexibility. This technology employs a newly designed optical modulator (n-hexadecane) infused into silica aerogel microparticles randomly distributed within a silicone elastomer (Figure 13i).^[167] The incorporation of a substantial volume (40%) of SiO₂ aerogel microparticles, fully saturated with n-hexadecane, results in an impressive ϵ_{LWIR} of 0.98. Simultaneously, the metamaterial effectively suppresses visible light scattering, achieving a high T_{vis} of $>91\%$. The underlying mechanism is revealed using the finite-difference time-domain (FDTD) simulation. When subjected to a plane wave with wavelengths of 400, 600, and 800 nm, the n-hexadecane within the SiO₂ aerogel microparticles can effectively transmit the incident visible light with little scattering due to the similar refractive indices of PDMS, SiO₂, and n-hexadecane (Figure 13j). In contrast, the introduction of air within the aerogel microparticles induces notable scattering of visible light. Due to its exceptional optical properties, when applied to the surface of Silicon solar cells, the metamaterial demonstrated temperature reductions of 9 and 8 °C for indoor and outdoor conditions, respectively.

6.2. Colored Radiative Cooling Materials

The color of a PDRC material when exposed to sunlight is typically determined by its absorption spectrum across solar wavelengths.^[170] However, the challenge arises as sunlight absorption tends to heat the PDRC material, making it difficult to maintain a specific color for aesthetic purposes while preserving RC ability.^[171] In this section, we will review specific strategies that have been developed to achieve PDRC with colorful visual appearances.

6.2.1. Photonic Colored Radiative Cooling Materials

Fan et al. pioneered a novel approach to lower the temperature of a structure while preserving its color under sunlight.^[172]

The original structure comprised a periodic array of silicon nanowires on an Al substrate, featuring square cross-section nanowires (100 nm side length) with a periodicity of 500 nm (Figure 14a). The original structure exhibited a near-zero emissivity, while there is substantial solar absorption present at optical wavelengths, corresponding to a light pink color. They designed a modified structure to achieve color preserving PDRC by adding a periodic array of quartz bars on top of the original structure. Each quartz bar had a square cross-section of 7 μm side length and a periodicity of 10 μm , and the quartz bars were orthogonal to the silicon nanowires. With the addition of the quartz array, the modified structure increases the ϵ_{LWIR} while maintaining its original color compared to the original structure (Figure 14b).

Subsequently, researchers employed meticulously designed multilayer structures on the top surface of PDRC material to manipulate its color. Li et al. proposed a colored radiative cooler assisted by optical Tamm resonance, demonstrating high-performance cooling effects and high-purity subtractive primary colors simultaneously.^[173] The colored radiative cooler consisted of an emitter layer composed of SiO₂, SiN, and SiO₂ from top to bottom, along with a color-controlling Tamm structure comprising two pairs of MgF₂/SiC (Figure 14c). By controlling the thicknesses of distributed Bragg reflector and Ag layers, the displayed colors of the radiative cooler could be easily regulated, and ϵ_{LWIR} is significantly increased, enabling effective RC. Meanwhile, Song et al. introduced a colored passive radiative cooler (CPRC) based on a thin-film optical resonator embedded in an efficient thermal emission structure.^[174] The CPRC comprised a selective emitter with a bilayer of SiO₂ (650 nm) and Si₃N₄ (910 nm), with thicknesses optimized through numerical analysis (Figure 14d). A metal reflector with an Ag film (100 nm) was deposited on a silicon substrate. Additional photonic nanostructures below the selective emitter generated vivid colors at specific desired areas, including a thin-film resonator composed of a metal-insulator-metal (MIM) structure (Figure 14d). The MIM structure determined each color through interference in the 1D stacked layers, precisely controlled by tuning the thickness of the insulator layer in the MIM. By following this design rule, CPRCs exhibited subtractive primary colors (cyan, magenta, and yellow) while maintaining PDRC ability.

One of the challenges of CPRC materials is the inevitable absorption of certain wavelengths in the visible spectrum, which reduces the cooling performance. Non-absorptive CPRC is an alternative strategy that relies on diffuse scattering instead of absorption to create color appearance, while maintaining high R_{sol} and cooling power.^[175] For instance, Wang et al. demonstrated a cooling structure that consists of three parts: a TiO₂/SiO₂ multilayer structure, a frosted glass disordered layer, and a reflective silver mirror.^[175a] The TiO₂/SiO₂ multilayer structure can effectively reflect the undesired light but has a high transmittance for the desired light. For example, a blue sample can strongly reflect yellow light (0.5–0.7 μm), and highly transmit blue light (0.4–0.5 μm). The frosted glass layer and silver mirror diffusively reflect the transmitted light in all directions, and then the light is further transmitted through the multilayer structure, creating a high saturation (100%) color appearance.

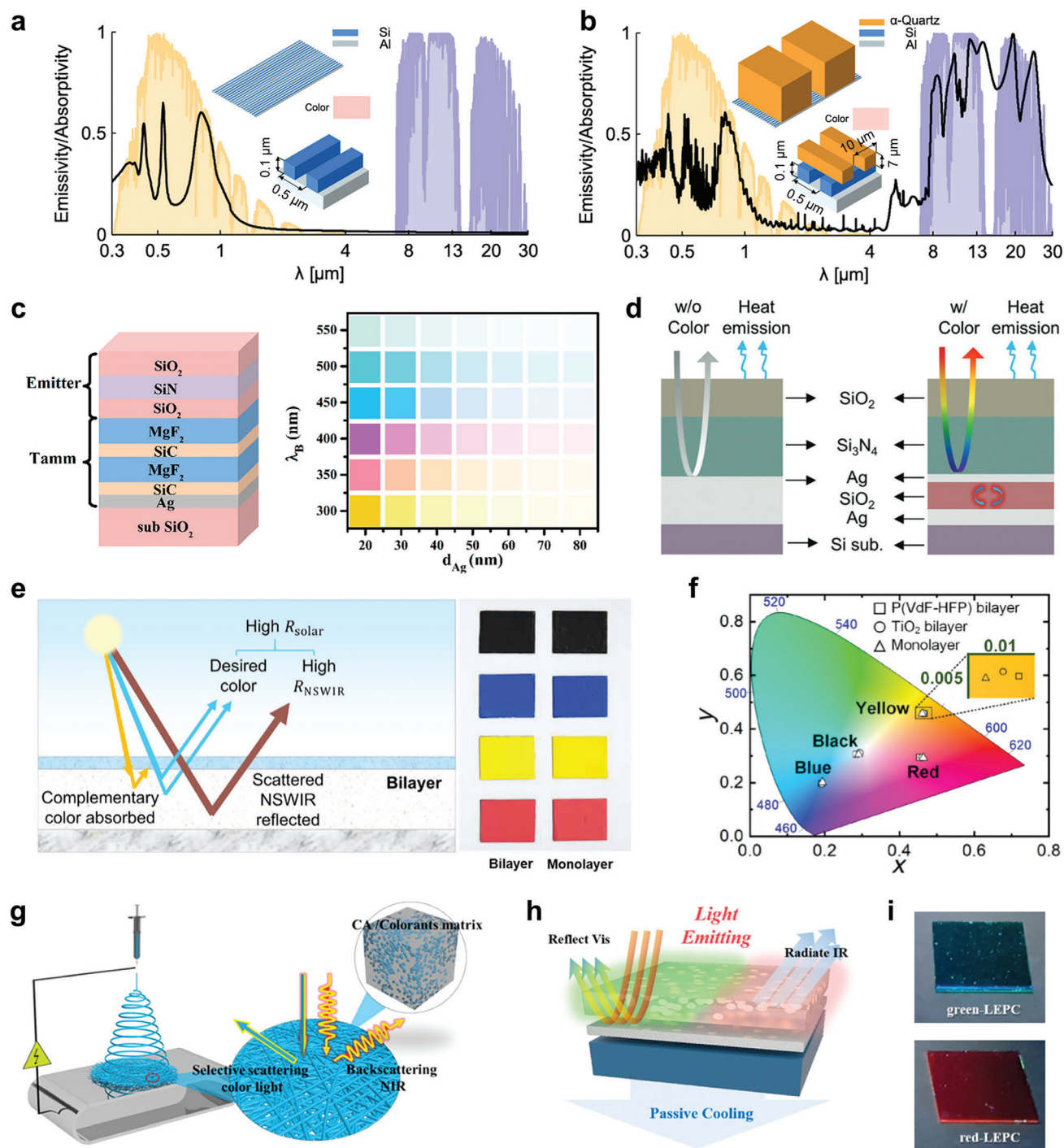


Figure 14. Colored RC materials. a) Emissivity/absorptivity spectrum of the original structure. The inset shows the original structure with a silicon nanowire array on top of an Al substrate. Color for sunlight reflection at normal incidence is displayed. b) Emissivity/absorptivity spectrum of the modified structure. Inset: Modified structure with a quartz bar array on top of the original structure of silicon nanowires. Reproduced with permission.^[172] Copyright 2013, American Institute of Physics. c) Structure diagram of the designed colored radiative cooler (left) and displayed colors achieved by changing the thicknesses of silver and DBR (right). Reproduced with permission.^[173] Copyright 2019, American Chemical Society. d) Cross-sectional schematic views of a conventional daytime radiative cooler (left) and CPRC (right). Reproduced with permission.^[174] Copyright 2018, John Wiley and Sons. e) Interaction of sunlight with a bilayer coating with higher NIR reflectance (left) and the real photographs of four differently colored pairs of P(VdF-HFP) bilayer (right). f) Chromaticity of black, blue, red, and yellow cooling paints shown in the CIE 1931 color space; inset shows an enlarged view of the chromaticity values of the yellow coatings. Reproduced with permission.^[176] Copyright 2020, AAAS. g) Schematic illustration of the preparation of the chromatic hierarchically structured CA nanofiber film. Inset: Photograph of the nanofiber-based film with different colors. Reproduced with permission.^[177] Copyright 2023, Royal Society of Chemistry. h) Schematic of a light-emitting and passive RC material. i) Photos of various RC devices under a UV lamp. Reproduced with permission.^[178] Copyright 2020, American Chemical Society.

6.2.2. Random Colored Radiative Cooling Materials

In contrast to photonic structures that require precision processing techniques, colored RC materials using random structures are generally considered more scalable for practical applications. Yang et al. introduced a paintable bilayer coating that seamlessly integrates color and efficient RC.^[176] The bilayer consists of a thin, visible absorptive layer overlaying a non-absorptive, solar-scattering underlayer. While the top layer showcases specific colors by absorbing distinctly visible wavelengths, the underlayer optimizes near-to-short wavelength infrared light reflection, minimizing solar heating (Figure 14e). Consequently, the bilayers exhibit near-identical colors and a visible appearance to commercial monolayer paint. As shown in Figure 14f, similar CIE x and y chromaticity values and small lightness differences are observed, confirming a close match in the visible spectra for the mono- and bilayers of each color. As a result, this design allows the bilayer paint to achieve both color and efficient RC in a simple, inexpensive, and scalable manner. Meanwhile, Qin et al. used nanofibers to achieve selective spectral absorption in a daytime radiative cooler through a dope-dyeing electrospinning technique.^[177] This approach selectively absorbs desired colors in the visible spectrum, while the nanofiber structure minimizes solar heating by providing robust visible and NIR light scattering (Figure 14g). The resulting colored PDRC materials demonstrate remarkable features, including ultra-high R_{NIR} (99%), high ϵ_{LWIR} (0.95), and vibrant colors. These optical properties lead to a notable reduction in maximum ambient temperature (3.2 °C) and a significant cooling power ($\approx 40 \text{ W m}^{-2}$) under a solar intensity of 700 W m^{-2} .

Besides, the integration of photoluminescent perovskite nanocrystals (NCs) with PDRC materials has led to the development of light-emitting and passive RC (LEPC) materials. These materials do not exhibit any color under visible spectra but emit bright, colorful light when exposed to UV or blue illumination (Figure 14h).^[178] The fabricated green and red LEPC devices are shown in Figure 14i. Under indoor light, all samples displayed the same silvery color, attributed to the high transparency of the radiative PDMS and silver reflection layers. However, under UV light illumination, the green and red LEPC devices showed strong green- and red-light-emitting properties due to the high photoluminescence quantum yield of perovskite NCs, with the generated light reflected outward by the silver layer. Similarly, Lee et al. fabricated silica-embedded perovskite nanocrystals to achieve white, green, and red emitters with excellent sub-ambient cooling ability.^[179] Furthermore, Cu-based quantum dots were deposited on white PDRC films to produce yellow, red, and brown colors while preserving high PDRC capability.^[180]

6.3. Self-Cleaning Radiative Cooling Materials

When RC materials are employed in outdoor environments, their surfaces inevitably contact with rain and dust, which can lead to increased solar absorption and compromised cooling efficiency. Thus, the design of self-cleaning surfaces becomes important for ensuring prolonged and effective utilization of RC materials.

Normally, there are two ways to introduce self-cleaning functionality into RC materials. The first approach is through con-

structing surface features and conducting surface modification. When water falls on a superhydrophobic surface (contact angle $>150^\circ$) with a slope, the water droplets adhere to solid contaminants on the surface, effectively cleaning it as they roll away. The structural design of superhydrophobic surfaces needs to meet the requirements of both RC functionality and the structural demands of micro-nanostructures to achieve superhydrophobic. Cai et al. reported a hierarchically patterned nanoporous composite (HPNC) RC material that incorporates solar reflection-enhancing ZnO nanoparticle fillers into a nanoporous PE (nanoPE) matrix.^[181] As depicted in Figure 15a, at the microscale, the film's surface is patterned and forms an array of microcolumns. The introduction of air into this array mitigates solid-liquid adhesion, endowing the film with superhydrophobic properties, giving a water contact angle of 158° (Figure 15b). Tso et al. converted the RC ceramic from superhydrophilic to superhydrophobic through impregnation with organosilicon compounds.^[5a] By treating with fluorosilane, a water contact angle of $\approx 150^\circ$ was achieved. The interconnected porous structure of the ceramic facilitated the penetration of the fluorosilane solution throughout the material, leading to water resistance on both the surface and the interior. On the other hand, Zheng et al. integrated a suspension of ethanol PTFE particles into the micropores of cellulose fibers, resulting in a superhydrophobic PTFE RC coating.^[74] This coating exhibits stain resistance and efficient water-based cleansing compared to conventional RC cellulose-based materials. Furthermore, due to the high R_{sol} of the PTFE particles, the composite achieves a RC power of 104 W m^{-2} under a solar radiation intensity of 671 W m^{-2} . Moreover, Hu et al. improved the soiling resistance of the RC glass by implementing a protective layer.^[5b] They added a low-melting point glass particles layer on top of the RC glass to produce a thin and dense surface. The dense structure of the transparent protective layer not only facilitates easy cleaning of liquid pollutants but also maintains a promising R_{sol} of 95%.

Another strategy to render RC materials with self-cleaning attributes by the incorporation of photocatalytic particles for pollutant degradation. For instance, when exposed to sunlight, TiO_2 undergoes a reaction with water and generates hydroxyl radicals that effectively decompose organic molecules and microorganisms adhered to the material's surface. Qiu et al. proposed a $\text{SiO}_2/\text{TiO}_2/\text{ZnO}$ stacked coating for energy-efficient windows with excellent RC performance.^[182] The self-cleaning performance of $\text{SiO}_2/\text{TiO}_2/\text{ZnO}$ was assessed through the degradation of methylene blue. The absorption spectroscopy test revealed a gradual reduction in the intensity of the absorption peak of methylene blue at 664 nm with increased UV irradiation time, providing evidence of the coating's self-cleaning capability.

6.4. Durable Radiative Cooling Materials

RC materials are subjected to prolonged exposure to sunlight, precipitation, surface dust accumulation, and the impact of external forces. These environmental factors can severely compromise their optical properties, diminish cooling efficiency, and curtail the materials' operational lifespan.^[183] Therefore, it is crucial to

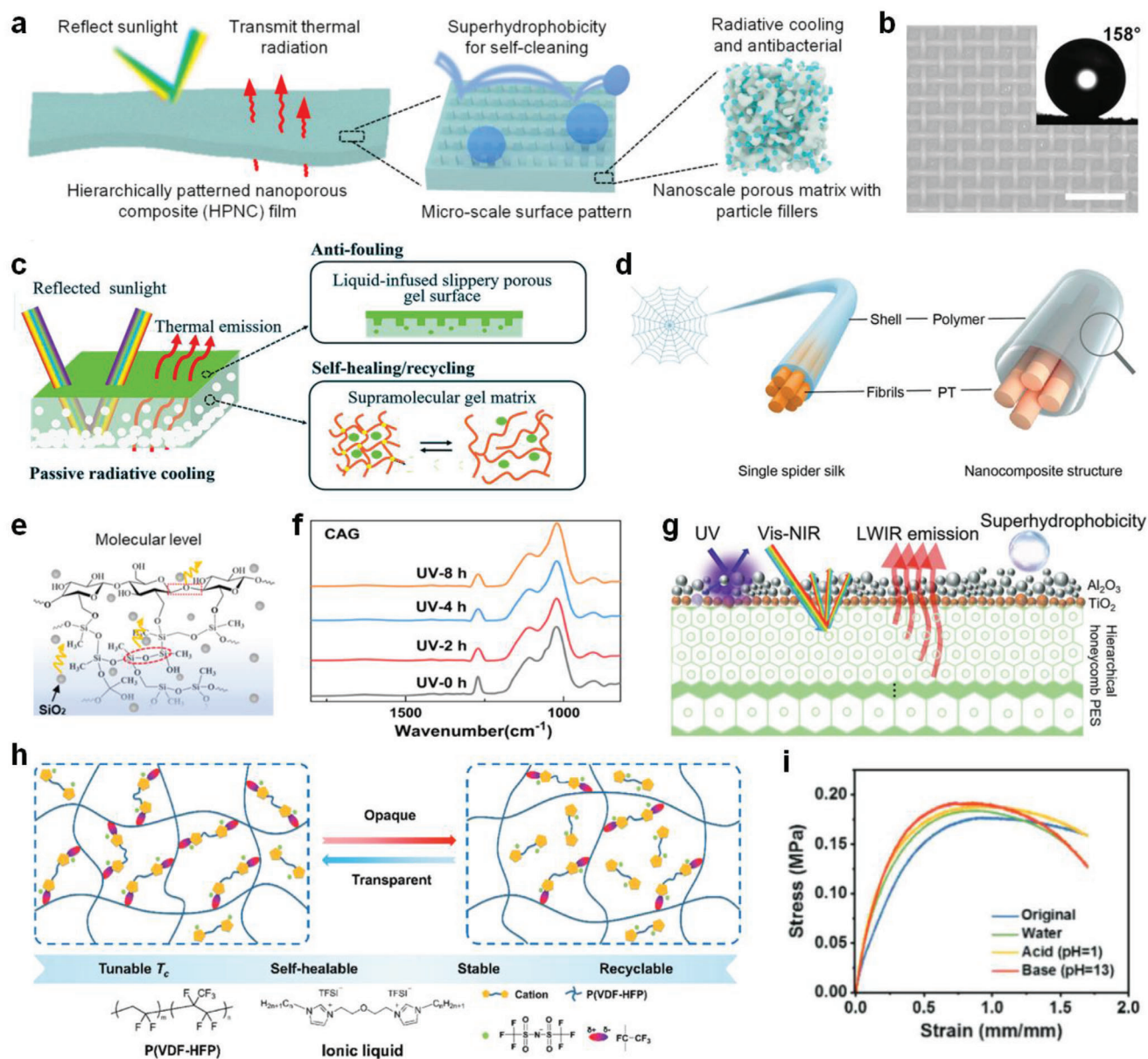


Figure 15. Self-cleaning and durable RC materials. a) The HPNC design for PDRC, with a microscale pillar array surface pattern that induces superhydrophobicity for a self-cleaning effect. b) Top-view optical image of HPNC. Inset: Image of the water contact angle. Reproduced with permission.^[181] Copyright 2023, American Chemical Society. c) Schematic illustration of the self-healing functional features of the passive RC polymer coatings (PRPCs). Reproduced with permission.^[184] Copyright 2022, American Chemical Society. d) The design of spider-silk-inspired nanocomposite PT@PEO with enhanced mechanical properties. Reproduced with permission.^[119] Copyright 2022, John Wiley and Sons. e) Schematic illustration of the chemical bonds of CAG. f) FTIR spectra of CAG before and after 2, 4, and 8 h of UV irradiation. No obvious C=O bonds were observed after 8 h UV radiation. Reproduced with permission.^[186] Copyright 2023, Elsevier. g) Cross-sectional schematic of the trilayer PES-TiO₂-Al₂O₃ cooler. Reproduced with permission.^[187] Copyright 2023, John Wiley and Sons. h) Chemical structure schematic of TPC, with a reversible phase transition behavior. i) The mechanical properties of the sample after immersion tests. Reproduced with permission.^[189] Copyright 2023, John Wiley and Sons.

consider their long-term resilience to weather conditions and the mechanical durability of RC materials during the design process. In this section, we present a review of recent advancements in the realm of durable RC materials. These advances include innovations in mechanical robustness, anti-corrosion properties, and resistance to UV exposure.

6.4.1. Mechanical Integrity and Anti-Abrasive Radiative Cooling Materials

RC materials are susceptible to mechanical failure when exposed to the rigors of outdoor environments. Mechanical abrasion or impact events may compromise the material's surface and

internal structure, or even lead to rupture, thereby diminishing R_{sol} and ϵ_{LWIR} , substantially impairing cooling performance.

To address this concern, Cui et al. developed a supramolecular polymer composite matrix utilizing a copolymer of PDMS and urea (uPDMS) in combination with Al_2O_3 particles, as illustrated in Figure 15c.^[184] The polymer matrix is engineered to form crosslinks through reversible hydrogen bonding between the urea units, imparting self-healing property to the RC material. Scratches on the sample surface vanish within 1–2 min at 70 °C. In addition, when the film is bisected, the two halves reunite into a complete bulk material after 4 h of annealing at 70 °C. Subsequent tensile tests confirm that the healed material retains 75% of its original toughness. Moreover, due to the near-unity emissivity of PDMS and promising reflection properties of Al_2O_3 , the material exhibits a ϵ_{LWIR} of 0.93 and a R_{sol} over 92%.

Furthermore, researchers have enhanced the mechanical properties of RC materials by formulating composites. Inspired by the layered and ordered structure of spider silk, Zhu et al. pioneered a composite polymer material based on PEO and PT nanofibers (Figure 15d).^[119] The incorporation of PT nanofibers, enveloped within the PEO matrix, establishes a nanocomposite structure PT@PEO. This structural enhancement results in a sevenfold increase in Young's modulus, as the nanofibers predominantly bear external forces. Simultaneously, the inherent RC properties of PEO remain intact with ϵ_{LWIR} of 0.95. This initiative further enhances PEO polymer's potential as a mechanically robust RC material for outdoor applications.

6.4.2. UV Resistance Radiative Cooling Materials

RC materials, especially polymers, are susceptible to damage from prolonged UV exposure due to their intrinsic vulnerability to photodegradation.^[185] Extended UV exposure not only alters the material's color and spectral characteristics but also compromises its mechanical properties, ultimately leading to inefficient cooling.

The deleterious effects of UV exposure on cellulose, specifically the breakdown of C—O—C bonds leading to C=O bond formation and oxidation, present a challenge to the UV stability of cellulose-based materials. Zhang et al. have developed a sustainable cellulose nanocrystalline aerogel grating (CAG) RC material that exhibits exceptional UV resistance.^[186] Benefits from the high bonding energy of Si—O chemical bonds (Figure 15e), the grating gives a high R_{UV} of 98%, and testing revealed no observable C=O bonding formation after 8 h of UV accelerated aging experiments (Figure 15f). Furthermore, CAG maintained its original cooling efficacy ($\epsilon_{\text{LWIR}} = 0.94$) after 6 months of outdoor exposure, affirming its promising UV resistance.

Huang et al. devised a multilayer RC material by coating a base polyethersulfone (PES) with Al_2O_3 NPs and TiO_2 NPs.^[187] As demonstrated in Figure 15g, Al_2O_3 acts as a reflective layer to reflect the most of incident UV light, while TiO_2 absorbs the rest UV radiation transmitted through Al_2O_3 . The designed PES- TiO_2 - Al_2O_3 cooler demonstrates sustained UV resistance, retaining its original color and spectral characteristics after 1000 h of UV irradiation. Deng et al. proposed an approach to enhance the

intrinsic UV resistance of RC materials by avoiding the use of polymeric binders.^[185] They assembled TiO_2 NPs by evaporation-driven assembly and the resulting material forms a hierarchical porous morphology, exhibiting a ϵ_{LWIR} of 0.97. Notably, the absence of a polymer binder in the coating not only completely mitigated UV degradation issues but also significantly enhanced light scattering ability due to the greater refractive index difference between TiO_2 and air than TiO_2 and polymer. Consequently, the solar reflectance of the resulting material remains at 99.5% of its original value after one year of outdoor exposure.

6.4.3. Anti-Corrosive Radiative Cooling Materials

RC materials often employ metals such as Al or steel as reflective substrates and polymeric-based overlayers, rendering them susceptible to corrosion upon contact with rainwater and pollutants. This susceptibility poses a challenge to their practical application. Consequently, the development of corrosion-resistant RC coatings has become an imperative research pursuit.

Yang et al. utilized a straightforward liquid scratch coating technique to form an RC coating based on tricyclodecane dimethanol diacrylate (DCPDA) on an Al substrate.^[188] This coating, characterized by a T_{vis} of 90% and a ϵ_{LWIR} of 0.95, underwent rigorous testing involving immersion in both highly acidic (pH = 2) and highly alkaline (pH = 13) environments. Remarkably, the coating retained its integrity following exposure to these extreme conditions, while pristine control samples suffered severe corrosion.

Furthermore, Wang et al. designed an anti-corrosive thermochromic polymer coating (TPC) by uniformly dispersing hydrophobic ionic liquids (IL) within a P(VdF-HFP) matrix.^[189] As depicted in Figure 15h, the TPC is rich in C—F bonds inherently resisting interaction with water molecules. This property endows the TPC with high stability in aqueous solutions. After 1 month of immersion in acidic (pH = 1) and strongly alkaline (pH = 13) solutions, TPC exhibited no significant degradation of mechanical properties, emphasizing its remarkable corrosion resistance (Figure 15i).

6.5. Thermal Insulated Radiative Cooling Materials

To optimize the net cooling capacity of RC materials, it is essential to minimize heat input from solar absorption and non-radiative heat flux, while concurrently enhancing heat dissipation through thermal radiation (Figure 16a).^[190] Traditional approaches to designing RC materials primarily focused on solar radiation suppression and infrared radiation enhancement while overlooking the impact of non-radiative heat fluxes originating from convection and heat transfer processes. During hot summer conditions, large temperature differences between indoor and outdoor environments, along with accompanied heat conduction, can significantly impair cooling performance and elevate energy consumption. Consequently, incorporating thermal insulation properties into RC materials offers a promising solution to address these challenges and improve cooling efficiency.

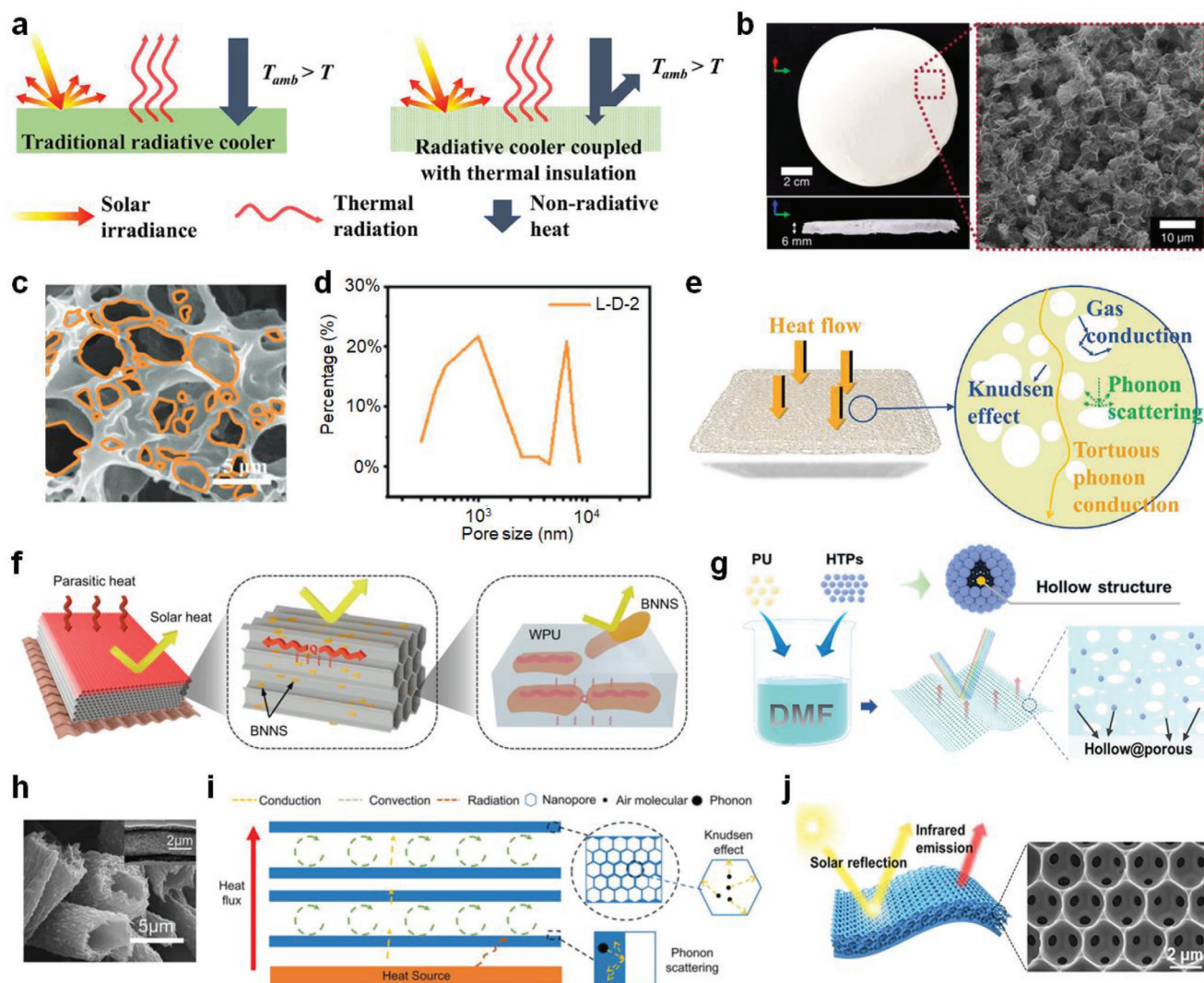


Figure 16. Thermal insulated RC materials. a) Traditional RC approach and RC coupled with thermal insulation. Reproduced with permission.^[190b] Copyright 2022, Elsevier. b) Optical image and SEM image of a 10-cm-diameter and 6-mm-thick PEA sample showing its internal porous structure. Reproduced with permission.^[191] Copyright 2019, AAAS. c) SEM images of L-D-2. Reproduced with permission.^[192] Copyright 2022, John Wiley and Sons. d) Pore size distribution of L-D-2. Reproduced with permission.^[192] Copyright 2022, Elsevier. e) Schematic illustration of thermal conductivity reduction by hierarchically porous structures of SHB-CAC. Reproduced with permission.^[190b] Copyright 2022, The Authors. f) Schematics of the multiscale structures responsible for ultralow k and high solar reflectance of the anisotropic aerogel panel. Reproduced with permission.^[15] Copyright 2022, The Authors. g) Schematic illustration of the hollow@porous structure contributed to ultralow thermal conductivity. Reproduced with permission.^[194] Copyright 2023, American Chemical Society. h) Microscopic architecture of the TIC. i) Schematic illustration of the heat transfer process of the TIC. Reproduced with permission.^[58] Copyright 2021, American Chemical Society. j) Schematic and SEM images of the IO-PMMA film, exhibiting 3D ordered micropores and highly interconnected nanopores. Reproduced with permission.^[195] Copyright 2023, American Chemical Society.

Characterized by its nanoporous network structure and dispersion of gaseous media within its pores, aerogel exhibits remarkable thermal insulation properties. Wang et al. introduced an IR-transparent thermal insulation overcoating based on PE aerogel (PEA) for RC materials.^[191] This overcoating features a highly porous structure and low density (Figure 16b). PEA achieves a low thermal conductivity of $28 \pm 5 \text{ mW m}^{-1} \text{ K}^{-1}$, on par with that of air ($k_{\text{air}} = 26 \text{ mW m}^{-1} \text{ K}^{-1}$), and a cooling power of 96 W m^{-2} . In a similar vein, Liu et al. developed a biomimetic hierarchically structured composite RC aerogel by combining PLA and cubic composite crystals, resulting in a finely

ordered hierarchical network with micro and nano-sized pores (Figure 16c,d).^[192] With nanopores of 500 nm and a porosity of up to 97.8%, this RC aerogel demonstrates a temperature reduction of 3.5°C during the day and 5.8°C at night. Qiu et al. isolated a cellulosic aerogel from waste paper, which demonstrated high R_{sol} (93%) and ϵ_{LWIR} (0.91).^[190b] Similar to other aerogels, solid-gas interfaces formed by the porous structure in the superhydrophobic cellulose aerogel cooler (SHB-CAC) facilitate phonon scattering and diffusive heat transport (Figure 16e). As a result, the thermal conductivity of SHB-CAC is ultimately reduced and becomes comparable to still air ($28 \text{ mW m}^{-1} \text{ K}^{-1}$).

Kim et al. harnessed boron nitride nanosheets (BNNS) within aligned waterborne polyurethane (WPU) pore walls through additive freeze-casting (Figure 16f), resulting in a thermally insulating RC cooler (TIC) with ultra-low thermal conductivity in the thickness direction.^[15] TIC achieves directional super-thermal insulation with a remarkable out-of-plane thermal conductivity value of $16.9 \text{ mW m}^{-1} \text{ K}^{-1}$ and provides an up to 7°C lower interior temperature than SiO_2 aerogel. Besides, Roehling et al. developed a simply fabricated aerogel-like foam structure using naturally abundant salts, namely sodium chloride (NaCl) and potassium chloride (KCl).^[193] The resulting porous structure of the foam exhibited low thermal conductivity ($37 \text{ mW m}^{-1} \text{ K}^{-1}$), minimizing parasitic heat transfer. Meanwhile, the salts provided high IR transmittance, allowing heat to escape. These foams were packaged into support frames and encapsulated with PE sheets to make foam panels, which demonstrated a maximum temperature reduction of 10°C within a 24-h period.

In addition to aerogel-based materials, researchers have explored the incorporation of hollow structures within matrices to increase air content and reduce thermal conductivity. Wu et al. proposed a novel hollow@porous RC film by incorporating hollow titanium dioxide microparticles (HTP) into polyurethane (Figure 16g).^[194] The hollow structure of HTP not only induces reflection and scattering phenomena within their internal structure but also creates more cavity volume. As a result, the fabricated hollow@porous RC film demonstrates exceptional optical and thermal properties, achieves a high R_{sol} of 93.7% and a strong ϵ_{LWIR} of 0.89, along with an ultralow thermal conductivity ($17.56 \text{ mW m}^{-1} \text{ K}^{-1}$). Additionally, Wang et al. designed a TIC characterized by layered hollow microfibers (Figure 16h).^[58] As shown in Figure 16i, the hollow structure disrupts solid heat conduction pathways in TIC. Moreover, phonon scattering at the solid–air interface and the presence of nanopores smaller than the mean free range of air further reduce its thermal conductivity. The TIC demonstrates a low thermal conductivity ($14 \text{ mW m}^{-1} \text{ K}^{-1}$), superior optical properties ($R_{\text{sol}} = 94\%$), and effective heat dissipation ($\epsilon_{\text{LWIR}} = 0.94$), making it highly effective in minimizing external heat input. Wu et al. developed a porous inverse opal poly(methyl methacrylate) (IO-PMMA) RC film comprising ordered micropores and highly interconnected nanopores (Figure 16j) via the sacrificial template method.^[195] Attribute to the high porosity of 74%, IO-PMMA demonstrated a thermal conductivity comparable to commercial silica aerogels ($44 \text{ mW m}^{-1} \text{ K}^{-1}$ for IO-PMMA and $20 \text{ mW m}^{-1} \text{ K}^{-1}$ for the aerogel). Simultaneously, it demonstrates high R_{sol} of 94% and ϵ_{LWIR} of 0.95.

In summary, various strategies are designed to provide multifunctionality to RC materials in specific scenarios, as outlined in Table 3. This section emphasizes the impact of environmental factors on outdoor RC materials, highlighting the crucial need for advancements in self-cleaning ability, mechanical robustness, anti-corrosion properties, and UV resistance. To optimize cooling efficiency, the reduction of both solar absorption and non-radiative heat flux, coupled with an increase in thermal radiation is deemed essential. The integration of thermal insulation properties into RC materials is underscored for improving cooling performance, particularly during hot summer conditions characterized by large temperature differences between indoor and outdoor.

7. Conclusions and Perspectives

This review provides a comprehensive summary of the design strategies for various RC materials and a systematic and up-to-date overview of the advancement in RC materials, including static-homogeneous, static-composite, dynamic, and multifunctional materials. A brief summary of these four categories of RC materials is outlined in Table 4. This review will be beneficial for understanding the process-structure-property relationships of various RC materials and provide guidance and insights for their future development.

Whilst RC materials have made considerable achievements so far, challenges and issues for practical applications remain. To foster the sustained development of RC materials, we outlined the major general challenges and future developments of RC materials in Figure 17.

- (1) Develop low-cost and eco-friendly fabrication processes to promote lab-to-fab transfer. Firstly, there is a need to explore simple and cost-effective large-area production techniques, such as printing or spraying, as alternatives to complex and expensive processes like vacuum deposition. Secondly, employing low-temperature or non-thermal processing techniques like solution bar-coating, self-assembly, or template-assisted methods, to overcome the limitations of high-temperature processing for fabricating RC materials with high durability and longevity, such as glass-based or ceramic materials.^[5] In addition, utilizing eco-friendly and biodegradable materials, e.g., cellulose and silk, to alleviate the potential environmental impact of waste plastics. Furthermore, leveraging materials with high cooling performance at an affordable cost or increasing the utilization and recycling rate of existing materials to cut material costs and drive sustainable development.
- (2) Improve the effectiveness of cooling performance. The spectral properties of static radiation cooling materials, such as solar reflectance and infrared emissivity, have currently almost reached their limits. To further enhance the effectiveness of the cooling effect, thermal insulating materials like aerogels are used to reduce the non-radiative heat exchange,^[15] and additionally coupled with evaporative cooling (generally hydrogels or hygroscopic salts) to overcome the challenges posed by the humid atmosphere.^[196] Notably, the mechanical integrity of materials, especially aerogels, still requires optimization for practical use.
- (3) Integrate with the existing functionalities and objects. The application of RC materials involves integration and coordination with other devices or systems, such as buildings, solar cells, etc. Therefore, RC materials need to be carefully designed based on practical factors, ensuring good compatibility and flexibility to adapt to diverse working conditions and multifaceted requirements in various scenarios. For example, seamlessly integrating into a building roof and exterior wall requires a comprehensive assessment of multiple practical factors that are not limited to adhesion, weatherability, anti-glare, color, processability, and cost.
- (4) Tackle the durability issue in outdoor environments. RC materials, when used over extended periods, face exposure to various harsh environmental factors, including UV

Table 3. Summary of multifunctional RC materials discussed above.

Type	Materials	Structure	Characteristic	Ref.
<i>Transparent and colored radiative cooling materials</i>				[24]
Transparent	PDMS, ITO and glass	Multilayer	Visibly transparent and NIR reflective	
	PDMS, ITO, SiO ₂ and glass	Multilayer		[168a]
	PDMS, a-Si:H and SiO ₂ and glass	Multilayer		[168b]
	TiO ₂ and SiO ₂ stacks	Multilayer		[168c]
	Regenerated silk cocoons	Homogeneous film	Vis-NIR transparent	[166]
	Silicone elastomer, silica aerogel microparticles	Metamaterial		[167]
Colored	SiO ₂ bars	Photonic structure	Maintain the pink color using a photonic structure	[172]
	MgF ₂ /SiC pairs	Multilayer	A colored radiative cooler assisted by the optical Tamm structure	[173]
	Ag/SiO ₂ /Ag	Meta-insulator-metal multilayer	Different colors can be easily controlled by embedding a color film composing a metal-insulator-metal structure	[174]
	Colored top layer/P(VdF-HFP) underlayer	Double layer	Top layer showcases specific colors and the underlayer optimizes near-to-short wavelength infrared light reflection	[176]
	CA nanofiber film	Nanofiber structure	The desired color is absorbed by colorant and nanofiber structures strongly scatter visible and NIR light.	[177]
	Light-emitting perovskite nanocrystals	Nanocrystal	Colorful light is emitted under UV or blue illumination	[178]
	Silica-embedded perovskite nanocrystals	Nanocrystal	Colorful light is emitted under UV or blue illumination	[179]
	Cu-based quantum dots	Quantum dots	The Cu-based quantum dots partially absorb light in the visible spectrum, allowing yellow, red, and brown colors	[180]

Type	Materials	Mechanism	Estimated durability time	R_{sol}	ϵ_{LWIR}	ΔT [°C]	Ref.
<i>Durable radiative cooling materials</i>							
Self-cleaning	PE/ZnO	Superhydrophobic surface	>1 month	94.6%	0.86	4.4	[181]
	Paper/PTFE	Superhydrophobic coating	>1 month	93%	0.92	7.1	[74]
	SiO ₂ /TiO ₂ /ZnO	Photocatalytic degradation of pollutants	/	55.4%	/	4.5	[182]
Mechanical durability	PDMS-urea copolymer/Al ₂ O ₃	Self-healing	/	92%	0.93	≈2	[184]
	PEO/K ₂ Ti ₆ O ₁₃	Doped nanofibers for improved mechanical strength	>30 days	94%	0.91	8	[119]
UV resistance	Cellulose nanocrystal-silane/SiO ₂	High UV reflectivity	>6 months	97.4%	0.94	10.5	[186]
	PES/Al ₂ O ₃ /TiO ₂	UV reflective and absorptive layers in series	>280 days	97%	0.92	3	[187]
Anti-corrosion	DCPDA	Anti-corrosion coating	>48 h	/	0.95	18.6	[188]
	P(VdF-HFP)/Ionic liquid	High bond energy of the C—F bonds	>1 month	83.4%	0.94	5.9	[189]

(Continued)

Table 3. (Continued)

Type	Materials	Mechanism	Thermal conductivity [mW m ⁻¹ K ⁻¹]	R _{sol}	ε _{LWIR}	ΔT [°C]	Ref.
<i>Thermal insulated radiative cooling materials</i>							
Thermal insulation	PEA	Highly porous structure	28	92.2%	/	13	[191]
	PLA	Hierarchy micro/nanostructure	37	89%	0.93	3.5	[192]
	Cellulose aerogel	Micro/nano-sized gaseous pores restrict the thermal conduction	28	93%	0.91	8.5	[190b]
	PU/BN	Anisotropic thermal conductivity	16.9	97%	0.88	7	[15]
	NaCl/KCl	Porous structure of foam	37	83%	/	10	[193]
	PU/TiO ₂	Embedding of hollow particles	17.56	93.7%	0.89	17.4	[194]
	PVDF/PVA	Embedding of hierarchically hollow microfibers	14.8	94%	0.94	9	[58]
	PMMA	Ordered micro/nanopores	44	94%	0.95	10.6	[195]

exposure, moisture, wind, raindrops, extreme weather, thermal cycling, and dust. These factors can result in performance degradation or even failure. To enhance weatherability, reduce maintenance costs, and prolong the material lifespan, it is advisable to utilize materials with robust corrosion resistance, such as ceramics. However, the high-temperature sintering involved in the ceramics process is always a hurdle for commercialization. Additionally, enhancing material stability can be achieved through surface treatment techniques (e.g., protective layers or additives), which could offer pollution resistance, oxidation resistance, and corrosion resistance.

(5) Develop dynamic switchable radiative coolers with controllability and adjustability in different seasons and regions. Static RC materials, while enabling stable passive cooling, suffer from the difficulty of adapting to dynamic climatic conditions. To overcome this limitation, a promising solution is to develop dynamic switchable radiative coolers, which can adjust their radiative properties according to different seasons and regions, and thus achieve a more efficient and flexible cooling performance. However, designing and fabricating such dynamic RC coolers is not an easy task, and the specific challenges of dynamic RC materials will be discussed below.

Table 4. The brief summary of static-homogeneous, static-composite, dynamic, and multifunctional RC materials and their challenges.

Type	Advantages	Disadvantages	Specific challenges/strategies	General challenges
Homogeneous	Simple composition	Performance confined by material's intrinsic properties	Explore more material candidates	Develop commercial fabrication processes
	Easy fabrication	Limited material selection	Microstructure manipulation	
	Potentially low cost			Improve cooling efficiency
Composite	Improved optical properties	Usually involves complex composition	Precise structural and composition design	
	Diversified combination	Potential compromised performance	Develop scalable fabrication	Ensure good integration and compatibility
	Customizable design			
Dynamic	Spectral dynamic adjustable	May need external energy input	Independent modulation of different bands	Tackle the durability issue
	Adaptability	Complex	Improve cycle stability	
	Year-round energy saving	Costly	Improve modulation/regulation performance	Develop dynamic radiative coolers
Multifunctional	Expand RC applications	May exist trade-offs or sacrifices between functions	Balance different functionalities	
	Improve the service life and reliability		Integration with other devices where heat management is essential	
	Reduce maintenance costs			

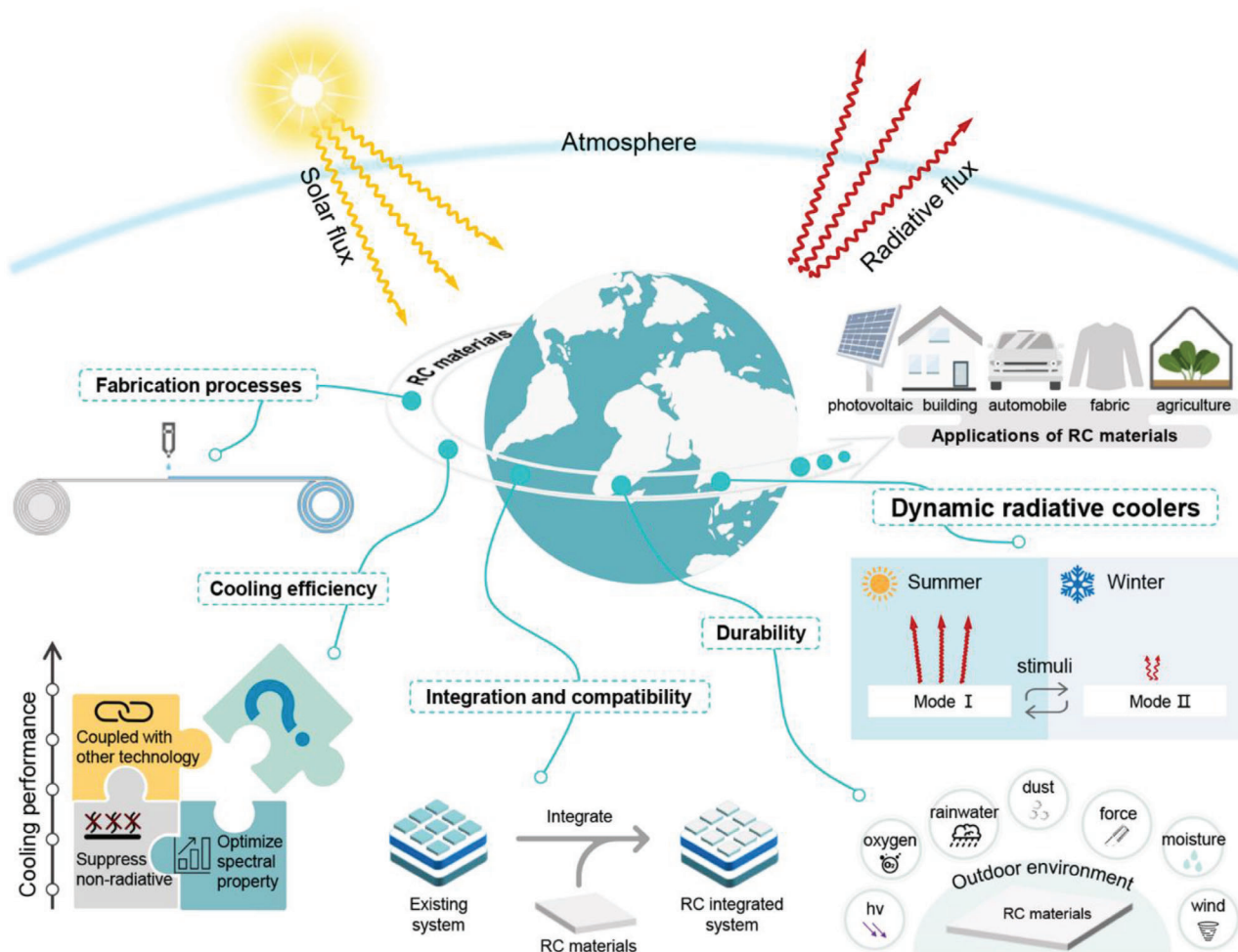


Figure 17. Illustration of prospects for RC materials.

However, in addition to the main general challenges mentioned above, different types of RC materials also have their own specific challenges that need to be addressed.

- **Static-homogeneous RC materials:** Homogeneous RC materials have simple compositions, and there are also works report on the successful preparation of large-area homogeneous RC materials by processes such as electrospinning^[197] and phase-inversion process.^[4c] However, the RC performance of homogeneous RC materials is limited by its intrinsic properties. Several strategies could be implemented to further improve the RC performance of static RC materials such as constructing nanostructure on materials' surface, introducing internal structures, and developing new candidates for RC applications.
- **Static-composite RC materials:** Composite RC materials can overcome the limitations of single component by combining the advantages of different components to achieve improved optical properties. Moreover, these materials can be customized by adjusting the composition or ratio of the components to meet the specific requirements of different applications. The major challenges come from the process of fabrica-

tion and composite design. During the large-area fabrication process, it is crucial to achieve the designed structure. Therefore, the process parameters need to be investigated in detail. On the other hand, different from static-homogeneous RC materials which have only one component, the components in the composite RC material could compromise overall RC performance. Therefore, a deeper understanding of the interaction of multiple components is needed to design suitable composite RC materials.

- **Dynamic RC materials:** Dynamic RC materials can adapt to different environmental conditions and demands, and they are the most ideal candidates for RC materials, except for extremely hot or cold regions. Recent innovations, such as VO₂ resonators^[11a] or electrochromic designs,^[10a] are capable of dynamic RC, but their modulation capability is limited, and large-scale, low-cost fabrication remains challenging. Therefore, one of the future research directions is to optimize the performance of dynamic RC materials, including the modulation range, energy-saving efficiency, durability, etc., and to develop a cost-effective fabrication process for lab-to-fab transfer. In addition, there is a strong need to develop RC materials that can independently modulate their thermal radiative properties

across the visible, NIR, and broadband IR or LWIR bands, particularly for applications where transparency is essential, such as windows.

- Multifunctional RC materials: Multifunctional RC materials are not merely capable of cooling features, but also possess other desirable functions, including but not limited to various colors, self-cleaning ability, and resistance to UV. These materials expand the application fields of RC and improve the adaptability and benefits of RC technology. However, designing such materials is a challenging task, as it involves taking into account multiple factors and optimizing different parameters. Furthermore, there may be conflicts or trade-offs among the functions, which require careful balancing and compromise. The integration of RC materials with various technologies where heat management is needed in the device design is another direction. Recent progress has been reported that RC materials could enhance the power conversion efficiency of solar cells.^[18,198] Thus, the integration of RC materials with other technologies could be one of the important research directions.

In summary, RC materials represent one of the greatest opportunities recently to address the urgent global warming issue. We hope this review serves as a platform for further development to bridge the gap between scientific research and commercialization.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

multifunctional materials, radiative cooling, static-cooling materials, stimuli-responsive materials, thermal management

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- [1] D. Zhao, A. Aili, Y. Zhai, S. Xu, G. Tan, X. Yin, R. Yang, *Appl. Phys. Rev.* **2019**, 6, 021306.
- [2] A. P. Raman, M. A. Anoma, L. Zhu, E. Rephaeli, S. Fan, *Nature* **2014**, 515, 540.
- [3] a) B. Bartoli, S. Catalanotti, B. Coluzzi, V. Cuomo, V. Silvestrini, G. Troise, *Appl. Energy* **1977**, 3, 267; b) C. G. Granqvist, A. Hjortsberg, *Appl. Phys. Lett.* **1980**, 36, 139; c) C. G. Granqvist, *Appl. Opt.* **1981**, 20, 2606.
- [4] a) T. Li, Y. Zhai, S. He, W. Gan, Z. Wei, M. Heidarinejad, D. Dalgo, R. Mi, X. Zhao, J. Song, J. Dai, C. Chen, A. Aili, A. Vellore, A. Martini,

- R. Yang, J. Srebric, X. Yin, L. Hu, *Science* **2019**, 364, 760; b) J. Li, Y. Liang, W. Li, N. Xu, B. Zhu, Z. Wu, X. Wang, S. Fan, M. Wang, J. Zhu, *Sci. Adv.* **2022**, 8, 9756; c) J. Mandal, Y. Fu, A. C. Overvig, M. Jia, K. Sun, N. N. Shi, H. Zhou, X. Xiao, N. Yu, Y. Yang, *Science* **2018**, 362, 315; d) D. Li, X. Liu, W. Li, Z. Lin, B. Zhu, Z. Li, J. Li, B. Li, S. Fan, J. Xie, J. Zhu, *Nat. Nanotechnol.* **2021**, 16, 153.
- [5] a) K. Lin, S. Chen, Y. Zeng, T. C. Ho, Y. Zhu, X. Wang, F. Liu, B. Huang, C. Y. Chao, Z. Wang, C. Y. Tso, *Science* **2023**, 382, 691; b) X. Zhao, T. Li, H. Xie, H. Liu, L. Wang, Y. Qu, S. C. Li, S. Liu, A. H. Brozena, Z. Yu, J. Srebric, L. Hu, *Science* **2023**, 382, 684.
- [6] Z. Chen, L. Zhu, A. Raman, S. Fan, *Nat. Commun.* **2016**, 7, 13729.
- [7] L. Zhu, A. P. Raman, S. Fan, *Proc. Natl. Acad. Sci. U. S. A.* **2015**, 112, 12282.
- [8] Y. Zhai, Y. Ma, S. N. David, D. Zhao, R. Lou, G. Tan, R. Yang, X. Yin, *Science* **2017**, 355, 1062.
- [9] T. Wang, Y. Wu, L. Shi, X. Hu, M. Chen, L. Wu, *Nat. Commun.* **2021**, 12, 365.
- [10] a) C. Sui, J. Pu, T. H. Chen, J. Liang, Y. T. Lai, Y. Rao, R. Wu, Y. Han, K. Wang, X. Li, V. Viswanathan, P. C. Hsu, *Nat. Sustainability* **2023**, 6, 428; b) Y. Jia, D. Liu, D. Chen, Y. Jin, C. Chen, J. Tao, H. Cheng, S. Zhou, B. Cheng, X. Wang, Z. Meng, T. Liu, *Nat. Commun.* **2023**, 14, 5087.
- [11] a) S. Wang, T. Jiang, Y. Meng, R. Yang, G. Tan, Y. Long, *Science* **2021**, 374, 1501; b) K. Tang, K. Dong, J. Li, M. P. Gordon, F. G. Reichertz, H. Kim, Y. Rho, Q. Wang, C.-Y. Lin, C. P. Grigoropoulos, A. Javey, J. J. Urban, J. Yao, R. Levinson, J. Wu, *Science* **2021**, 374, 1504.
- [12] X. Li, B. Sun, C. Sui, A. Nandi, H. Fang, Y. Peng, G. Tan, P.-C. Hsu, *Nat. Commun.* **2020**, 11, 6101.
- [13] W. Meng, A. J. J. Kragt, Y. Gao, E. Brembilla, X. Hu, J. S. van der Burgt, A. P. H. J. Schenning, T. Klein, G. Zhou, E. R. van den Ham, L. Tan, L. Li, J. Wang, L. Jiang, *Adv. Mater.* **2024**, 36, 2304910.
- [14] J. Mandal, Y. Yang, N. Yu, A. P. Raman, *Joule* **2020**, 4, 1350.
- [15] K.-Y. Chan, X. Shen, J. Yang, K.-T. Lin, H. Venkatesan, E. Kim, H. Zhang, J.-H. Lee, J. Yu, J. Yang, J.-K. Kim, *Nat. Commun.* **2022**, 13, 5553.
- [16] S. Zeng, S. Pian, M. Su, Z. Wang, M. Wu, X. Liu, M. Chen, Y. Xiang, J. Wu, M. Zhang, Q. Cen, Y. Tang, X. Zhou, Z. Huang, R. Wang, A. Tunuhe, X. Sun, Z. Xia, M. Tian, M. Chen, X. Ma, L. Yang, J. Zhou, H. Zhou, Q. Yang, X. Li, Y. Ma, G. Tao, *Science* **2021**, 373, 692.
- [17] S.-Y. Heo, G. J. Lee, D. H. Kim, Y. J. Kim, S. Ishii, M. S. Kim, T. J. Seok, B. J. Lee, H. Lee, Y. M. Song, *Sci. Adv.* **2020**, 6, 1906.
- [18] Z. Wang, D. Kortge, J. Zhu, Z. Zhou, H. Torsina, C. Lee, P. Bermel, *Joule* **2020**, 4, 2702.
- [19] B. Zhu, W. Li, Q. Zhang, D. Li, X. Liu, Y. Wang, N. Xu, Z. Wu, J. Li, X. Li, P. B. Catrysse, W. Xu, S. Fan, J. Zhu, *Nat. Nanotechnol.* **2021**, 16, 1342.
- [20] A. P. Raman, W. Li, S. Fan, *Joule* **2019**, 3, 2679.
- [21] I. Haechler, H. Park, G. Schnoering, T. Gulich, M. Rohner, A. Tripathy, A. Milionis, T. M. Schutzius, D. Poulikakos, *Sci. Adv.* **2021**, 7, 3978.
- [22] a) Z. H. Li, Q. Y. Chen, Y. Song, B. Zhu, J. Zhu, *Adv. Mater. Technol.* **2020**, 5, 1901007; b) X. Yu, J. Chan, C. Chen, *Nano Energy* **2021**, 88, 106259; c) M. Lee, G. Kim, Y. Jung, K. R. Pyun, J. Lee, B. W. Kim, S. H. Ko, *Light: Sci. Appl.* **2023**, 12, 134; d) J. Liu, H. Tang, C. Jiang, S. Wu, L. Ye, D. Zhao, Z. Zhou, *Adv. Funct. Mater.* **2022**, 32, 2206962; e) K. Bu, X. Huang, X. Li, H. Bao, *Adv. Funct. Mater.* **2023**, 33, 2307191; f) W. Gao, Y. Chen, *Small* **2023**, 19, 2206145; g) J. Liu, Y. Zhang, S. Li, C. Valenzuela, S. Shi, C. Jiang, S. Wu, L. Ye, L. Wang, Z. Zhou, *Chem. Eng. J.* **2023**, 453, 139739; h) W. Xie, C. Xiao, Y. Sun, Y. Fan, B. Zhao, D. Zhang, T. Fan, H. Zhou, *Adv. Funct. Mater.* **2023**, 33, 2305734; i) D. Zhang, H. Zhang, Z. Xu, Y. Zhao, *Materials* **2023**, 16, 3677; j) X. Li, Z. Ding, L. Kong, X. Fan, Y. Li, J. Zhao, L. Pan, D. S. Wiersma, L. Pattelli, H. Xu, *Mater. Adv.* **2023**, 4, 804; k) S. Wu, Y. Cao, Y. Li, W. Sun, *Adv. Opt. Mater.* **2023**, 11, 2202163; l) M. Hu, B. Zhao, Suhendri, X.

- Ao, J. Cao, Q. Wang, S. Riffat, Y. Su, G. Pei, *Renewable Sustainable Energy Rev.* **2022**, 160, 112304; m) M. Zeyghami, D. Y. Goswami, E. Stefanakos, *Sol. Energy Mater. Sol. Cells* **2018**, 178, 115; n) Suhendri, M. Hu, Y. Su, J. Darkwa, S. Riffat, *Buildings* **2020**, 10, 215; o) J. Chai, J. Fan, *Adv. Energy Mater.* **2023**, 13, 2202932; p) Y. An, Y. Fu, J.-G. Dai, X. Yin, D. Lei, *Cell Rep. Phys. Sci.* **2022**, 3, 101098; q) M. He, B. Zhao, X. Yue, Y. Chen, F. Qiu, T. Zhang, *Nano Energy* **2023**, 116, 108821; r) K.-T. Lin, J. Han, K. Li, C. Guo, H. Lin, B. Jia, *Nano Energy* **2021**, 80, 105517; s) J. Liu, Z. Zhou, J. Zhang, W. Feng, J. Zuo, *Mater. Today Phys.* **2019**, 11, 100161; t) S. So, J. Yun, B. Ko, D. Lee, M. Kim, J. Noh, C. Park, J. Park, J. Rho, *Adv. Sci.* **2024**, 11, 2305067; u) X. Yin, R. Yang, G. Tan, S. Fan, *Science* **2020**, 370, 786.
- [23] J. L. Miller, *Principles of Infrared Technology*, Van Nostrand Reinhold, New York **1994**.
- [24] Z. Zhou, X. Wang, Y. Ma, B. Hu, J. Zhou, *Cell Rep. Phys. Sci.* **2020**, 1, 100231.
- [25] Z. Tong, J. Peoples, X. Li, X. Yang, H. Bao, X. Ruan, *Mater. Today Phys.* **2022**, 24, 100658.
- [26] X. Li, J. Peoples, P. Yao, X. Ruan, *ACS Appl. Mater. Interfaces* **2021**, 13, 21733.
- [27] E. O. Filatova, A. S. Konashuk, *J. Phys. Chem. C* **2015**, 119, 20755.
- [28] Z. Cheng, F. Wang, D. Gong, H. Liang, S. Yong, *Sol. Energy Mater. Sol. Cells* **2020**, 213, 110563.
- [29] H. Tang, S. Li, Y. Zhang, Y. Na, C. Sun, D. Zhao, J. Liu, Z. Zhou, *J. Cleaner Prod.* **2022**, 380, 135035.
- [30] D. Chae, H. Lim, S. So, S. Son, S. Ju, W. Kim, J. Rho, H. Lee, *ACS Appl. Mater. Interfaces* **2021**, 13, 21119.
- [31] L. Zhou, H. Song, J. Liang, M. Singer, M. Zhou, E. Stegenburgs, N. Zhang, C. Xu, T. Ng, Z. Yu, B. Ooi, Q. Gan, *Nat. Sustainability* **2019**, 2, 718.
- [32] R. Landauer, *IBM J. Res. Dev.* **2000**, 44, 251.
- [33] C.-L. Luo, L.-X. Zheng, J.-Y. Jiao, W.-G. Yan, J. Zhao, G.-Z. Jia, Z.-F. Liu, Z.-B. Liu, J.-G. Tian, *Opt. Mater.* **2023**, 138, 113710.
- [34] M. Gao, L. Zhu, C. K. Peh, G. W. Ho, *Energy Environ. Sci.* **2019**, 12, 841.
- [35] W. Zhu, B. Drogue, Q. Shen, Y. Zhang, T. G. Parton, X. Shan, R. M. Parker, M. F. L. De Volder, T. Deng, S. Vignolini, T. Li, *Adv. Sci.* **2022**, 9, 2202061.
- [36] C. G. Ribbing, E. Wackelgard, *Thin Solid Films* **1991**, 206, 312.
- [37] J. Jaramillo-Fernandez, G. L. Whitworth, J. A. Pariente, A. Blanco, P. D. Garcia, C. Lopez, C. M. Sotomayor-Torres, *Small* **2019**, 15, 1905290.
- [38] a) C. Lin, Y. Li, C. Chi, Y. S. Kwon, J. Huang, Z. Wu, J. Zheng, G. Liu, C. Y. Tso, C. Y. H. Chao, B. Huang, *Adv. Mater.* **2022**, 34, 2109350; b) S. Atiganyanun, J. B. Plumley, S. J. Han, K. Hsu, J. Cytrynbaum, T. L. Peng, S. M. Han, S. E. Han, *ACS Photonics* **2018**, 5, 1181; c) J. J. Greffet, R. Carminati, K. Joulain, J. P. Mulet, S. Mainguy, Y. Chen, *Nature* **2002**, 416, 61.
- [39] H. C. Van de Hulst, *Light Scattering by Small Particles*, Dover, New York **1957**.
- [40] a) D. Xie, Z. Yang, X. Liu, S. Cui, H. Zhou, T. Fan, *Soft Matter* **2019**, 15, 4294; b) H. Zhang, K. C. S. Ly, X. Liu, Z. Chen, M. Yan, Z. Wu, X. Wang, Y. Zheng, H. Zhou, T. Fan, *Proc. Natl. Acad. Sci. U. S. A.* **2020**, 117, 14657; c) N. N. Shi, C. C. Tsai, F. Camino, G. D. Bernard, N. Yu, R. Wehner, *Science* **2015**, 349, 298.
- [41] S. Li, W. Zhao, Y. Sun, N. Dai, G. Hu, *AIP Adv.* **2020**, 10, 055224.
- [42] H. Bartsch, S. Lange, P. Frach, K. Goedicke, *Surf. Coat. Technol.* **2004**, 180, 616.
- [43] S. Taylor, Y. Yang, L. Wang, *J. Quant. Spectrosc. Radiat. Transfer* **2017**, 197, 76.
- [44] D. Liu, Y. Xu, Y. Xuan, *Appl. Opt.* **2020**, 59, 6861.
- [45] K. Ikeda, H. T. Miyazaki, T. Kasaya, K. Yamamoto, Y. Inoue, K. Fujimura, T. Kanakugi, M. Okada, K. Hatade, S. Kitagawa, *Appl. Phys. Lett.* **2008**, 92, 021117.
- [46] a) J.-W. Cho, S.-J. Park, S.-J. Park, Y.-B. Kim, Y.-J. Moon, S.-K. Kim, *Nano Lett.* **2021**, 21, 3974; b) I.-H. Lee, D. Yoo, P. Avouris, T. Low, S.-H. Oh, *Nat. Nanotechnol.* **2019**, 14, 313.
- [47] R. Zhu, D. Hu, Z. Chen, X. Xu, Y. Zou, L. Wang, Y. Gu, *Nano Lett.* **2020**, 20, 6974.
- [48] a) S. Y. Jeong, C. Y. Tso, Y. M. Wong, C. Y. H. Chao, B. Huang, *Sol. Energy Mater. Sol. Cells* **2020**, 206, 110296; b) J. Sun, X. Wang, J. Wu, C. Jiang, J. Shen, M. A. Cooper, X. Zheng, Y. Liu, Z. Yang, D. Wu, *Sci. Rep.* **2018**, 8, 5438; c) R. H. Siddique, G. Gomard, H. Holscher, *Nat. Commun.* **2015**, 6, 6909.
- [49] M. M. Hossain, M. Gu, *Adv. Sci.* **2016**, 3, 1500360.
- [50] F. Trombe, *Rev. Gen. Therm.* **1967**, 6, 1285.
- [51] P. Grenier, *Rev. Phys. Appl.* **1979**, 14, 87.
- [52] B. Orel, M. K. Gunde, A. Krainer, *Sol. Energy* **1993**, 50, 477.
- [53] E. M. Lushiku, C.-G. Granqvist, *Appl. Opt.* **1984**, 23, 1835.
- [54] a) W. Gao, Z. Lei, K. Wu, Y. Chen, *Adv. Funct. Mater.* **2021**, 31, 2100535; b) W. Huang, Y. Chen, Y. Luo, J. Mandal, W. Li, M. Chen, C.-C. Tsai, Z. Shan, N. Yu, Y. Yang, *Adv. Funct. Mater.* **2021**, 31, 2010334.
- [55] Y. Ding, Z. Pang, K. Lan, Y. Yao, G. Panzarasa, L. Xu, M. L. Ricco, D. R. Rammer, J. Y. Zhu, M. Hu, X. Pan, T. Li, I. Burgert, L. Hu, *Chem. Rev.* **2023**, 123, 1843.
- [56] R. Mi, T. Li, D. Dalgo, C. Chen, Y. Kuang, S. He, X. Zhao, W. Xie, W. Gan, J. Zhu, J. Srebric, R. Yang, L. Hu, *Adv. Funct. Mater.* **2019**, 30, 1907511.
- [57] L. Zhou, J. Zhao, H. Huang, F. Nan, G. Zhou, Q. Ou, *ACS Photonics* **2021**, 8, 3301.
- [58] H. Zhong, Y. Li, P. Zhang, S. Gao, B. Liu, Y. Wang, T. Meng, Y. Zhou, H. Hou, C. Xue, Y. Zhao, Z. Wang, *ACS Nano* **2021**, 15, 10076.
- [59] R. A. Yalcin, E. Blandre, K. Joulain, J. Drévilion, *Sol. Energy Mater. Sol. Cells* **2020**, 206, 110320.
- [60] Y. Li, Z. An, X. Liu, R. Zhang, *Sol. Energy Mater. Sol. Cells* **2022**, 246, 111918.
- [61] a) K. Yao, H. Ma, M. Huang, H. Zhao, J. Zhao, Y. Li, S. Dou, Y. Zhan, *ACS Appl. Nano Mater.* **2019**, 2, 5512; b) H. Ma, K. Yao, S. Dou, M. Xiao, M. Dai, L. Wang, H. Zhao, J. Zhao, Y. Li, Y. Zhan, *Sol. Energy Mater. Sol. Cells* **2020**, 212, 110584; c) A. Kong, B. Cai, P. Shi, X.-C. Yuan, *Opt. Express* **2019**, 27, 30102.
- [62] a) D. Chae, M. Kim, P.-H. Jung, S. Son, J. Seo, Y. Liu, B. J. Lee, H. Lee, *ACS Appl. Mater. Interfaces* **2020**, 12, 8073; b) S. Y. Jeong, C. Y. Tso, J. Ha, Y. M. Wong, C. Y. H. Chao, B. Huang, H. Qiu, *Renewable Energy* **2020**, 146, 44; c) K. Sun, C. A. Riedel, Y. Wang, A. Urbani, M. Simeoni, S. Mengali, M. Zalkovskij, B. Bilenberg, C. H. de Groot, O. L. Muskens, *ACS Photonics* **2018**, 5, 495.
- [63] a) Y. Fu, J. Yang, Y. S. Su, W. Du, Y. G. Ma, *Sol. Energy Mater. Sol. Cells* **2019**, 191, 50; b) D. Lee, M. Go, S. Son, M. Kim, T. Badloe, H. Lee, J. K. Kim, J. Rho, *Nano Energy* **2021**, 79, 105426.
- [64] M. M. Hossain, B. Jia, M. Gu, *Adv. Opt. Mater.* **2015**, 3, 1047.
- [65] C. Zou, G. Ren, M. M. Hossain, S. Nirantar, W. Withayachumnankul, T. Ahmed, M. Bhaskaran, S. Sriram, M. Gu, C. Fumeaux, *Adv. Opt. Mater.* **2017**, 5, 1700460.
- [66] Y. Yang, L. Long, S. Meng, N. Denisuk, G. Chen, L. Wang, Y. Zhu, *Sol. Energy Mater. Sol. Cells* **2020**, 211, 110548.
- [67] D. Zhao, H. Tang, *Science* **2023**, 382, 644.
- [68] T. Li, H. Sun, M. Yang, C. Zhang, S. Lv, B. Li, L. Chen, D. Sun, *Chem. Eng. J.* **2023**, 452, 139518.
- [69] J. Zhao, Q. Meng, Y. Li, Z. Yang, J. Li, *ACS Appl. Mater. Interfaces* **2023**, 15, 47286.
- [70] a) N. Yang, Y. Fu, X. Xue, D. Lei, J.-G. Dai, *EcoMat* **2022**, 5, e12284; b) J. Huang, D. Fan, Q. Li, *Mater. Today Energy* **2022**, 25, 100955.
- [71] J. S. Dolado, G. Goracci, S. Arrese-Igor, A. Ayuela, A. Torres, I. Liberal, M. Beruete, J. J. Gaitero, M. Cagnoni, F. Cappelluti, *ACS Appl. Opt. Mater.* **2023**.
- [72] D. Liu, A. Kaja, J. C. O. Zepper, D. Fan, D. Zhang, H. J. H. Brouwers, Q. Yu, *Energy Build.* **2023**, 285, 112909.

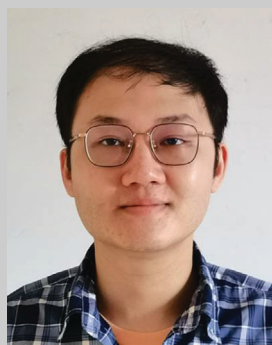
- [73] M. Cagnoni, A. Tibaldi, J. S. Dolado, F. Cappelluti, *iScience* **2022**, 25, 105320.
- [74] Y. Tian, H. Shao, X. Liu, F. Chen, Y. Li, C. Tang, Y. Zheng, *ACS Appl. Mater. Interfaces* **2021**, 13, 22521.
- [75] Y. Ji, Y. Sun, J. Muhammad, X. Li, Z. Liu, P. Tu, Y. Wang, Z. Cai, B. Xu, *Macromol. Mater. Eng.* **2022**, 307, 2100795.
- [76] X. Wu, Y. Wang, X. Liang, Y. Zhang, P. Bi, M. Zhang, S. Li, H. Liang, S. Wang, H. Wang, H. Lu, Y. Zhang, *Adv. Funct. Mater.* **2023**, 34, 2313539.
- [77] Z. Chen, Y. Jiang, Z. Yang, M. Zhang, P. Yao, T. Liu, W. Li, B. Zhu, *ACS Appl. Opt. Mater.* **2023**.
- [78] P. Xu, B. Xiang, W. Zhong, Y. Wu, Y. Zhang, Y. Chang, W. Lei, R. Zhang, *Sol. Energy Mater. Sol. Cells* **2023**, 253, 112209.
- [79] Y. Xu, X. Zhang, Y. Li, Y. Zhang, T. Zhao, Y. Zeng, *Chem. Eng. J.* **2024**, 481, 148722.
- [80] H. Jiang, S. Zhao, H. Ju, Z. Li, J. Chen, N. Li, X. Liu, *Chem. Eng. J.* **2023**, 473, 145432.
- [81] S. Son, Y. Liu, D. Chae, H. Lee, *ACS Appl. Mater. Interfaces* **2020**, 12, 57832.
- [82] B. Liu, C. Xue, H. Zhong, X. Guo, H. Wang, H. Li, M. Du, M. Huang, R. Wei, L. Song, B. Chang, Z. Wang, *J. Mater. Chem. A* **2021**, 9, 24276.
- [83] F. Nan, Y. Zhu, H. Wei, Y. Lin, B. Fan, L. Zhou, *Opt. Express* **2022**, 30, 6003.
- [84] H. Wang, C. Xue, Z. Ji, M. Huang, Z. Jiang, B. Liu, F. Deng, Q. An, X. Guo, *ACS Appl. Mater. Interfaces* **2022**, 14, 51307.
- [85] Q. Tian, X. Tu, L. Yang, H. Liu, Y. Zhou, Y. Xing, Z. Chen, S. Fan, J. Evans, S. He, *Small* **2022**, 18, 2205091.
- [86] Z. Yang, J. Zhang, *ACS Appl. Mater. Interfaces* **2021**, 13, 43387.
- [87] M. Yang, W. Zou, J. Guo, Z. Qian, H. Luo, S. Yang, N. Zhao, L. Pattelli, J. Xu, D. S. Wiersma, *ACS Appl. Mater. Interfaces* **2020**, 12, 25286.
- [88] Y. Song, Y. Li, D. Yan, J. Lei, Z. Li, *Composites, Part A* **2020**, 130, 105738.
- [89] U. Banik, A. Agrawal, H. Meddeb, O. Sergeev, N. Reininghaus, M. Götz-Köhler, K. Gehrke, J. Stührenberg, M. Vehse, M. Sznajder, C. Agert, *ACS Appl. Mater. Interfaces* **2021**, 13, 24130.
- [90] M. Chen, D. Pang, J. Mandal, X. Chen, H. Yan, Y. He, N. Yu, Y. Yang, *Nano Lett.* **2021**, 21, 1412.
- [91] S. Meng, L. Long, Z. Wu, N. Denisuk, Y. Yang, L. Wang, F. Cao, Y. Zhu, *Sol. Energy Mater. Sol. Cells* **2020**, 208, 110393.
- [92] X. Huang, J. Mandal, J. Xu, A. P. Raman, *Joule* **2022**, 6, 2762.
- [93] A. Aili, Z. Y. Wei, Y. Z. Chen, D. L. Zhao, R. G. Yang, X. B. Yin, *Mater. Today Phys.* **2019**, 10, 100127.
- [94] K.-T. Lin, X. Nian, K. Li, J. Han, N. Zheng, X. Lu, C. Guo, H. Lin, B. Jia, *eLight* **2023**, 3, 22.
- [95] J. Zhou, T. G. Chen, Y. Tsurimaki, A. Hajj-Ahmad, L. Fan, Y. Peng, R. Xu, Y. Wu, S. Assaworrorarit, S. Fan, M. R. Cutkosky, Y. Cui, *Joule* **2023**, 7, 2830.
- [96] D. Zhao, A. Aili, Y. Zhai, J. Lu, D. Kidd, G. Tan, X. Yin, R. Yang, *Joule* **2019**, 3, 111.
- [97] S. Zhang, Z. Wu, Z. Liu, H. Lin, Z. Lin, J. Li, S. Kong, Z. Hu, *Mater. Today Commun.* **2022**, 31, 103398.
- [98] Y. Liu, S. Son, D. Chae, P.-H. Jung, H. Lee, *Sol. Energy Mater. Sol. Cells* **2020**, 213, 110561.
- [99] A. R. Gentle, G. B. Smith, *Nano Lett.* **2010**, 10, 373.
- [100] Y. Liu, A. Bai, Z. Fang, Y. Ni, C. Lu, Z. Xu, *Materials* **2019**, 12, 1208.
- [101] X. Xue, M. Qiu, Y. Li, Q. M. Zhang, S. Li, Z. Yang, C. Feng, W. Zhang, J. G. Dai, D. Lei, W. Jin, L. Xu, T. Zhang, J. Qin, H. Wang, S. Fan, *Adv. Mater.* **2020**, 32, 1906751.
- [102] J. Li, Y. Fu, J. Zhou, K. Yao, X. Ma, S. Gao, Z. Wang, J. G. Dai, D. Lei, X. Yu, *Sci. Adv.* **2023**, 9, 1837.
- [103] L. Xiong, Y. Wei, C. Chen, X. Chen, Q. Fu, H. Deng, *Nat. Commun.* **2023**, 14, 6129.
- [104] X. Liu, C. Xiao, P. Wang, M. Yan, H. Wang, P. Xie, G. Liu, H. Zhou, D. Zhang, T. Fan, *Adv. Opt. Mater.* **2021**, 9, 2101151.
- [105] X. Wang, X. Liu, Z. Li, H. Zhang, Z. Yang, H. Zhou, T. Fan, *Adv. Funct. Mater.* **2020**, 30, 1907562.
- [106] K. Zhou, W. Li, B. B. Patel, R. Tao, Y. Chang, S. Fan, Y. Diao, L. Cai, *Nano Lett.* **2021**, 21, 1493.
- [107] L. Hu, C. Xue, B. Liu, X. Guo, J. Wang, F. Deng, *ACS Appl. Polym. Mater.* **2022**, 4, 3343.
- [108] B. Xiang, R. Zhang, Y. Luo, S. Zhang, L. Xu, H. Min, S. Tang, X. Meng, *Nano Energy* **2021**, 81, 105600.
- [109] C. Chen, X. Jia, X. Li, M. Shi, J. Hu, M. Song, S. Wu, H. Dai, X. Wang, H. Geng, *Chem. Eng. J.* **2023**, 475, 146307.
- [110] H. Wang, C. Xue, X. Guo, B. Liu, Z. Ji, M. Huang, S. Jia, *Appl. Mater. Today* **2021**, 24, 101100.
- [111] C. Park, C. Park, S. Park, J. Lee, Y. S. Kim, Y. Yoo, *Chem. Eng. J.* **2023**, 459, 141652.
- [112] R. Liu, Z. Zhou, X. Mo, P. Liu, B. Hu, J. Duan, J. Zhou, *ACS Appl. Mater. Interfaces* **2022**, 14, 46972.
- [113] Z. Cheng, H. Han, F. Wang, Y. Yan, X. Shi, H. Liang, X. Zhang, Y. Shuai, *Nano Energy* **2021**, 89, 106377.
- [114] S. Feng, Y. Zhou, X. Chen, S. Shi, C. Liu, T. Zhang, *J. Mater. Chem. A* **2021**, 9, 25178.
- [115] S. Feng, Y. Zhou, C. Liu, T. Zhang, X. Bu, Y. Huang, M. He, *Chem. Eng. J.* **2023**, 452, 139377.
- [116] X. Huang, N. Li, J. Wang, D. Liu, J. Xu, Z. Zhang, M. Zhong, *ACS Appl. Mater. Interfaces* **2020**, 12, 2252.
- [117] P. Yang, J. He, Y. Ju, Q. Zhang, Y. Wu, Z. Xia, L. Chen, S. Tang, *Adv. Sci.* **2023**, 10, 2206176.
- [118] J. Wang, Y. Zhang, L. Tian, H. Wang, T. Wang, C. Feng, J. Xia, S. Zhang, *Optik* **2020**, 223, 165547.
- [119] P. Yao, Z. Chen, T. Liu, X. Liao, Z. Yang, J. Li, Y. Jiang, N. Xu, W. Li, B. Zhu, J. Zhu, *Adv. Mater.* **2022**, 34, 2208236.
- [120] B. Xiang, X. Yin, J. Zhang, *Compos. Sci. Technol.* **2017**, 145, 149.
- [121] X. Li, J. Peoples, Z. Huang, Z. Zhao, J. Qiu, X. Ruan, *Cell Rep. Phys. Sci.* **2020**, 1, 100221.
- [122] C. Feng, P. Yang, H. Liu, M. Mao, Y. Liu, T. Xue, J. Fu, T. Cheng, X. Hu, H. J. Fan, K. Liu, *Nano Energy* **2021**, 85, 105971.
- [123] W. Zhu, Y. Zhang, C. Zhang, X. Shan, A. K. Rao, S. L. Pitts, T. J. Woodbury, T. S. Masnyk, D. Derome, D. M. Warsinger, X. Ruan, L. J. Mauer, J. Carmeliet, T. Li, *Commun. Eng.* **2023**, 2, 35.
- [124] L. Cai, A. Y. Song, W. Li, P.-C. Hsu, D. Lin, P. B. Catrysse, Y. Liu, Y. Peng, J. Chen, H. Wang, J. Xu, A. Yang, S. Fan, Y. Cui, *Adv. Mater.* **2018**, 30, 1802152.
- [125] D. Miao, N. Cheng, X. Wang, J. Yu, B. Ding, *Nano Lett.* **2022**, 22, 680.
- [126] P. Li, A. Wang, J. Fan, Q. Kang, P. Jiang, H. Bao, X. Huang, *Adv. Funct. Mater.* **2021**, 32, 2109542.
- [127] Z. Yang, T. Jiang, J. Zhang, *Sol. Energy Mater. Sol. Cells* **2021**, 219, 110783.
- [128] Y. Zhang, X. Tan, G. Qi, X. Yang, D. Hu, P. Fyffe, X. Chen, *Sol. Energy Mater. Sol. Cells* **2021**, 229, 111129.
- [129] M. M. Qazilbash, M. Brehm, B.-G. Chae, P.-C. Ho, G. O. Andreev, B.-J. Kim, S. J. Yun, A. V. Balatsky, M. B. Maple, F. Keilmann, H.-T. Kim, D. N. Basov, *Science* **2007**, 318, 1750.
- [130] X. Ao, B. Li, B. Zhao, M. Hu, H. Ren, H. Yang, J. Liu, J. Cao, J. Feng, Y. Yang, Z. Qi, L. Li, C. Zou, G. Pei, *Proc. Natl. Acad. Sci. U. S. A.* **2022**, 119, e2120557119.
- [131] M. Liu, X. Li, L. Li, L. Li, S. Zhao, K. Lu, K. Chen, J. Zhu, T. Zhou, C. Hu, Z. Lin, C. Xu, B. Zhao, G. Zhang, G. Pei, C. Zou, *ACS Nano* **2023**, 17, 9501.
- [132] K. Sun, W. Xiao, C. Wheeler, M. Simeoni, A. Urbani, M. Gaspari, S. Mengali, C. H. d. Groot, O. L. Muskens, *Nanophotonics* **2022**, 11, 4101.
- [133] S. Bhupathi, S. Wang, G. Wang, Y. Long, *Nanophotonics* **2024**, 13, 711.

- [134] S. Abdollahramezani, O. Hemmatyar, H. Taghinejad, A. Krasnok, Y. Kiarashinejad, M. Zandehshahvar, A. Alù, A. Adibi, *Nanophotonics* **2020**, 9, 1189.
- [135] Y. Qu, Q. Li, K. Du, L. Cai, J. Lu, M. Qiu, *Laser Photonics Rev.* **2017**, 11, 1700091.
- [136] J. H. Ko, D. H. Kim, S.-H. Hong, S.-K. Kim, Y. M. Song, *iScience* **2023**, 26, 105780.
- [137] W. Zha, Y. Zhu, B. Ma, J. Yu, P. Ghosh, M. Qiu, Q. Li, *ACS Photonics* **2023**, 10, 2165.
- [138] Q. Zhang, Y. Lv, Y. Wang, S. Yu, C. Li, R. Ma, Y. Chen, *Nat. Commun.* **2022**, 13, 4874.
- [139] Q. Zhang, Y. Wang, Y. Lv, S. Yu, R. Ma, *Proc. Natl. Acad. Sci. U. S. A.* **2022**, 119, e2207353119.
- [140] C. Lin, J. Hur, C. Y. H. Chao, G. Liu, S. Yao, W. Li, B. Huang, *Sci. Adv.* **2022**, 8, 7359.
- [141] M. Li, D. Liu, H. Cheng, L. Peng, M. Zu, *Sci. Adv.* **2020**, 6, 3494.
- [142] Y. Rao, J. Dai, C. Sui, Y.-T. Lai, Z. Li, H. Fang, X. Li, W. Li, P.-C. Hsu, *ACS Energy Lett.* **2021**, 6, 3906.
- [143] W. T. Neo, M. H. Chua, J. W. Xu, *Electrochromic Smart Materials: Fabrication and Applications*, The Royal Society of Chemistry, United Kingdom **2019**, p. 22.
- [144] D. Banerjee, T. Hallberg, S. Chen, C. Kuang, M. Liao, H. Kariis, M. P. Jonsson, *Cell Rep. Phys. Sci.* **2023**, 4, 101274.
- [145] B. Wang, G. Xu, S. Song, Z. Ren, D. Liu, L. Hao, F. Xing, H. Wei, L. Zhang, Y. Li, *Chem. Eng. J.* **2022**, 445, 136819.
- [146] T.-H. Chen, Y. Hong, C.-T. Fu, A. Nandi, W. Xie, J. Yin, P.-C. Hsu, *PNAS Nexus* **2023**, 2, 165.
- [147] Z. Wu, Z. Lian, S. Yan, J. Li, J. Xu, S. Chen, Z. Tang, S.-P. Wang, K. W. Ng, *ACS Nano* **2022**, 16, 13199.
- [148] J. Mandal, S. Du, M. Dontigny, K. Zaghib, N. Yu, Y. Yang, *Adv. Funct. Mater.* **2018**, 28, 1802180.
- [149] T. Bai, W. Li, G. Fu, Y. Shen, Q. Zhang, J. Liu, P. Xue, K. Zhou, H. Wang, *ACS Appl. Mater. Interfaces* **2022**, 14, 52193.
- [150] K. C. S. Ly, H. Zhang, X. Liu, D. Tang, J. Xu, X. Fu, H. Zhou, T. Fan, *J. Am. Ceram. Soc.* **2021**, 104, 2143.
- [151] C. Xu, G. T. Stiubianu, A. A. Gorodetsky, *Science* **2018**, 359, 1495.
- [152] E. M. Leung, M. Colorado Escobar, G. T. Stiubianu, S. R. Jim, A. L. Vyatskikh, Z. Feng, N. Garner, P. Patel, K. L. Naughton, M. Follador, E. Karshalev, M. D. Trexler, A. A. Gorodetsky, *Nat. Commun.* **2019**, 10, 1947.
- [153] H. Fang, W. Xie, X. Li, K. Fan, Y.-T. Lai, B. Sun, S. Bai, W. J. Padilla, P.-C. Hsu, *Nano Lett.* **2021**, 21, 4106.
- [154] Z. Zhou, Y. Fang, X. Wang, E. Yang, R. Liu, X. Zhou, Z. Huang, H. Yin, J. Zhou, B. Hu, *Nano Energy* **2022**, 93, 106865.
- [155] T. Castle, D. M. Sussman, M. Tanis, R. D. Kamien, *Sci. Adv.* **2016**, 2, e1601258.
- [156] D. Jung, C. Lim, H. J. Shim, Y. Kim, C. Park, J. Jung, S. I. Han, S.-H. Sunwoo, K. W. Cho, G. D. Cha, D. C. Kim, J. H. Koo, J. H. Kim, T. Hyeon, D.-H. Kim, *Science* **2021**, 373, 1022.
- [157] H.-R. Lim, H. S. Kim, R. Qazi, Y.-T. Kwon, J.-W. Jeong, W.-H. Yeo, *Adv. Mater.* **2020**, 32, 1901924.
- [158] S. Wang, Y. Dong, Y. Li, K. Ryu, Z. Dong, J. Chen, Z. Dai, Y. Ke, J. Yin, Y. Long, *Mater. Horiz.* **2023**, 10, 4243.
- [159] H. Yin, X. Zhou, Z. Zhou, R. Liu, X. Mo, Z. Chen, E. Yang, Z. Huang, H. Li, H. Wu, J. Zhou, Y. Long, B. Hu, *Research* **2023**, 6, 0103.
- [160] X. A. Zhang, S. Yu, B. Xu, M. Li, Z. Peng, Y. Wang, S. Deng, X. Wu, Z. Wu, M. Ouyang, Y. Wang, *Science* **2019**, 363, 619.
- [161] J. Mandal, M. Jia, A. Overvig, Y. Fu, E. Che, N. Yu, Y. Yang, *Joule* **2019**, 3, 3088.
- [162] Y. Ke, Y. Li, L. Wu, S. Wang, R. Yang, J. Yin, G. Tan, Y. Long, *ACS Energy Lett.* **2022**, 7, 1758.
- [163] S. Wang, Y. Zhou, T. Jiang, R. Yang, G. Tan, Y. Long, *Nano Energy* **2021**, 89, 106440.
- [164] M. Shi, Z. Song, J. Ni, X. Du, Y. Cao, Y. Yang, W. Wang, J. Wang, *ACS Nano* **2023**, 17, 2029.
- [165] a) B. Zhao, C. Wang, M. Hu, X. Ao, J. Liu, Q. Xuan, G. Pei, *Energy* **2022**, 238, 121761; b) Z. Yi, Y. Lv, D. Xu, J. Xu, H. Qian, D. Zhao, R. Yang, *Energy Built Environ.* **2021**, 2, 214.
- [166] Y.-H. Chen, C.-W. Hwang, S.-W. Chang, M.-T. Tsai, K. N. Jayakumaran, L.-C. Yang, Y.-C. Lo, F.-H. Ko, H.-C. Wang, H.-L. Chen, D. Wan, *Adv. Funct. Mater.* **2023**, 33, 2301924.
- [167] K. W. Lee, W. Lim, M. S. Jeon, H. Jang, J. Hwang, C. H. Lee, D. R. Kim, *Adv. Funct. Mater.* **2021**, 32, 2105882.
- [168] a) X. Zhang, X. Li, F. Wang, W. Yuan, Z. Cheng, H. Liang, Y. Yan, *Adv. Opt. Mater.* **2022**, 10, 2202031; b) M. Kim, D. Lee, S. Son, Y. Yang, H. Lee, J. Rho, *Adv. Opt. Mater.* **2021**, 9, 2002226; c) Y. Zhu, H. Luo, C. Yang, B. Qin, P. Ghosh, S. Kaur, W. Shen, M. Qiu, P. Belov, Q. Li, *Light: Sci. Appl.* **2022**, 11, 122.
- [169] C. Zhou, I. Julianri, S. Wang, S. H. Chan, M. Li, Y. Long, *ACS Mater. Lett.* **2021**, 3, 883.
- [170] a) B. Xie, Y. Liu, W. Xi, R. Hu, *Mater. Today Energy* **2023**, 34, 101302; b) W. Li, Y. Shi, Z. Chen, S. Fan, *Nat. Commun.* **2018**, 9, 4240.
- [171] Z. Ding, X. Li, X. Fan, M. Xu, J. Zhao, Y. Li, H. Xu, *Carbon Capture Sci. Technol.* **2022**, 4, 100066.
- [172] L. Zhu, A. Raman, S. Fan, *Appl. Phys. Lett.* **2013**, 103, 223902.
- [173] C. Sheng, Y. An, J. Du, X. Li, *ACS Photonics* **2019**, 6, 2545.
- [174] G. J. Lee, Y. J. Kim, H. M. Kim, Y. J. Yoo, Y. M. Song, *Adv. Opt. Mater.* **2018**, 6, 1800707.
- [175] a) W. Wang, H. Xing, X. Shu, X. Zhao, X. Yan, B. Hong, L. Sun, W. Zhang, G. P. Wang, *Optica* **2023**, 10, 1059; b) Z. Ding, L. Pattelli, H. Xu, W. Sun, X. Li, L. Pan, J. Zhao, C. Wang, X. Zhang, Y. Song, J. Qiu, Y. Li, R. Yang, *Small* **2022**, 18, 2202400.
- [176] Y. Chen, J. Mandal, W. Li, A. Smith-Washington, C. C. Tsai, W. Huang, S. Shrestha, N. Yu, R. P. S. Han, A. Cao, Y. Yang, *Sci. Adv.* **2020**, 6, 5413.
- [177] X. Li, H. Xu, Y. Yang, F. Li, S. Ramakrishna, J. Yu, D. Ji, X. Qin, *Mater. Horiz.* **2023**, 10, 2487.
- [178] S. Jeon, S. Son, S. Y. Lee, D. Chae, J. H. Bae, H. Lee, S. J. Oh, *ACS Appl. Mater. Interfaces* **2020**, 12, 54763.
- [179] S. Son, S. Jeon, D. Chae, S. Y. Lee, Y. Liu, H. Lim, S. J. Oh, H. Lee, *Nano Energy* **2021**, 79, 105461.
- [180] T. Y. Yoon, S. Son, S. Min, D. Chae, H. Y. Woo, J.-Y. Chae, H. Lim, J. Shin, T. Paik, H. Lee, *Mater. Today Phys.* **2021**, 21, 100510.
- [181] K. Zhou, X. Yan, S. J. Oh, G. Padilla-Rivera, H. A. Kim, D. M. Cropek, N. Miljkovic, L. L. Cai, *Nano Lett.* **2023**, 23, 3669.
- [182] G. Wei, D. Yang, T. Zhang, X. Yue, F. Qiu, *Sol. Energy Mater. Sol. Cells* **2019**, 202, 110125.
- [183] M.-C. Huang, M. Yang, X.-J. Guo, C.-H. Xue, H.-D. Wang, C.-Q. Ma, Z. Bai, X. Zhou, Z. Wang, B.-Y. Liu, Y.-G. Wu, C.-W. Qiu, C. Hou, G. Tao, *Prog. Mater. Sci.* **2023**, 137, 101144.
- [184] S. Dong, Q. Wu, W. Zhang, G. Xia, L. Yang, J. Cui, *ACS Appl. Mater. Interfaces* **2022**, 14, 4571.
- [185] J. Song, W. Zhang, Z. Sun, M. Pan, F. Tian, X. Li, M. Ye, X. Deng, *Nat. Commun.* **2022**, 13, 4805.
- [186] C. Cai, W. Chen, Z. Wei, C. Ding, B. Sun, C. Gerhard, Y. Fu, K. Zhang, *Nano Energy* **2023**, 114, 108625.
- [187] M. Li, C. Lin, K. Li, W. Ma, B. Doppoophya, Y. Li, B. Huang, *Small* **2023**, 19, 2301159.
- [188] Y. Tu, X. Tan, G. Qi, X. Yang, X. Ouyang, W. Yan, W. Hu, J. Geng, R. Yang, *Mater. Today Phys.* **2023**, 34, 101070.
- [189] Y. Liu, Y. Zhang, T. Chen, Z. Jin, W. Feng, M. Li, L. Chen, C. Wang, *Adv. Funct. Mater.* **2023**, 33, 2307240.
- [190] a) E. Cuce, P. M. Cuce, C. J. Wood, S. B. Riffat, *Renewable Sustainable Energy Rev.* **2014**, 34, 273; b) X. Yue, H. Wu, T. Zhang, D. Yang, F. Qiu, *Energy* **2022**, 245, 123287.

- [191] A. Leroy, B. Bhatia, C. C. Kelsall, A. Castillejo-Cuberos, H. M. Di Capua, L. Zhao, L. Zhang, A. M. Guzman, E. N. Wang, *Sci. Adv.* **2019**, 5, 9480.
- [192] X. Liu, M. Zhang, Y. Hou, Y. Pan, C. Liu, C. Shen, *Adv. Funct. Mater.* **2022**, 32, 2207414.
- [193] M. D. R. Batista, A. L. Troksa, H. V. Eshelman, M. Bagge-Hansen, J. D. Roehling, *Mater. Horiz.* **2023**, 10, 5694.
- [194] J. Zhou, C. Ding, X. Zhang, D. Li, D. Yang, B. You, L. Wu, *ACS Appl. Mater. Interfaces* **2023**, 15, 57514.
- [195] Y. Zhang, T. Wang, X. Mei, M. Chen, L. Wu, *ACS Photonics* **2023**, 10, 3124.
- [196] R. H. Galib, Y. Tian, Y. Lei, S. Dang, X. Li, A. Yudhanto, G. Lubineau, Q. Gan, *Nat. Commun.* **2023**, 14, 6707.
- [197] X. Wu, J. Li, Q. Jiang, W. Zhang, B. Wang, R. Li, S. Zhao, F. Wang, Ya Huang, P. Lyu, Y. Zhao, J. Zhu, R. Zhang, *Nat. Sustainability* **2023**, 6, 1446.
- [198] S. Y. Heo, D. H. Kim, Y. M. Song, G. J. Lee, *Adv. Energy Mater.* **2022**, 12, 2103258.



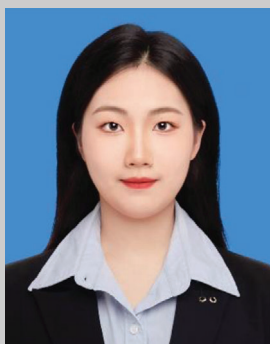
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